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Machine Learning for Satellite Link Fade Nowcasting

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Abstract

Satellite communication links may experience fade events that degrade the received signal and require the activation of a backup communication resource. In this setting, the relevant objective is not only to predict the future signal accurately, but to support a timely switch decision using only information available at the current time. This thesis studies satellite link fade nowcasting as a decision-oriented time-series problem.

The work develops a reproducible signal-only machine-learning framework for satellite fade data acquired at the Fucino ground station. The raw files are standardized, cleaned, segmented, and converted into task-specific supervised datasets. An external holdout source is kept separate from the development data to evaluate generalization to an unseen acquisition file. A common Perfect Switch reference and shared post-processing procedure are used to compare models through operational switch metrics rather than only through native prediction losses.

Four task formulations are investigated: autoregressive forecasting of the future signal trajectory, current-level persistence regression, long-fade detection, and survival persistence modeling. The evaluated model families include GRU forecasting models, PatchTST, Chronos zero-shot forecasting, XGBoost baselines, learnable shapelet models, temporal convolutional networks, XGBoost-AFT, and a discrete-time survival TCN.

The results show that native predictive accuracy alone is not sufficient to assess operational usefulness. Forecast errors, duration errors, classification scores, and survival metrics are useful diagnostics, but final switch quality depends on the formulation, conversion rule, and post-processing. XGBoost-based current-level persistence provides a precise and conservative switch policy, while XGBoost-AFT survival persistence offers a stronger recall-oriented residual-duration formulation. PatchTST is the strongest autoregressive switch model, whereas long-fade detection is more appropriate as a high-recall warning mechanism than as a final controller. Overall, satellite fade nowcasting should be evaluated as a complete decision pipeline in which modeling choices and operational switch metrics are inseparable.

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Chapter 1

Introduction

Satellite communication links are affected by propagation impairments that can temporarily reduce the received signal quality. This problem is especially relevant at high frequencies, where atmospheric phenomena such as rain can produce strong fades on Earth-space paths [1, 2]. In operational systems, a fade is not only a physical phenomenon to be described after it occurs. It is also a decision problem: when the primary link degrades, an operator or automated controller may need to activate a backup communication resource. Activating this resource too late may lead to service degradation, while activating it too often or for too long may waste capacity and increase operational cost.

The central question of this thesis is therefore not simply whether the future satellite signal can be predicted. The more relevant question is whether information available at the current time can support a timely and reliable switch decision. This distinction changes how the problem should be modelled and evaluated. A model with a low forecast error is not necessarily a good switch controller if its errors occur around the operational threshold. A model that estimates a duration or a probability may be more useful than a trajectory forecaster if that quantity is more directly connected to the decision. For this reason, the thesis treats satellite fade nowcasting as a decision-oriented time-series problem.

1.1 Operational Motivation

Fade mitigation techniques are commonly used to preserve link availability under adverse propagation conditions. Depending on the system, mitigation may involve power control, adaptive coding and modulation, site diversity, or the activation of an alternative link. The specific mitigation mechanism is outside the scope of this thesis. The focus here is the upstream decision that such a mechanism requires: given the

signal observed up to the current timestamp, should the system switch to a backup resource?

This decision has an intrinsically temporal structure. A very short threshold crossing may not justify a switch, because the system could recover before the backup activation becomes useful. Conversely, a fade that persists for several minutes may justify switching even if the instantaneous signal is not at its worst point yet. The operational objective is therefore tied to persistence, residual duration, and event evolution, not only to the instantaneous signal level. This is consistent with the broader propagation literature, where fade dynamics and time-series behavior are treated as relevant design quantities for Earth-space communication systems [3, 4].

This thesis was carried out at M.B.I. S.r.l. [5] within the broader context of the NEFOCAST project, for which M.B.I. is the lead company. NEFOCAST investigates an integrated technological platform for estimating rainfall fields from satellite-signal attenuation measurements [6]. This industrial and research setting provides the application context for the decision-oriented nowcasting problem addressed in this work.

The data used in this work consist of signal time series acquired at the Fucino ground station from multiple files and time periods. The available files are heterogeneous: some are uplink or downlink beacon measurements, some are associated with different satellites, and only a subset contains auxiliary columns. To make the experimental comparison reproducible and fair, the thesis adopts a strict signal-only policy. Every dataset is standardized to a common time and signal representation, and all models use only the historical signal as input. This avoids comparing models that have access to different physical variables and keeps the focus on what can be learned from the signal evolution itself.

1.2 Satellite Link Fade Nowcasting

Nowcasting refers to short-term prediction at or near the present time. In this work, the prediction time is a timestamp t in a satellite signal time series. The information available to a model is the recent past and present signal history, while the target concerns the future behavior of the same signal or of the ongoing fade event. No future observation can be used as an input. This no-leakage constraint is essential because the final goal is an operational decision that could be made online.

The thesis investigates four task formulations that all serve the same final switching objective but estimate different intermediate quantities:

- autoregressive forecasting, where the model predicts a future signal trajectory;

- current-level persistence, where the model estimates how long the signal will remain close to its current level;
- long-fade detection, where the model estimates whether the current grouped fade event will last longer than an operational duration threshold;
- survival persistence, where the model estimates a residual-duration survival function for the ongoing fade event.

These formulations are not interchangeable. They answer different operational questions and produce different native outputs: trajectories, scalar durations, class probabilities, or survival curves. To compare them, the outputs must be converted into binary switch sequences and evaluated against a common reference. This thesis therefore builds a shared Perfect Switch reference, applies common switch post-processing, and reports operational metrics in addition to native model metrics.

1.3 Research Questions

The work is guided by four research questions.

1. **Can task-specific machine-learning models improve operational switch decisions for satellite link fades?** The objective is to evaluate whether learned models produce switch sequences that are closer to the reference decision than simple threshold reasoning.
2. **Which predictive formulation is most useful for the switching objective?** The thesis compares trajectory forecasting, duration regression, event-level classification, and survival modeling because the best formulation is not obvious a priori.
3. **Do native predictive metrics explain operational usefulness?** Forecast RMSE, duration MAE, classification AUPRC, and survival metrics are informative, but the final decision depends on thresholding, post-processing, and event timing. The thesis therefore studies how native metrics relate to switch metrics.
4. **How important are common reference construction and post-processing?** Since different tasks produce outputs on different timestamp domains and with different semantics, the comparison must separate modeling differences from evaluation artifacts.

These questions deliberately place the switch decision at the centre of the study. The intermediate predictions remain important, but they are evaluated as components of a decision pipeline rather than as isolated forecasting outputs.

1.4 Contributions

The main methodological and empirical contributions of the thesis are the following.

1. It formulates satellite fade nowcasting as a decision-oriented problem through four complementary predictive tasks: autoregressive forecasting, current-level persistence, long-fade detection, and survival persistence. Each task estimates a different intermediate quantity but is connected explicitly to the same operational switch objective.
2. It establishes a shared switch-evaluation framework with a common Perfect Switch reference, shared post-processing, task-native comparisons, and a cross-task reference-grid comparison. This framework makes models with different native outputs operationally comparable.
3. It evaluates a range of model families under these formulations, including GRU models, PatchTST, Chronos zero-shot forecasting, XGBoost baselines, learnable shapelet models, temporal convolutional networks, XGBoost-AFT, and a discrete-time survival TCN.
4. It provides an empirical analysis of the relationship between native prediction quality and operational switch quality. The results show that the best native predictor is not always the best switch controller and that task formulation strongly affects the precision-recall trade-off.
5. To support reproducibility, it develops a signal-only preprocessing pipeline for heterogeneous satellite fade and beacon files. The pipeline validates and standardizes raw data, handles gaps conservatively, constructs task-specific datasets, and preserves the metadata required for event-level evaluation.

The source-code repository for the experimental framework is hosted on GitHub: <https://github.com/Giusgarus/Nowcasting-project>. It contains the task implementations, experiment configurations, and evaluation scripts used in this work.

1.5 Thesis Structure

Chapter 2 introduces the technical background needed for the work. It covers time-series data, machine-learning models used in the thesis, survival analysis, nowcasting and validation principles, and the satellite communication context.

Chapter 3 describes the data preparation and reference-decision framework. It explains the available signal files, the signal-only policy, parsing and validation, sampling gaps, conservative imputation, event construction, the external holdout split, the Perfect Switch reference, and the shared switch post-processing rules.

Chapter 4 presents the four modeling formulations. For each formulation, it defines the input representation, target construction, model families, native output, and switch-conversion rule.

Chapter 5 describes the experimental protocol, including dataset construction, train-validation-test usage, hyperparameter search, native metrics, switch metrics, intra-task comparisons, cross-task comparisons, and reproducibility settings.

Chapter 6 reports and discusses the empirical results. It first analyses the native predictive performance within each task and then focuses on the switch metrics and event-level behavior that determine operational usefulness.

Chapter 7 summarizes the main findings, answers the research questions, discusses limitations, and outlines future work.

Chapter 2

Background

Before the individual nowcasting tasks can be defined, two questions have to be made explicit: what kind of object a time series is, and what assumptions are introduced when it is transformed into supervised learning data. The background therefore starts from time-indexed observations, with emphasis on ordering, sampling, windowing, and the risk of temporal leakage. It then reviews the machine-learning model families used later in the thesis at the level of their modeling mechanisms rather than as task-specific experimental choices. On this basis, a separate section defines the nowcasting problem and the validation principles needed to assess temporal generalization. The final part returns to the satellite-link setting, where the technical modeling problem becomes an operational decision about whether a backup link should be activated. Detailed model instantiations are left to chapter 4, while training, selection, and evaluation procedures are left to the experimental setup chapter.

2.1 Time Series as Data

2.1.1 Time Index and Information Order

A time series is not merely a table with a timestamp column. It is a realization of a temporal process observed at ordered time instants [7, 8]. In the setting considered in this thesis, each observation consists of a timestamp t_i and a scalar signal value S_i :

$$\{(t_i, S_i)\}_{i=1}^N, \quad t_1 < t_2 < \dots < t_N.$$

Equivalently, the measured signal can be interpreted as samples from an underlying process $S(t)$, observed only at the finite set of times $\{t_i\}_{i=1}^N$. The temporal index is therefore part of the data definition. Two sequences containing the same numerical values but arranged in a different order can describe different physical evolutions.

Let $\mathcal{F}_i$ denote the information available at time t_i . Formally, it is the sigma-algebra generated by $S_1, \dots, S_i$: the smallest sigma-algebra with respect to which these random variables are measurable. A causal nowcasting model must base its decision at t_i only on $\mathcal{F}_i$. This notation is useful because it separates the offline construction of labels from the online information set. Future observations may be used after the fact to define a supervised target, namely the outcome the model is trained to predict, but they are not part of the information available when the operational decision is made.

This makes time-series data different from ordinary exchangeable tabular data. In a tabular classification problem, the rows are often treated as independent or at least exchangeable after conditioning on their features. In a time series, row $i + 1$ is not an arbitrary new case: it is the continuation of the same temporal evolution observed at row i . Models, preprocessing steps, and validation protocols must respect this direction of time.

2.1.2 Temporal Dependence and Non-Stationarity

Consecutive samples in a time series are statistically dependent. In a satellite-link signal, the value at t_i is informative about nearby future values because atmospheric and equipment states evolve continuously rather than being redrawn independently at each timestamp. This temporal dependence is the basic reason why forecasting methods use past observations as input features [9].

For a regularly sampled univariate series, second-order temporal dependence is often summarized through the autocovariance and autocorrelation functions [8]. If the process is weakly stationary, with constant mean μ and variance, the lag- k autocovariance is

$$\gamma(k) = \text{Cov}(S_i, S_{i-k}) = \mathbb{E}[(S_i - \mu)(S_{i-k} - \mu)],$$

and the autocorrelation is

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)}.$$

Positive autocorrelation at small lags means that recent values carry predictive information about the next samples. For fade nowcasting, this local dependence is central: an above-threshold sample is operationally more important when it occurs inside a persistent fade trajectory than when it is an isolated fluctuation.

Stationarity is a useful reference concept, but real measurement series often violate it. A strictly stationary process has joint distributions that are invariant under time shifts, while weak stationarity requires constant first and second moments

and autocovariances that depend only on lag. Satellite-link data can depart from these assumptions because weather regimes, seasonal effects, receiver conditions, maintenance interventions, and acquisition quality can change over long periods. A model trained on one part of the acquisition history may therefore face a different distribution later in time.

Fade-related signals also combine dense ordinary behavior with sparse operationally relevant events. Most timestamps may correspond to acceptable link conditions, while the samples near or above an operational threshold determine whether a backup link should be activated. This creates an imbalance between global signal reconstruction and decision relevance. A small numerical error far from the threshold may have little operational effect, whereas a similar error near the threshold can change the switch decision.

2.1.3 Sampling Cadence, Gaps, and Acquisition Segments

Time-series models often operate on index steps, but the physical process evolves in elapsed time. Let

$$\Delta_i = t_i - t_{i-1}$$

be the interval between two consecutive observations. When Δ_i is approximately constant, a context of L samples corresponds to a stable look-back duration of roughly $L\Delta_i$. When the intervals are irregular or large gaps appear, the same number of samples can correspond to very different amounts of physical time.

This distinction matters for both inputs and targets. If a gap of several minutes is treated as a normal one-step transition, the model receives a distorted representation of the signal dynamics. The same issue affects label construction: a later observation after a long acquisition gap is not evidence that the signal persisted continuously through the missing interval. Missing observations are therefore not only a data-cleaning inconvenience; they change what can be inferred about duration, recovery, and event continuity.

Within the scope of this thesis we distinguish sample order from continuous acquisition. A continuous acquisition segment is a portion of the series where the sampling cadence is regular enough to support windowed modeling and target construction. Context windows must remain inside such segments. Conservative short-gap imputation can be useful when the missing interval is small and explicitly bounded, but it should not bridge long gaps or merge distinct acquisition segments. This principle is methodological rather than model-specific: every downstream task depends on the same interpretation of time.

Sampling cadence also determines the temporal meaning of a prediction horizon. A horizon of ten samples is not intrinsically a ten-step operational forecast; under a 30-second cadence, it corresponds to approximately five minutes. The same horizon would have a different meaning under a different cadence. For this reason, time-series experiments must report both sample-based and time-based interpretations when the operational decision depends on elapsed duration.

2.1.4 Windowed Supervised Learning

In supervised learning, a dataset consists of input–target pairs (X_i, Y_i) . The input X_i contains the information supplied to the model, while the target Y_i provides the observed outcome, or the available information about that outcome, used to supervise learning. The task definition determines how the target is constructed from the recorded data.

A time series is converted into such examples by applying a sliding-window transformation to the ordered observations and associating each window with its target. For a decision made at index i , a context of length L is

$$X_i = [S_{i-L+1}, \dots, S_i].$$

The last value S_i is the newest signal value available when the decision is made. Given this input, a model produces a prediction through the mapping

$$z_i = f_{\vartheta}(X_i),$$

where z_i is the task-specific output and ϑ denotes learned or pretrained parameters. The output may be a future trajectory, a scalar duration, a class probability, or a survival curve.

The prediction z_i and the target Y_i have distinct roles: the former is computed by the model, while the latter supplies the supervision. During training, a task-specific loss function measures how well the model output agrees with the target. Their representations need not be identical. A binary classifier can output a probability while its target is a class label; a survival model can output a curve while its target records a follow-up duration and whether the event was observed or censored.

For example, in multi-step forecasting with horizon h , the target is the observed future trajectory,

$$Y_i = [S_{i+1}, S_{i+2}, \dots, S_{i+h}].$$

The corresponding model output z_i is a predicted trajectory, denoted by $\hat{Y}_i$, which is

compared with Y_i during training and evaluation. If a regularly sampled segment contains N_s observations, the number of valid fully observed forecasting windows with unit stride is

$$N_{\text{win}} = N_s - L - h + 1,$$

provided $N_s \geq L + h$. This simple expression hides an important statistical fact: adjacent windows are strongly overlapping. X_i and X_{i+1} share $L - 1$ observations, so the resulting supervised rows are not independent even if they are stored in a matrix.

The target may depend on future observations during offline dataset generation, but the input must not. This is the central no-leakage constraint:

$$X_i \text{ and all input features at } t_i \text{ may depend only on } \{S_j : j \leq i\}.$$

Future observations can define supervision for offline learning and evaluation, but they cannot enter the input features used for a decision at t_i . Fitted preprocessing parameters and model selection must also respect the separation of training, validation, and test data discussed in section 2.3.2. The same input-time restriction applies to derived features: a rolling mean, slope, threshold occupancy, or event-state feature is valid only if it can be computed from the past and present observations available at t_i .

For multi-step forecasting, several strategies are possible [10]. A recursive strategy trains a one-step-ahead predictor and applies it repeatedly, feeding previous predictions back as inputs. A direct strategy trains one predictor per forecast horizon. A multiple-output strategy predicts the whole future vector in one pass. These strategies make different assumptions about how errors propagate and about whether future horizons should be modelled independently or jointly. The choice is therefore not just an implementation detail: it determines the relation between the learned model, the prediction horizon, and the operational switch rule applied later.

2.2 Machine Learning for Time Series

This section presents the theoretical foundations of the machine-learning model families used in the experiments of this thesis. Its scope follows the implemented approaches: gradient-boosted decision trees with XGBoost, GRU recurrent networks, patch-based Transformers through PatchTST, the pretrained Chronos forecasting model, learnable shapelet models, and temporal convolutional networks. It also introduces the survival-analysis concepts underlying the XGBoost-AFT and discrete-time TCN survival models.

For each family, the following subsections explain its mathematical formulation, how it represents temporal information, and the assumptions that shape its predictions, with reference to the relevant literature. These foundations provide the terminology and notation for chapter 4, where the models are assigned to the predictive tasks and their inputs, outputs, and switch-conversion rules are specified. Training procedures, hyperparameter selection, and evaluation protocols are described in chapter 5.

2.2.1 Gradient-Boosted Decision Trees

Gradient boosting builds an additive model by fitting weak learners sequentially. The general idea was formalized by Friedman as gradient descent in function space [11]. XGBoost is a regularized and scalable implementation of boosted decision trees [12]. A tree ensemble can be written as

$$F(\phi_i) = \sum_{m=1}^M f_m(\phi_i),$$

where ϕ_i is a tabular feature vector and f_m is an individual decision tree. Each tree partitions the feature space into leaves and assigns a constant score to each leaf. If $q_m(\phi_i)$ denotes the leaf reached by sample i in tree m , then $f_m(\phi_i) = w_{q_m(\phi_i)}$, where w_j is the learned score of leaf j .

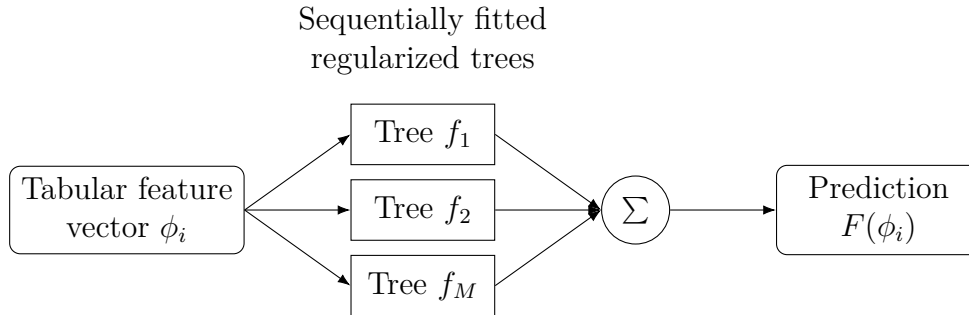


Figure 2.1: Schematic representation of an XGBoost tree ensemble. A tabular feature vector is evaluated by multiple decision trees, and their leaf scores are added to form the final prediction.

Training minimizes a predictive loss plus a regularization term on tree complexity:

$$\mathcal{L} = \sum_i \ell(y_i, F(\phi_i)) + \sum_{m=1}^M \Omega(f_m).$$

In XGBoost, the regularization term penalizes both the number of leaves and the

magnitude of leaf scores:

$$\Omega(f_m) = \gamma T_m + \frac{1}{2} \lambda \sum_{j=1}^{T_m} w_j^2,$$

where T_m is the number of leaves, γ penalizes additional leaves, and λ shrinks leaf weights. This is important because boosted trees can otherwise fit highly specific partitions of the training data, especially when many lag features are derived from overlapping time windows.

At boosting iteration m , the new tree is fitted to improve the current prediction $\hat{y}_i^{(m-1)}$. XGBoost uses a second-order Taylor approximation of the loss around the current prediction:

$$\mathcal{L}^{(m)} \simeq \sum_i \left[g_i f_m(\phi_i) + \frac{1}{2} h_i f_m(\phi_i)^2 \right] + \Omega(f_m),$$

where

$$g_i = \partial_{\hat{y}_i^{(m-1)}} \ell(y_i, \hat{y}_i^{(m-1)}), \quad h_i = \partial_{\hat{y}_i^{(m-1)}}^2 \ell(y_i, \hat{y}_i^{(m-1)}).$$

The first derivative indicates the direction in which the prediction should be corrected, while the second derivative scales that correction by local curvature. For a candidate split, XGBoost compares the score of the parent node with the scores of the two child nodes. If G and H are the sums of gradients and Hessians in a node, the split gain has the form

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right] - \gamma.$$

Splits are therefore selected only when they reduce the regularized objective, not merely when they improve the training loss locally.

For time-series data, a boosted tree does not process temporal order directly. The ordered context must first be converted into a feature vector, for example by flattening lagged signal values and adding scalar summaries of the recent window:

$$\phi_i = [S_{i-L+1}, \dots, S_i, \eta_1(X_i), \dots, \eta_p(X_i)],$$

where $\eta_1, \dots, \eta_p$ are optional summaries computed only from the available context, such as recent slopes or window statistics. In this representation, temporal order is encoded by feature position: the model can distinguish S_i from S_{i-10} , but it has no built-in notion that these features are adjacent samples. This is both a limitation and a strength. It is a limitation because the model does not share parameters across

time and does not learn a recurrent state. It is a strength because decision trees can create piecewise rules involving specific lags, levels, and interactions, such as a large current signal combined with a positive recent slope.

The same ensemble mechanism can be paired with different losses. Squared-error or absolute-error variants support scalar regression; logistic objectives support binary event classification; and the accelerated-failure-time objective extends the framework to right-censored duration targets [13, 14]. In a nowcasting study, this makes XGBoost useful in two roles. First, it is a strong tabular baseline that tests how much information is already present in lagged signal values and simple context summaries. Second, it provides a robust comparison point for neural sequence models, because it avoids recurrent or attention-based optimization while still modeling nonlinear threshold effects.

2.2.2 Recurrent Sequence Models and GRUs

Recurrent neural networks process a sequence one element at a time while maintaining a hidden state.

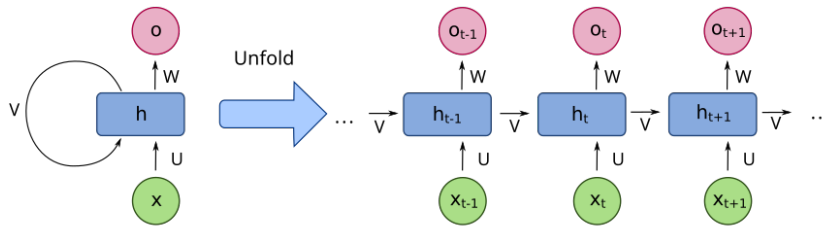


Figure 2.2: Unfolded recurrent neural network processing a sequence through a hidden state. Source: [15].

The basic recurrent idea is to represent time implicitly through feedback in the hidden representation [16]. At sequence position ℓ , a recurrent model can be written as

$$h_\ell = \psi(x_\ell, h_{\ell-1}),$$

where x_ℓ is the current input and $h_{\ell-1}$ summarizes the previous part of the sequence. The hidden state is therefore a learned memory of the past. A forward unidirectional recurrence processes the input window from its oldest observation toward its most recent one.

A bidirectional recurrent network processes the supplied sequence in both directions and combines the forward and backward representations [17]. For a pre-

diction issued at t_i , both directions can operate on the fully observed context $X_i = [S_{i-L+1}, \dots, S_i]$. The backward recurrence then uses observations that are later than an internal window position, but all of them are already available at t_i . Bidirectional encoding of this past context is therefore compatible with the no-leakage constraint. The relevant boundary is the time at which the prediction is issued: using S_j with $j > i$ to make that prediction would violate the constraint. Likewise, a representation computed using the entire window cannot be treated as available at an earlier position inside that window if it depends on observations acquired after that position.

Plain recurrent networks can have difficulty learning long dependencies because gradients may vanish or explode during backpropagation through time. Gated architectures address this by controlling how information is stored and updated. Long short-term memory networks introduced explicit memory cells and gates [18]. Gated recurrent units simplify this idea while retaining gates that regulate state updates.

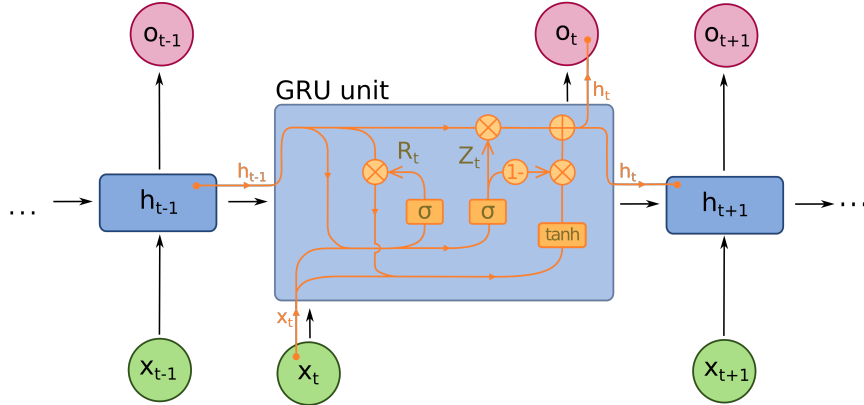


Figure 2.3: Structure of a gated recurrent unit. Source: [19].

GRUs were introduced in the context of encoder-decoder sequence learning [20] and later evaluated as general sequence modeling units [21].

Throughout this thesis, $\sigma(\cdot)$ denotes the logistic sigmoid function,

$$\sigma(z) = \frac{1}{1 + \exp(-z)},$$

applied elementwise when its argument is a vector.

For a univariate input value x_t , a GRU updates its hidden state h_t using an

update gate z_t , a reset gate r_t , and a candidate state $\tilde{h}_t$:

$$\begin{aligned} z_t &= \sigma(W_z x_t + U_z h_{t-1} + b_z), \\ r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r), \\ \tilde{h}_t &= \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h), \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \end{aligned}$$

The update gate controls how much of the candidate state replaces the previous state, while the reset gate controls how much previous memory is used to form the candidate. When z_t is small, the unit mostly preserves its previous state; when z_t is large, it updates the memory toward the new candidate. When r_t is small, the candidate is computed with little dependence on older memory, allowing the unit to react to a local change. When r_t is large, the candidate can use a longer effective history. In a fade-related signal, these gates let the model represent whether the recent pattern resembles a rising fade, a plateau, an oscillation, or a recovery without requiring manually defined event states.

The hidden dimension controls the capacity of this memory. A very small hidden state may compress different temporal patterns into similar representations, whereas a large hidden state can overfit repeated windows from a limited number of fade episodes. Stacking multiple GRU layers increases representational power: lower layers can encode local signal transitions, while upper layers combine those transitions into a coarser sequence representation. Dropout [22] and early stopping [23] are often used to limit overfitting, but they do not replace the need for chronological validation because neighboring windows remain highly correlated.

Two recurrent structures are common in multi-step forecasting. A sequence-to-vector model encodes the whole context and maps the final hidden state directly to an output vector:

$$\hat{Y}_i = W_o a_L + b_o.$$

This is a direct multiple-output forecast: all horizons are predicted from the same encoded context. The advantage is that errors made for early horizons do not have to be fed back into later horizons. The limitation is that the forecast horizon is represented mainly through the output layer, so the model does not explicitly simulate a future recurrent trajectory.

An encoder-decoder model instead uses one recurrent network to encode the observed context and another to decode future values step by step, following the broader sequence-to-sequence framework [24]. The decoder state evolves over the

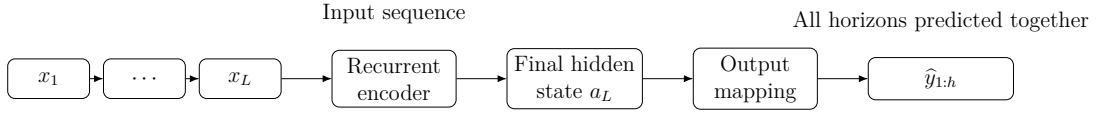


Figure 2.4: Generic sequence-to-vector recurrent forecaster. The input sequence is encoded once, and the final hidden state is mapped directly to a multi-step output vector.

prediction horizon,

$$d_k = \psi_{\text{dec}}(\xi_k, d_{k-1}), \quad \hat{S}_{i+k} = W_d d_k + b_d,$$

where ξ_k is the decoder input at future step k . During training, ξ_k may be the previous true target value, a previous prediction, or a mixture of the two, depending on the teacher-forcing strategy. During evaluation, future true values are unavailable, so decoded predictions must be generated using only the encoded context and previously generated outputs. This gives the forecast horizon an explicit temporal structure, but it can also propagate early prediction errors.

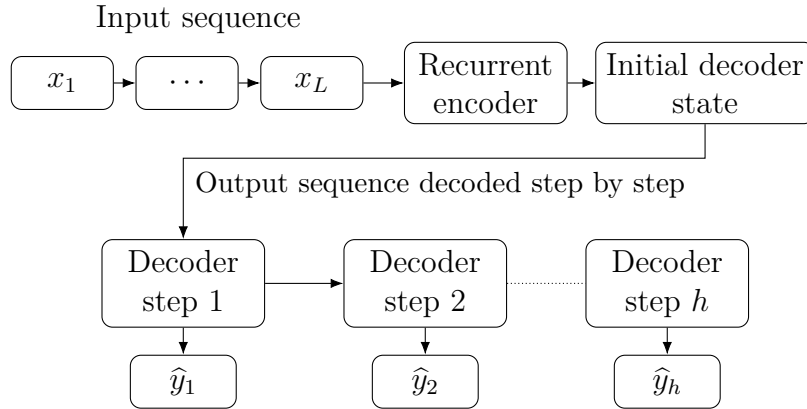


Figure 2.5: Generic encoder-decoder recurrent forecaster. The encoder summarizes the input sequence, while the decoder evolves over the forecast horizon and emits one output at each step.

Compared with tree-based lag models, GRUs encode temporal order directly and share recurrent parameters across positions in the window. This learned memory of the available signal history makes them a natural baseline for testing whether explicit sequence modeling improves over tabular lag features in satellite fade nowcasting.

2.2.3 Transformers and PatchTST

Transformers represent sequences through attention rather than recurrence. The core operation is scaled dot-product attention [25]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

Queries, keys, and values are learned projections of input tokens. Attention allows each token representation to combine information from other tokens in the context.

This mechanism can now be compared with the GRU encoders introduced above. A GRU updates its hidden state sequentially across context positions, whereas a Transformer encoder can compute representations for all observed context tokens in parallel within each layer [25]. Self-attention also connects distant positions directly, while a recurrent encoder propagates information through successive state updates. The GRU forecasting structures described above summarize the context through a final recurrent state; a Transformer encoder retains a sequence of contextualized token representations that can be combined by the output head. The computational trade-off depends on the context length and model dimensions, so greater parallelism alone does not establish which model is faster for the short windows considered in this thesis.

PatchTST adapts this architecture to time-series forecasting through two design choices: patch-based tokenization and channel-independent processing [26, 27]. Instead of using one token per timestamp, the input context is segmented into short subseries.

For a context of length L , patch length P , and stride s , the number of complete patches is

$$N_p = 1 + \left\lfloor \frac{L - P}{s} \right\rfloor,$$

and the j -th patch is

$$p_j = [x_{1+(j-1)s}, \dots, x_{1+(j-1)s+P-1}] \in \mathbb{R}^P.$$

Each patch is projected into a d -dimensional token and combined with positional information:

$$e_j = W_p p_j + b_p + r_j,$$

where W_p and b_p define the patch embedding and r_j encodes the position of the patch inside the look-back window. The ordered token sequence $[e_1, \dots, e_{N_p}]$ is then processed by Transformer encoder blocks, whose self-attention layers allow the

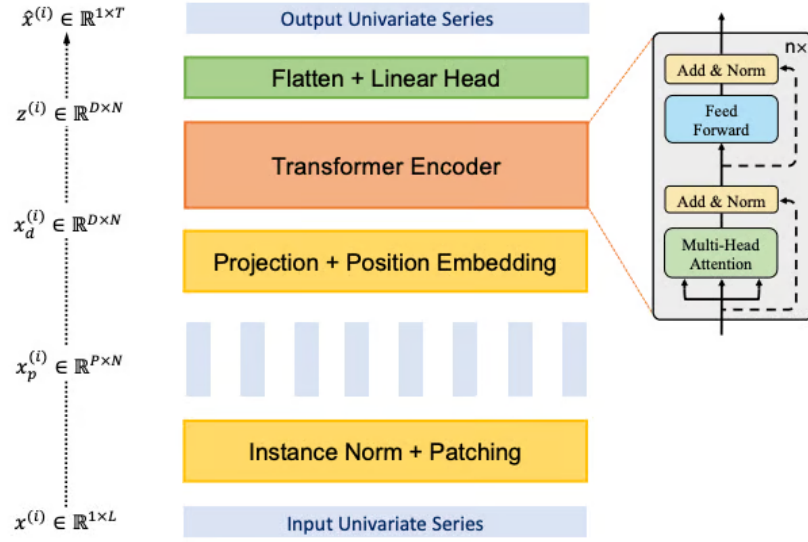


Figure 2.6: PatchTST architecture for a univariate time series. The input sequence is normalized and divided into patches, embedded with positional information, processed by Transformer encoder blocks, and mapped to the output series by a linear head. Source: [26].

representation of one patch to depend on other patches in the context. A forecasting head maps the encoded representation to the prediction horizon, for example through a direct multi-output projection

$$\hat{Y} = W_o \text{vec}(H) + b_o,$$

where H denotes the encoded patch sequence.

Patching changes the granularity at which attention is applied. A token no longer represents an individual sample; it represents a local temporal segment, such as a short rise, plateau, oscillation, or recovery. This has two practical effects. First, the embedding can preserve local shape information before the global attention layers are applied. Second, the attention sequence length is reduced from L to N_p , lowering the cost of self-attention for the same look-back window. The trade-off is that the patch length and stride become important modeling choices: large or non-overlapping patches can smooth rapid changes, while very short or highly overlapping patches reduce the computational advantage and may behave closer to timestamp-level tokenization.

The second defining idea of PatchTST is channel independence. In a multivariate series, each variable is treated as a separate univariate stream, with shared embedding and Transformer weights across channels. This avoids forcing early cross-channel

mixing and improves scalability when many variables are present [26]. In the signal-only setting of this thesis, there is only one channel, so this design reduces to a purely univariate patch encoder. This is consistent with the no-leakage and signal-only data policy: the model can learn temporal structure in the recent signal history without receiving additional covariates.

Practical implementations expose additional architectural and optimization choices. The NeuralForecast implementation, for example, parameterizes the forecast horizon and input size, the number of encoder layers and attention heads, embedding size, patch length, stride, dropout, learnable positional embeddings, residual attention, and reversible instance normalization [27]. These options are not part of the task definition itself; they control how the PatchTST family is instantiated and tuned. For satellite fade nowcasting, the relevant motivation is that patch-level attention can compare local fade patterns across the recent context while still producing a direct forecast over the operational horizon.

2.2.4 Foundation Time-Series Models

Foundation time-series models are pretrained on broad collections of time series and then applied to new forecasting problems with little or no task-specific training. Chronos is one such model family [28]. It scales and quantizes real-valued time-series values into discrete tokens and uses a language-model-style architecture to generate future tokens autoregressively.

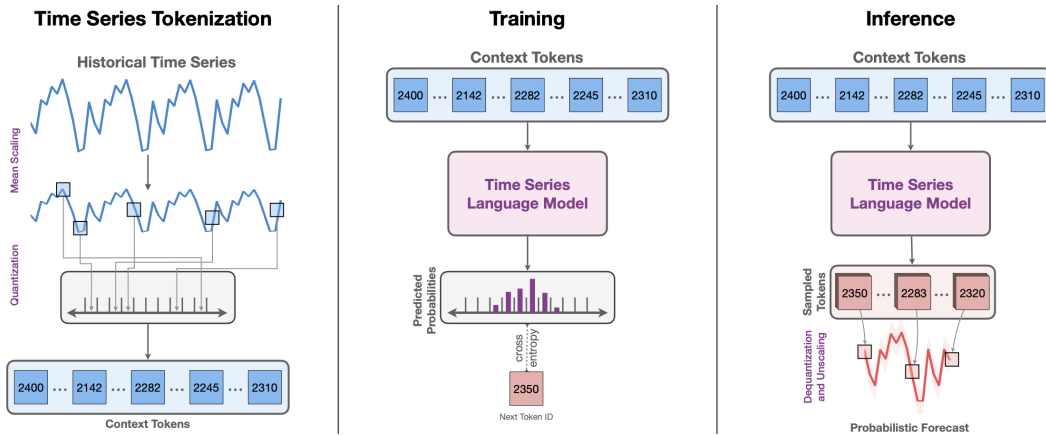


Figure 2.7: High-level Chronos forecasting pipeline. The input time series is scaled and quantized into tokens, processed by a language-model architecture, and sampled autoregressively before being mapped back to numerical trajectories. Multiple sampled trajectories define a predictive distribution. Source: [28].

In zero-shot forecasting, a pretrained model receives only the past context from

the target series and produces a predictive distribution over future trajectories:

$$p(\hat{Y}_i \mid X_i).$$

If multiple forecast samples are generated, a deterministic forecast can be obtained by taking a statistic such as the median at each horizon. The main advantage is that no project-specific parameter fitting is required. The main limitation is domain transfer: the sampling cadence, scale, noise level, and event structure of a satellite fade signal may differ from the data distribution seen during pretraining.

2.2.5 Temporal Convolutional Networks

Temporal convolutional networks use one-dimensional convolutions over ordered sequences. A temporal convolution applies the same local filter at different positions in the context, so the model can detect repeated patterns regardless of where they occur inside the look-back window. This parameter sharing gives a different inductive bias from lag-based trees, where each lag is a separate feature, and from recurrent networks, where a hidden state is updated sequentially.

Let $x_t \in \mathbb{R}^{C_{\text{in}}}$ be the input vector at sequence position t . A causal convolution with kernel size K , dilation d , and output channel r can be written as

$$y_{t,r} = b_r + \sum_{c=1}^{C_{\text{in}}} \sum_{a=0}^{K-1} W_{a,c,r} x_{t-ad,c}.$$

The index $t - ad$ is never greater than t . The representation at position t therefore depends only on current and past samples, which is the key causality requirement for online nowcasting. In practice, causal convolution is implemented with left padding or by trimming the output so that future samples are never included in the receptive field.

The dilation factor spaces the filter taps apart. With $d = 1$, the convolution uses adjacent samples $(x_t, x_{t-1}, \dots)$. With $d = 2$, it uses every second sample $(x_t, x_{t-2}, \dots)$, and so on. Dilated causal convolutions were popularized in sequence generation models such as WaveNet [29]. Their advantage is that the receptive field can grow rapidly without requiring very large kernels or very deep recurrent unrolling.

The receptive field is the part of the past that can influence an output. For a stack of dilated convolutional layers with kernel sizes K_ℓ and dilations d_ℓ , assuming unit stride, the receptive field length is

$$R = 1 + \sum_{\ell} (K_\ell - 1)d_\ell.$$

If a TCN block contains two convolutions with a common kernel size K , and the blocks use exponentially increasing dilations $d_b = 2^b$ for $b = 0, \dots, B-1$, the receptive field becomes

$$R = 1 + 2(K - 1) \sum_{b=0}^{B-1} 2^b = 1 + 2(K - 1)(2^B - 1).$$

This formula makes an important modeling constraint explicit: a TCN can use only the history covered by its receptive field. If $R < L$, the earliest part of a length- L context cannot affect the final output. If R is much larger than the context, part of the nominal receptive field is filled by padding rather than observed data.

A standard TCN is built from residual temporal blocks [30]. A block receives a sequence representation $H^{(b-1)} \in \mathbb{R}^{L \times C_b}$, applies causal dilated convolutions, nonlinearities, normalization, and dropout, and adds the result back to the input:

$$H^{(b)} = H^{(b-1)} + f_b(H^{(b-1)}).$$

When the number of channels changes, a 1×1 convolution can project the residual path to the correct dimension before the addition. The residual connection helps optimization because each block learns a correction to the previous representation instead of having to relearn the entire transformation. Dropout and weight normalization are commonly used to stabilize training and reduce overfitting in stacked convolutional sequence models.

For a sequence-level prediction, the final temporal representation must be converted into a fixed-dimensional vector. Common choices include using the last causal position, applying average or maximum pooling over time, or concatenating several pooled summaries. The last-position representation is the most directly causal for a decision at the end of the context, while temporal pooling can make the model more sensitive to patterns that occurred anywhere in the look-back window. For pointwise sequence outputs, such as a hazard at each future time bin in a discrete-time survival model, the convolutional representation can instead be mapped to a vector of horizon-specific logits.

TCNs occupy an intermediate position between recurrent and attention-based models. Like GRUs, they can be made causal and are well suited to online prediction. Unlike GRUs, they process all positions in parallel during training and do not compress the entire past through a single recurrent state. Like Transformers, they support parallel computation over the sequence, but their dependency pattern is local and controlled by the receptive field rather than by global self-attention. This makes them attractive when the relevant information is expected to lie in local

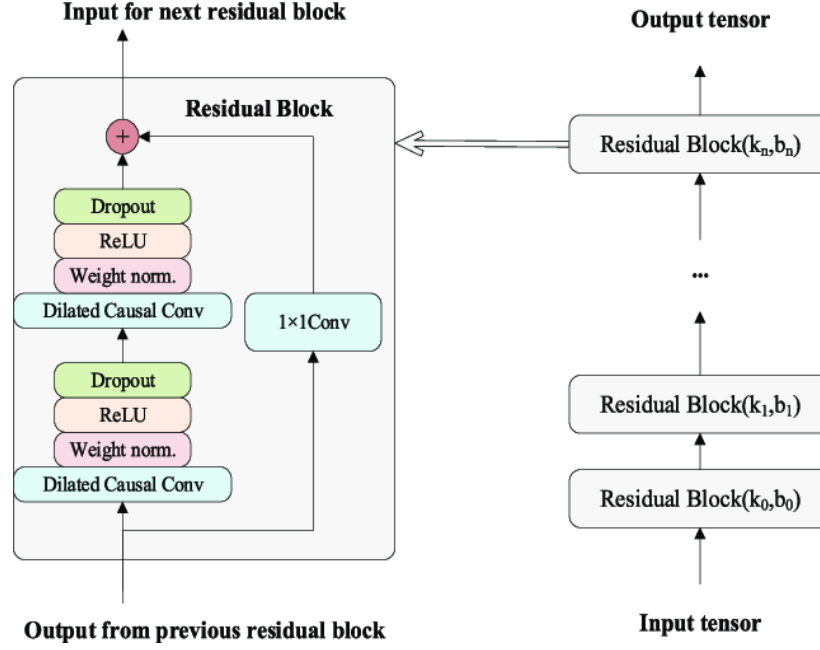


Figure 2.8: Example architecture of a temporal convolutional network. Temporal convolutional blocks combine one-dimensional convolutions, nonlinear transformations, and residual connections to build sequence representations over a finite receptive field. Source: [31].

temporal shapes, slopes, threshold crossings, and short persistence patterns. Their main limitation is that long-range dependence must be covered deliberately through the choice of kernel size, dilation schedule, and number of blocks; otherwise, distant samples are simply outside the model's effective view.

In satellite fade nowcasting, this inductive bias is useful because many operational cues are local but not instantaneous. A single sample above a threshold may be ambiguous, whereas a short pattern of rising signal values, oscillation near the threshold, or sustained exceedance can indicate a different event regime. A causal TCN can learn filters for these patterns and combine them across multiple temporal scales while preserving the no-leakage constraint. This is why TCNs are suitable both for event classification and for discrete-time hazard prediction, where the model must summarize recent history before estimating the probability of future persistence or recovery.

2.2.6 Shapelet-Based Time-Series Modeling

A time-series shapelet is a short subsequence whose presence, absence, or distance to a larger series is informative for prediction. Shapelets were introduced as discriminative subsequences for interpretable time-series data mining [32]. Shapelet transformations

then made it possible to use distances to selected shapelets as features for downstream classifiers [33].

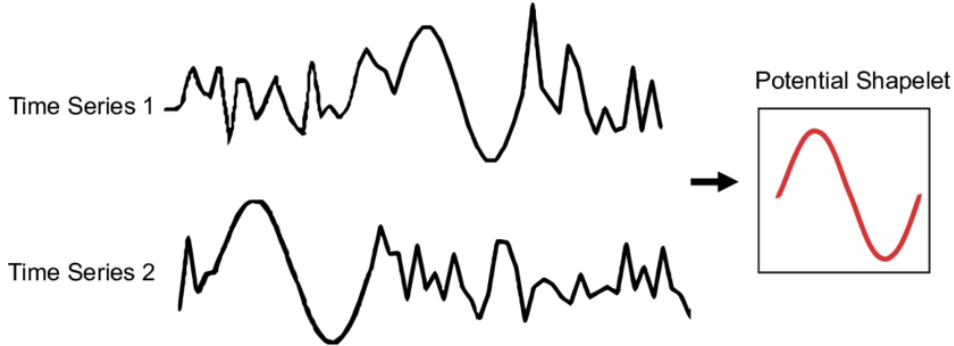


Figure 2.9: Illustration of time-series shapelets as local subsequences that are informative for distinguishing temporal patterns. Source: [34].

Given a shapelet $w \in \mathbb{R}^\ell$ and a context $x \in \mathbb{R}^L$, a standard squared-distance response is

$$d(x, w) = \min_{a=1, \dots, L-\ell+1} \sum_{r=1}^{\ell} (x_{a+r-1} - w_r)^2.$$

The minimum over positions makes the feature approximately phase independent: the pattern may appear early or late in the context and still produce a small distance. This is useful for fade-related modeling because rapid rises, temporary plateaus, partial recoveries, or oscillations around a threshold can be informative even when they are not perfectly aligned across events.

The original shapelet approach searched over candidate subsequences and selected discriminative patterns. Learnable shapelets instead treat the patterns as differentiable model parameters optimized jointly with the prediction head [35]. Multiscale learnable shapelets extend this idea by using several shapelet lengths, allowing the model to represent both short local transitions and longer local persistence patterns.

2.2.7 Survival Analysis for Time-to-Recovery Modeling

Survival analysis studies time-to-event outcomes in the presence of censoring. Classical foundations include the Kaplan-Meier estimator for incomplete observations [36] and Cox regression models for life-table data [37]. Let T be the true event time and C the censoring time. The censoring time is the last time up to which the sample can be followed before observation stops or before the event status becomes unobservable. In this thesis, for example, a fade recovery is censored when the acquisition segment

ends before the end of the ongoing fade can be seen. The observed time is

$$Y = \min(T, C),$$

and the event indicator is

$$\Delta = \mathbb{I}[T \leq C].$$

If $\Delta = 0$, the event has not been observed by time Y , so the valid statement is that $T > Y$. This is right censoring.

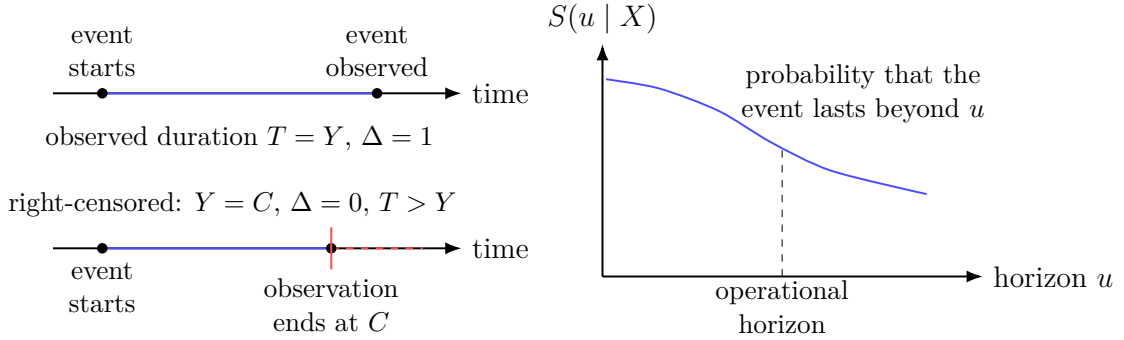


Figure 2.10: Basic survival-analysis quantities. Observed samples have a known event time, while right-censored samples are only known to survive beyond the last observable time. The survival curve maps each future horizon to the probability that the event has not yet ended.

The survival function is

$$S(u | X) = P(T > u | X),$$

which gives the probability that the event time exceeds horizon u given the available input X . In a satellite fade setting, the event of interest can be interpreted as recovery from an ongoing fade rather than failure or death. This makes survival analysis suitable for residual-duration questions, because a survival curve can answer whether the current event is likely to last beyond an operational horizon.

Accelerated Failure Time Models and XGBoost-AFT

Accelerated failure time (AFT) models are a parametric alternative to proportional-hazards models. Instead of modeling the hazard ratio directly, they model the logarithm of the event time as a location model [38, 14]:

$$\log T_i = \mu(X_i) + \tau\epsilon_i,$$

where X_i is the available input for sample i , $\mu(X_i)$ is a location term, $\tau > 0$ is a scale parameter, and ϵ_i follows a chosen distribution such as normal, logistic, or extreme value. In a classical linear AFT model, $\mu(X_i) = \beta^\top X_i$. A positive change in $\mu(X_i)$ multiplies the typical event time by a factor on the original time scale, which motivates the term “accelerated” failure time.

The AFT formulation is useful for censored duration targets because each sample can be represented by lower and upper bounds on its event time. If the recovery time is observed exactly, both bounds correspond to the observed duration, $L_i = U_i = T_i$. If the sample is right-censored at the end of the observable segment, the only valid statement is $T_i > C_i$, which can be encoded as $L_i = C_i$ and $U_i = \infty$. More generally, the likelihood contribution of an interval-censored observation is the model probability that the event time lies inside the interval:

$$P(L_i < T_i \leq U_i \mid X_i) = F_\epsilon \left(\frac{\log U_i - \mu(X_i)}{\tau} \right) - F_\epsilon \left(\frac{\log L_i - \mu(X_i)}{\tau} \right),$$

with the appropriate density limit used for exact observations and with the upper cumulative term equal to one when $U_i = \infty$. The negative log-likelihood derived from these terms trains the model without discarding censored samples.

XGBoost-AFT keeps the same censored-label likelihood but replaces the linear location term with a boosted tree ensemble:

$$\mu(X_i) = F(\phi_i) = \sum_{m=1}^M f_m(\phi_i),$$

where ϕ_i is the tabular representation of the available context [13, 14]. This is important for the present thesis because the survival-persistence task uses past signal windows and scalar summaries as input, while the target is a residual fade duration that may be censored by the end of an acquisition segment. XGBoost-AFT can therefore learn a nonlinear mapping from the current context to the location of the log-duration distribution while still using both observed and right-censored training samples.

Once $\mu(X_i)$, the scale parameter, and the error distribution are fixed, the model implies a full conditional survival curve:

$$S(u \mid X_i) = P(T_i > u \mid X_i) = 1 - F_\epsilon \left(\frac{\log u - \mu(X_i)}{\tau} \right).$$

Thus XGBoost-AFT does not output a binary switch by itself. It first defines a distribution over the residual duration; an operational switch decision is then

obtained by evaluating the survival probability at a chosen horizon and applying a configured probability threshold.

Discrete-time neural survival models partition time into ordered bins and predict a hazard probability for each bin. If $h_k(X)$ is the conditional event probability in bin k , then the discrete survival probability after K bins is

$$S_K(X) = \prod_{k=1}^K (1 - h_k(X)).$$

Neural discrete-time survival models can be trained with likelihood terms for both observed and censored samples [39]. When the hazard model is implemented with a sequence architecture, the input representation can still be restricted to the past context while the output retains a time-to-event interpretation.

2.3 Nowcasting Problem and Validation Principles

The model families described above provide different ways to represent temporal information. Defining a prediction problem additionally requires specifying when a prediction is issued, which observations are available at that time, and which unknown outcome the prediction should describe. These choices establish the nowcasting setting and determine what constitutes a meaningful evaluation.

2.3.1 Prediction at the Current Time

Nowcasting refers to very short-range prediction from the current and recent state of a system. In meteorology, the World Meteorological Organization describes nowcasting as detailed local forecasting from the present to a few hours ahead using rapidly updated observations [40]. The term is used here for signal-based estimates of the near-future evolution or persistence of an ongoing satellite-link degradation, issued to support an operational decision in the next few minutes.

At a decision time t_i , the observed history supplies the information set $\mathcal{F}_i$, and the past context X_i is a finite representation of that information. The target Y_i specifies the outcome to be estimated from this information. It may concern future signal values, a remaining duration, or an event property whose definitive label requires later observations. With model parameters fixed for evaluation, the prediction $z_i = f_\vartheta(X_i)$ must be computable from information available at t_i . The target can be established retrospectively when the relevant future has been observed, or represented as incomplete supervision when observation ends first.

The look-back duration and the prediction horizon have different roles. The context specifies how much past information is supplied to the model; the horizon specifies how far ahead an outcome is assessed. For a regularly sampled forecast with interval Δ , the h -step horizon corresponds to $h\Delta$ units of elapsed time. A duration or survival model can instead describe persistence over a broader range while still informing a decision at a fixed short operational horizon. Thus, a common decision horizon does not require every predictive task to have the same target representation.

The operational question in this thesis is whether the evolution or persistence inferred at t_i justifies activating a backup communication link. Observing that degradation is present does not by itself answer that question: the unobserved continuation determines whether the interruption will be brief or sustained. The predictive output therefore supplies evidence for a decision, through an explicitly defined conversion rule. The satellite context is developed in section 2.4, and the alternative targets and conversion rules are formalized in chapter 4.

2.3.2 Temporal Generalization and Validation

Evaluation must reflect the information constraints of this prediction problem. Two distinct requirements arise: each input must respect its decision-time boundary, and the evaluation data must test the intended form of generalization. A past-only input is insufficient if the training and evaluation sets contain near-duplicate windows from the same episode.

For example, a random split may place X_i in training and X_{i+1} in testing even though they share $L - 1$ observations. Target horizons may overlap as well. Such a split can primarily measure interpolation within already represented episodes, whereas the operational question concerns performance on episodes or acquisition periods that were not used to fit the model.

This does not imply that standard cross-validation is invalid for every time series problem. Bergmeir et al. establish conditions under which it can be used for autoregressive prediction, including uncorrelated model errors [41]. Those conditions should not be assumed merely because a time series has been stored as a table. Temporal dependence, distribution changes, and repeated windows from the same physical episode motivate chronological or event-grouped development splits. Holding out an entire source dataset addresses the additional question of transfer to a different acquisition record.

Training data are used to fit parameters. Validation data guide model and hyperparameter selection, early stopping, and any learned decision threshold. Test data assess the resulting fixed procedure. Transformations with fitted parameters

must follow the same separation. Changing a decision rule after inspecting test performance makes that comparison exploratory rather than an independent final evaluation. The concrete external-holdout construction and experimental protocol are specified in chapters 3 and 5.

2.4 Satellite and Operational Background

2.4.1 Earth-Space Propagation and Fade Events

An Earth-space communication link is affected by deterministic link-budget terms and by time-varying propagation impairments. The deterministic terms include transmit power, antenna gains, free-space path loss, and receiver noise characteristics. The time-varying terms include atmospheric gases, clouds, rain, tropospheric scintillation, depolarization, and low-elevation effects. These effects become increasingly important at high carrier frequencies. In Ku-, Ka-, and V-band satellite systems, atmospheric attenuation is one of the main limits to link availability and therefore a central design and operational concern [1, 42].

The International Telecommunication Union provides several recommendations for Earth-space propagation engineering. Recommendation ITU-R P.618 gives methods for propagation data and prediction in Earth-space telecommunication systems [2]. Recommendation ITU-R P.837 provides precipitation statistics used by propagation models [43]. Recommendation ITU-R P.1623 focuses on fade dynamics, including fade duration and fade slope on Earth-space paths [3], while Recommendation ITU-R P.1853 addresses synthesis of tropospheric impairment time series [4]. These documents are important because they show that attenuation is not only a static exceedance probability problem. Its temporal structure, including how long fades last and how quickly they evolve, matters for systems that react dynamically.

Rain attenuation is especially relevant for short-term availability. Rain cells can produce strong and localized signal degradation, and their movement through the slant path causes attenuation to rise, fluctuate, and recover over time. A classical example of dynamic rain-attenuation modeling is the stochastic model of Maseng and Bakken [44]. This thesis does not build a physical rain model and does not use rain-rate measurements as inputs. The propagation literature nevertheless motivates the modeling premise that recent signal history contains information about the near-future state of the link, while the useful information is local, noisy, and event dependent.

In the data representation used in this work, the measured column S_i is treated as a signal or fade-level quantity. Larger values correspond to worse link conditions.

A fade region is therefore defined by an operational threshold

$$S_i \geq \theta.$$

Real fade events are not perfectly smooth: the signal can cross below and above the threshold several times during one physical degradation episode. Operational event definitions therefore need persistence, grouping, or stable-recovery rules rather than relying only on isolated threshold crossings.

2.4.2 Fade Mitigation Techniques

Satellite systems can respond to fading through several fade mitigation techniques. Examples include static fade margins, uplink power control, adaptive coding and modulation, bandwidth or data-rate adaptation, site diversity, and switching traffic to an alternative path or backup link [1, 42]. These techniques have different physical and operational costs. Increasing transmit power consumes power budget and may be limited by hardware or regulatory constraints. Adaptive coding and modulation protects the link by lowering spectral efficiency. Site diversity or backup-link switching consumes shared network resources and can require non-negligible activation time.

Because mitigation has a cost, the operational objective is not to react to every threshold crossing. A short isolated fade may be tolerable, while a fade that is expected to persist for several minutes may justify early activation of a backup resource. This turns attenuation monitoring into a short-term decision problem: the system must use recent observations to decide whether the current or imminent fade is persistent enough to justify action.

2.4.3 Switch Decisions and Operational Metrics

The final operational decision considered in the thesis is binary: switch or do not switch. Different predictive tasks may estimate different intermediate quantities before that decision is made. A forecasting model estimates future signal values, a duration model estimates persistence time, a classifier estimates an event probability, and a survival model estimates a residual duration curve. These outputs are not directly comparable until each is converted into a binary switch sequence.

This separation between model output and switch rule is important. A model can have a low native predictive loss but still produce poor operational behavior if its errors occur near the threshold, if it triggers too many short switch islands, or if it activates too late inside a fade. Conversely, a model with moderate native error can be operationally useful if it makes reliable decisions near the switch boundary.

For this reason, this thesis distinguishes between native predictive metrics and operational switch metrics. Native metrics evaluate the intermediate quantity predicted by a model. Switch metrics evaluate the binary decision sequence after task-specific conversion and common post-processing.

For real-valued forecasting or duration regression, the main error measures are mean absolute error and root mean squared error. Given targets y_i and predictions $\hat{y}_i$, they are

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|,$$

and

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}.$$

MAE gives the average absolute error in the physical unit of the target. RMSE penalizes large errors more strongly because the residuals are squared before averaging. In multi-step forecasting, these quantities can also be computed by horizon step, replacing the single target y_i with $y_{i,k}$, $k = 1, \dots, h$.

For binary classification, let $q_i \in [0, 1]$ be the predicted probability and $y_i \in \{0, 1\}$ the true class. A probability threshold converts q_i into a class prediction. Precision, recall, and F1 are then defined from the confusion counts

$$\begin{aligned} TP &= \sum_i \mathbb{I}[\hat{y}_i = 1 \wedge y_i = 1], & FP &= \sum_i \mathbb{I}[\hat{y}_i = 1 \wedge y_i = 0], \\ TN &= \sum_i \mathbb{I}[\hat{y}_i = 0 \wedge y_i = 0], & FN &= \sum_i \mathbb{I}[\hat{y}_i = 0 \wedge y_i = 1]. \end{aligned}$$

The corresponding metrics are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

and

$$F_1 = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall}}.$$

The area under the receiver operating characteristic curve, AUROC, evaluates the ranking of positive samples above negative samples across all possible thresholds. The area under the precision-recall curve, AUPRC, evaluates the precision-recall trade-off across thresholds and is often more informative when the positive class is rare.

For survival models, the output is not a point estimate or a class probability but a survival curve $S(u | X)$. The negative log-likelihood measures how much probability the model assigns to the observed time-to-event information. For interval survival

labels $[L_i, U_i]$, used for accelerated failure time models, a generic contribution is

$$-\log P(L_i \leq T_i \leq U_i \mid X_i),$$

with $U_i = \infty$ for right-censored samples. In that case, the likelihood term becomes the probability of surviving beyond the censoring lower bound. Harrell’s concordance index evaluates ranking quality: it is the fraction of comparable sample pairs for which the model assigns shorter predicted survival to the sample with the earlier observed event time. A value of 0.5 is approximately random ranking, while a value near 1 indicates strong ordering.

The Brier score evaluates the calibration and discrimination of survival probabilities at a fixed horizon u . In its simplest uncensored form, it is

$$\text{BS}(u) = \frac{1}{N} \sum_{i=1}^N \left(\mathbb{I}[T_i > u] - \hat{S}(u \mid X_i) \right)^2.$$

With censoring, inverse-probability-of-censoring weighting is commonly used to avoid treating censored event status as known. The integrated Brier score aggregates the horizon-wise Brier score over a range of horizons,

$$\text{IBS} = \frac{1}{u_{\max} - u_{\min}} \int_{u_{\min}}^{u_{\max}} \text{BS}(u) du,$$

or the corresponding discrete average over evaluated time bins.

Operational switch metrics reuse the same confusion-count logic, but the positive class is now a switch decision rather than a native task label. Let $m_i \in \{0, 1\}$ denote the post-processed model switch and $p_i \in \{0, 1\}$ the aligned Perfect Switch reference. The switch-level confusion counts are

$$\begin{aligned} TP &= \sum_i \mathbb{I}[m_i = 1 \wedge p_i = 1], & FP &= \sum_i \mathbb{I}[m_i = 1 \wedge p_i = 0], \\ TN &= \sum_i \mathbb{I}[m_i = 0 \wedge p_i = 0], & FN &= \sum_i \mathbb{I}[m_i = 0 \wedge p_i = 1]. \end{aligned}$$

In addition to precision, recall, and F1, switch evaluation reports intersection over union,

$$\text{IoU} = \frac{TP}{TP + FP + FN},$$

balanced accuracy,

$$\text{BalancedAccuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right),$$

false positive rate and false negative rate,

$$\text{FPR} = \frac{FP}{FP + TN}, \quad \text{FNR} = \frac{FN}{FN + TP}.$$

Undefined divisions are handled explicitly rather than silently converted to optimistic values.

Pointwise switch metrics are complemented by duration and event-level behavior metrics. Active duration counts how many samples the model keeps the switch active,

$$D_m = \sum_i m_i,$$

or $D_m \Delta t$ in seconds when the sampling time is Δt . The Perfect Switch duration is $D_p = \sum_i p_i$, and the outage-mask duration is the number of samples for which the observed signal itself is above the threshold. The number of model switch islands is

$$N_m = \sum_i \mathbb{1} [m_i = 1 \wedge (i = 1 \vee m_{i-1} = 0)].$$

Detected events count Perfect Switch events that overlap at least one model switch island. Missed events count Perfect Switch events without model overlap. False switch events count model switch islands that do not overlap a Perfect Switch event. When the onset of both the reference and the model switch island is available, onset delay measures the model onset time relative to the reference onset. Duration similarity is reported as

$$100 \frac{D_p - |D_m - D_p|}{D_p}.$$

These metrics connect statistical modeling to the practical behavior of a fade mitigation system: they quantify not only whether the model is correct at individual timestamps, but also whether it activates too long, misses events, fragments one event into many switch islands, or reacts too late.

Chapter 3

Data and Reference Decision Framework

The operational problem addressed in this thesis starts from a stream of timestamped satellite signal measurements and asks whether the link condition justifies activating a backup resource. Before any model can be trained, this problem requires a precise data and decision framework: the raw files must be converted into a common signal representation, invalid observations and gaps must be handled consistently, continuous acquisition segments must be defined, and the reference switch sequence must be constructed from explicit rules.

This chapter defines that framework. Its purpose is not only to prepare data for modeling, but to fix the operational meaning of the quantities evaluated later in the thesis. The same data layer determines what counts as an available observation at prediction time, how heterogeneous sources are reduced to the signal-only schema, how candidate fade periods are identified, and how model outputs are compared against a common Perfect Switch reference. The experimental split and validation protocol then follow from this framework, because model comparisons are only meaningful when they share the same cleaned data, timestamp grid, reference decision, and post-processing policy.

3.1 Data Preparation Pipeline Overview

The complete preparation flow is summarized in fig. 3.1. The diagram is intended as a map for the rest of the chapter: each block is introduced here once, and the following sections describe the corresponding assumptions and implementation choices in detail.

The key point is that the pipeline is shared up to the point where each modeling

Common signal-only preparation: all tasks start from the same cleaned univariate signal representation, and no future observations are used as model inputs.

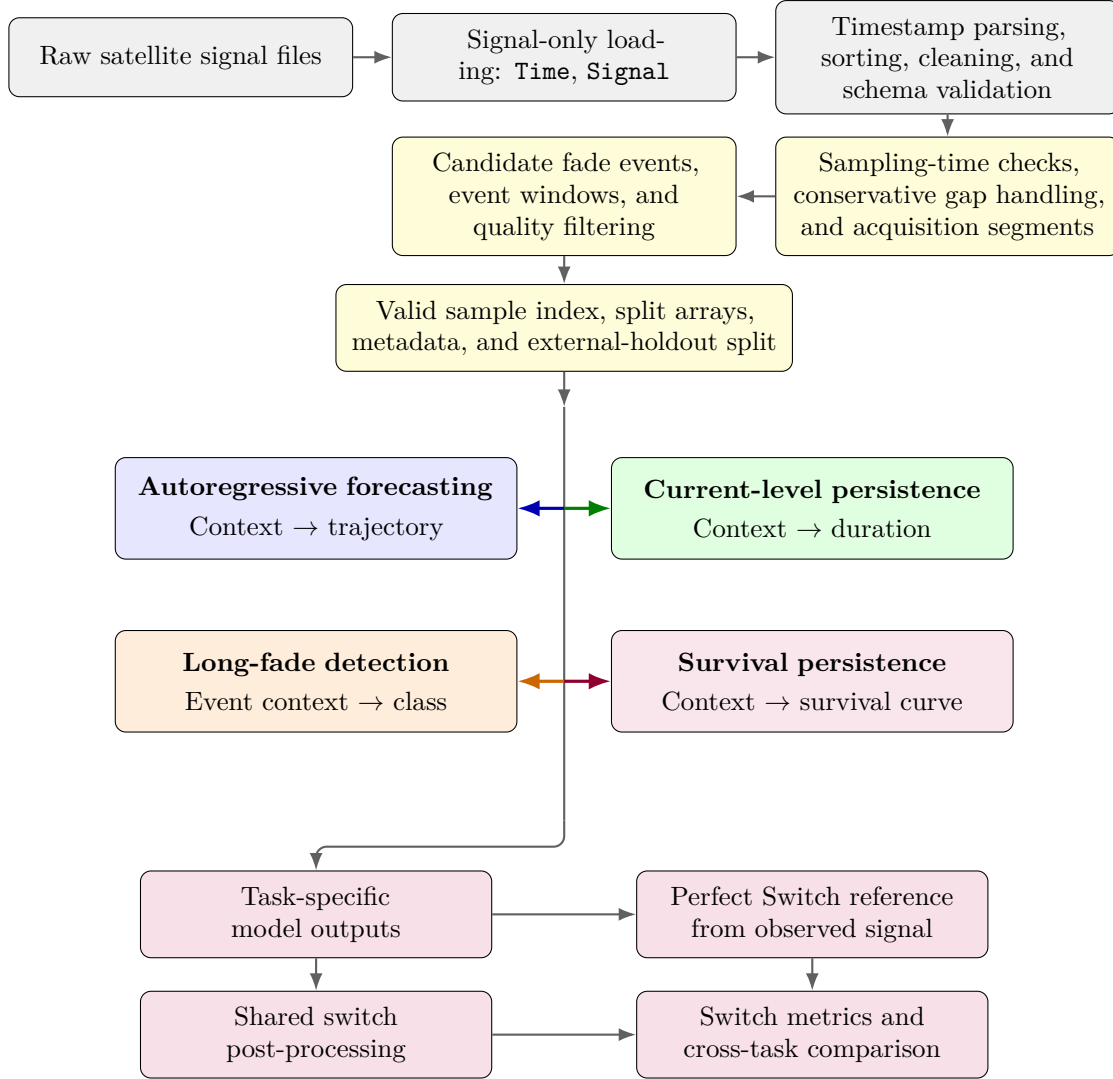


Figure 3.1: Overview of the signal-only data preparation and reference-decision pipeline. Gray and yellow blocks are shared preprocessing steps; colored blocks are task-specific dataset targets and downstream switch-evaluation inputs.

task requires a different target. Autoregressive forecasting needs past-future signal windows, current-level persistence needs residual duration relative to the current signal level, long-fade detection needs a grouped-event class label, and survival persistence needs observed or right-censored remaining event durations. The raw data cleaning, segment boundaries, external holdout policy, and switch reference construction remain common, so differences in the results can be attributed to the task formulation and model family rather than to incompatible preprocessing.

3.2 Available Signal Datasets

The project uses a collection of satellite fade and beacon signal files associated with the Fucino ground segment in Italy and with GX4/GX5 satellite links [45, 46, 47]. The files were made available for this thesis as project data rather than as a public benchmark dataset. They cover different sources and acquisition periods. Each file contains a sequence of timestamped observations of the link signal or attenuation level, but the raw format is not homogeneous across sources. Some files include explicit column headers, while others are plain delimited files without column names. Some files contain only timestamp and signal columns, whereas a small subset includes additional environmental columns. The first requirement of the data layer is therefore to map these heterogeneous files into a common representation before any task-specific dataset is built.

The raw inventory used in the final experiments contains eight loaded signal datasets. Their time coverage ranges from several weeks to several months, with the earliest acquisition starting in July 2020 and the latest ending in July 2021. Table 3.1 summarizes the loaded files at the level needed for the experimental protocol.

Table 3.1: Raw signal datasets used in the project inventory. Row counts refer to valid rows after the common loading and validation stage.

Dataset	Valid rows	Duration (hours)
uplink_beacon_data_fucino_GX4.csv	481715	4081.5
dati_meteo_beacon.csv	484031	4081.5
fc-uplink-fade.csv	378191	3218.0
fc-downlink-fade.csv	380566	3215.8
UPLINK-FADE_Fucino_GX4_AprJun2021.csv	259992	2171.7
UPLINK-FADE_Fucino_GX5_AprJun2021.csv	256519	2161.0
DOWNLINK-FADE_Fucino_GX4_MarJun2021.csv	338281	2826.0
DOWNLINK-FADE_Fucino_GX5_AprJun2021.csv	256519	2161.0

All datasets used in this work are associated with measurements from the Fucino

ground segment in Italy. Fucino is one of Telespazio’s main space centres and teleports [45]. The satellite identifiers in the file names refer to GX4 and GX5, corresponding to the Inmarsat Global Xpress spacecraft Inmarsat GX4, catalogued as NORAD 42698, and Inmarsat GX5, catalogued as NORAD 44801 [46, 47]. Viasat completed its acquisition of Inmarsat in 2023, so these spacecraft belong to the current Viasat/Inmarsat provider context [48]. The historical Inmarsat GX names are nevertheless retained in this thesis because they are the identifiers used in the raw files and in public satellite catalogues.

Within this context, several files are also related by link direction. In the dataset names, *uplink* denotes the ground-to-satellite propagation path, while *downlink* denotes the satellite-to-ground propagation path. Thus, uplink and downlink files can be interpreted as physically paired measurements or derived fade series for opposite directions of the same broader communication link, especially when they refer to the same site and payload identifier. In this thesis, however, these files are not combined into a multivariate paired representation. Each file is treated as an independent univariate signal series, because the final modeling policy requires the same single input variable, **Signal**, to be available for every dataset.

The datasets differ not only in their acquisition period, but also in their maximum observed signal excursions and in the amount of invalid or missing source rows. They also differ in the availability of auxiliary variables. Table 3.2 reports the files whose raw format contains more than the timestamp and signal fields used by the models.

Table 3.2: Additional source columns found in the raw files and ignored by the signal-only data policy.

Dataset	Ignored source columns
uplink beacon GX4	Temperature, Pressure, Humidity
meteo beacon	airt_bt, pres_bt, relh_bt, temp_bt
fc-uplink fade	third trailing column, empty in the inspected rows
fc-downlink fade	third trailing column, empty in the inspected rows

The environmental columns would be potentially useful in a broader multivariate nowcasting study, since atmospheric conditions can affect attenuation dynamics. They are not used here because they are not available in a consistent form across all datasets. Including them would force the experiments to either discard most datasets or compare models trained with different input information. By contrast, several of the Fucino 2021 files contain exactly two columns and are loaded through positional column selection. This variability motivates the strict loading policy described in the

next section: downstream models should not depend on source-specific formatting decisions.

The final experimental protocol treats `fc-uplink-fade.csv` as an external hold-out. This file is included in the project inventory and in the final processed datasets, but it is reserved for test evaluation. All remaining datasets form the development pool used for training and validation. Exact supervised sample counts are reported with the experimental setup, because they depend on the task-specific target and validity rules.

3.3 Signal-Only Policy

All models in this thesis follow a strict signal-only policy. After loading and validation, every dataset is standardized to exactly two columns:

Time, Signal.

The **Time** column identifies when the observation was acquired. It is used for chronological ordering, gap detection, elapsed-duration computation, and splitting into train, validation, and test sets. It is not used directly as a model input. The **Signal** column is the only measured variable made available to the models. In the raw files, this quantity represents the signal or fade level used to describe the link degradation. The exact source formatting may differ, but after the common loading step all downstream tasks operate on the same semantic object: a univariate time series of signal values.

This policy is deliberately restrictive. Some raw files contain environmental measurements, such as temperature, pressure, or humidity, and these variables could plausibly improve attenuation nowcasting in a dedicated multivariate study. They are excluded here because they are not available consistently across the full dataset collection. If such variables were used, the experimental comparison would no longer isolate the effect of the modeling formulation. Some models would exploit richer inputs, while others would be trained on signal-only data or on a reduced subset of files. The resulting differences could then reflect input availability rather than modeling quality.

The signal-only policy also reduces the risk of uncontrolled leakage. Metadata columns, dataset identifiers, event labels, final event durations, and future event properties are never passed as model inputs. The timestamp is not used as a standalone calendar covariate: it is not converted into hour-of-day, day-of-year, or

month features. This avoids letting the models exploit dataset-specific acquisition periods or seasonal correlations that may not transfer to a new deployment period.

The derived predictors allowed by the final datasets must be observable at the decision time. Most of them are deterministic functions of the past and present signal window, such as a recent mean, a local range, a recent slope, or a value expressed relative to the current signal or to the fade threshold. Some task-specific datasets may also include online elapsed-time quantities defined by a threshold state machine, such as time since the first crossing of the current grouped event. These features still obey the signal-only policy because the state machine is driven only by past and current `Signal` values and their timestamps. They do not include calendar predictors, source-specific metadata, future event ends, final event durations, or labels.

In practical terms, this means that all task formulations are evaluated under the same information constraint. Autoregressive forecasting, current-level persistence, long-fade detection, and survival persistence ask different questions about the same observed signal history, but none of them receives additional external predictors. This makes the experimental comparison less ambitious than a full operational system with meteorological inputs, but it makes the methodological comparison cleaner and reproducible across all available files.

3.4 Parsing, Cleaning, and Validation

The first processing stage converts each raw file into a validated `Time/Signal` table. This stage is intentionally separated from all modeling scripts. Its purpose is to make every subsequent task start from the same chronological signal representation, independently of the formatting details of the original file.

Raw files may or may not contain column headers. When headers are available, the configured timestamp and signal columns are selected by name when possible. When headers are absent, or when positional loading is explicitly required, the first column is interpreted as `Time` and the second column as `Signal`. If the file contains fewer than two columns, loading fails with an explicit error. If the file contains more than two columns, the extra columns are ignored and recorded in the loading metadata.

The parser performs only strict conversion. Timestamps are parsed into datetime values and signal entries are converted to numeric values. Rows with missing or unparseable timestamps are excluded, because they cannot be placed in the time series. Rows with missing, non-numeric, or non-finite signal values are also excluded,

because they cannot be used as observed signal measurements. These rows are not silently hidden: the corresponding counts are saved for each dataset and used in the inventory summaries.

Some raw files may also contain suspicious finite values that are technically numeric but likely represent default missing-data codes. In this project, signal values at or below -10 are reported as suspected sentinel values. They are not removed automatically at the parsing stage, because doing so would embed a domain assumption directly into the generic loader. Instead, they are kept as observations and made visible through the data-quality metadata. This keeps the cleaning policy auditable and avoids silently changing the empirical signal distribution.

After parsing, each dataset is cleaned and temporally segmented. The common cleaning stage applies the following operations:

- sort each dataset chronologically;
- resolve duplicate timestamps according to the configured strategy;
- assign a deterministic `dataset_id` and `row_id`;
- compute the elapsed time from the previous observation;
- assign a continuous acquisition `segment_id`;
- save the cleaned signal table and the loading metadata.

Duplicate timestamps are handled before segment construction. In the canonical configuration, duplicate observations with the same timestamp are collapsed by averaging their signal values. Alternative strategies, such as keeping the first value, keeping the last value, or raising an error, are supported by the preprocessing layer, but the final experiments use one configured policy consistently. This is important because duplicate handling can otherwise change threshold crossings and event boundaries.

The `segment_id` marks continuous acquisition stretches. It is created from the chronological time differences: whenever the elapsed time between two successive observations exceeds the configured gap threshold, a new segment is started. At this stage no missing samples are inserted and no signal values are interpolated. The cleaned tables therefore remain faithful to the observed timestamps, while still making acquisition breaks explicit for later no-leakage checks.

Validation is performed through explicit metadata rather than implicit assumptions. For every raw file, the pipeline records the number of raw rows, valid rows,

invalid timestamp rows, invalid signal rows, suspected sentinel rows, duplicate timestamps, ignored columns, and whether positional fallback was used. These diagnostics make it possible to identify datasets that require special attention before they influence model training or evaluation.

The output of this stage is the common clean signal layer stored under the intermediate data directory. Downstream task builders reuse this layer instead of reading raw CSV files independently. This design prevents each task from implementing its own parser, duplicate policy, or timestamp convention. As a result, differences between autoregressive forecasting, current-level persistence, long-fade detection, and survival persistence are due to the task definition and model family, not to inconsistent low-level data handling.

3.5 Sampling Time, Gaps, and Conservative Imputation

The final experiments assume a nominal sampling interval $\Delta t = 30$ seconds. This interval is used to interpret context lengths, forecast horizons, switch-time constraints, and duration thresholds. For example, a context length of $L = 30$ corresponds to approximately 15 minutes of past observations, while a prediction horizon of $h = 10$ corresponds to approximately 5 minutes into the future. These conversions are meaningful only if the sequence is approximately regular, so the treatment of missing timestamps is part of the experimental methodology rather than a minor implementation detail.

Let an observed signal segment be represented as an ordered sequence

$$(t_1, s_1), (t_2, s_2), \dots, (t_n, s_n),$$

where t_i is the timestamp and s_i is the corresponding signal value. For two consecutive observations in the same acquisition segment, define

$$d_i = t_i - t_{i-1}.$$

If $d_i \leq 1.5\Delta t$, the pair is considered close enough to the expected sampling rate and no missing sample is inferred. If $d_i > 1.5\Delta t$, the number of missing samples is estimated as

$$m_i = \text{round}\left(\frac{d_i}{\Delta t}\right) - 1.$$

This rule identifies gaps that are longer than the nominal interval but still compatible

with a small number of missing 30-second samples.

The conservative imputation policy used in the final datasets is:

- expected sampling interval: 30 seconds;
- maximum gap eligible for imputation: 90 seconds;
- maximum number of inserted samples in one gap: two;
- interpolation method: linear interpolation in time;
- allowed scope: only inside the same continuous acquisition segment.

Therefore, a gap is imputed only if $m_i > 0$, $m_i \leq 2$, and $d_i \leq 90$ seconds. For each missing index $k = 1, \dots, m_i$, the inserted timestamp and signal value are

$$\begin{aligned}\tilde{t}_{i,k} &= t_{i-1} + k\Delta t, \\ \tilde{s}_{i,k} &= s_{i-1} + \frac{k}{m_i + 1}(s_i - s_{i-1}).\end{aligned}$$

The reconstructed value is thus a straight-line interpolation between the previous and next observed samples. No model is used to infer the missing signal. The method does not attempt to reconstruct hidden attenuation peaks, fast rain-cell dynamics, or any physical process during the missing interval. It only repairs very short holes that are consistent with one or two omitted regular samples.

The most important constraint is that imputation is never allowed across different acquisition segments. Segment boundaries are defined before imputation from gaps exceeding the configured continuity threshold. Once two observations belong to different `segment_id` values, no interpolation can connect them. A later observation after such a break is therefore not treated as evidence that the signal stayed above or below any threshold during the unobserved period.

This has direct consequences for target construction. In autoregressive forecasting, a context-target pair is valid only if the required past and future observations can be represented on the prepared event-window grid. In current-level persistence, a recovery observed only after an unbridgeable gap cannot be used as the true recovery time for a sample before the gap. In survival persistence, if the end of the fade event is not observed before the continuous segment ends, the corresponding sample is right-censored rather than assigned an artificial observed residual duration. The policy therefore prevents missing data from being converted into false certainty about event persistence or recovery.

There are differences across tasks, but not across models within the same task. The imputation policy is applied before model training or evaluation, so all models belonging to the same task consume the same prepared dataset. GRU, PatchTST, autoregressive XGBoost, and Chronos are evaluated on the same autoregressive windows for a given context length and horizon. The current-level shapelet models and current-level XGBoost share the same current-level dataset. The long-fade classifiers share the same grouped-event classification dataset, and the survival models share the same survival-persistence dataset with observed and censored targets.

The task-level usage is as follows. The autoregressive preparation first constructs event-centered windows and stores a prepared signal version with small recoverable gaps filled. This prepared signal is also used by task builders that start from the shared event-window artifacts, such as current-level persistence and long-fade detection. Duration-oriented builders also apply the same small-gap rule to the full cleaned signal inside each continuous segment when they need to search for a future recovery or event end. Survival persistence applies the same rule before running its event state machine, so that very short missing intervals do not artificially fragment an otherwise continuous fade episode. In every case, imputation remains local, auditable, and bounded by the same numerical constraints.

Event-window quality is derived from this gap analysis. Windows with acceptable coverage and only recoverable small gaps can be marked as usable or warning-level, depending on the number and length of the gaps. Windows with insufficient coverage, excessive missing runs, or gaps that cannot be repaired are excluded from the final supervised window index. The retained datasets also store imputation summaries, including how many rows were inserted, how many gaps were filled, and the largest imputed gap per group.

The policy is intentionally conservative. It may discard some potentially informative samples, but it avoids a more serious failure: constructing inputs, targets, or switch references from time intervals where the signal was not actually observed. This trade-off is consistent with the operational objective of the thesis, where reliable switch behavior is more important than maximizing the number of training windows at any cost.

3.6 Candidate Events and Acquisition Segments

The modeling tasks do not sample uniformly from the complete raw files. Most timestamps correspond to normal or weakly attenuated propagation conditions,

whereas the operational decision is relevant around fade episodes. The common preparation stage therefore introduces two related, but distinct, concepts: *acquisition segments* and *candidate fade events*.

An acquisition segment is a continuous portion of one loaded signal file. After the signal has been sorted by time, a new segment is started whenever the time distance between two consecutive valid observations is larger than the maximum allowed continuity gap. In the canonical configuration this limit is 90 seconds. Formally, if $d_i = t_i - t_{i-1}$, then observations $i - 1$ and i belong to the same segment only when $d_i \leq 90$ seconds. The resulting `segment_id` is used as a guardrail throughout the pipeline: input contexts must not cross segment boundaries, and future targets must not be resolved by jumping over an unobserved acquisition break.

Candidate fade events are instead a practical sampling device. For each dataset, the preparation code first identifies threshold-crossing timestamps,

$$\mathcal{C}_\theta = \{t_i : S(t_i) \geq \theta\},$$

with $\theta = 10.0$ dB in the main experiments. The condition is deliberately non-strict, so a sample exactly equal to 10.0 dB is treated as belonging to the threshold region. Crossings are then grouped chronologically. A group starts at its first crossing time t_0 , and subsequent crossings remain in the same candidate event while they occur within the configured grouping horizon from t_0 . The default horizon is three hours. A later crossing outside that horizon starts a new candidate event.

Each candidate event is anchored at the first threshold crossing, called the `event_timestamp`, and is expanded into an event-centered analysis window,

$$[t_0 - 1.5\text{h}, t_0 + 1.5\text{h}],$$

using the canonical configuration. This window is not intended to say that the fade lasts three hours. It is only a local region that contains pre-event context, the threshold-crossing period, and post-event recovery or follow-up samples when they are observed. The window may therefore contain both above-threshold and below-threshold samples.

These candidate events should not be confused with the final reference switch, nor with the task-specific event definitions used later. They are used to make the raw time series operationally manageable, to compute coverage and gap quality checks, and to decide which local portions of the files can generate supervised samples. Window quality is assessed after bounded imputation: the pipeline records coverage, maximum gap length, number of inferred missing samples, number of imputed

samples, duplicate timestamps, missing values, and the number of points above and below the threshold. Events with insufficient coverage or unrecoverable gaps are excluded from the final window index, while warning-level events can be retained when this is explicitly enabled.

Task-specific datasets are then built on top of this common signal layer:

- autoregressive forecasting uses event-window indices to build past context and future trajectory pairs. A training sample is valid only when the full context and the full prediction horizon are available on the prepared time grid;
- current-level persistence searches the full cleaned signal for the first future recovery below a level relative to the current signal. The candidate window can determine which timestamps are considered, but the target duration is not truncated at the event-window boundary;
- long-fade detection groups threshold crossings within a configured temporal horizon and assigns one binary label to all timestamps in the grouped fade episode. Temporary below-threshold oscillations inside the grouped event do not split the event;
- survival persistence uses a state machine with activation, candidate recovery, stable recovery, and right-censoring at segment boundaries. Its target is the remaining duration until the end of the whole fade event, not the duration of the current signal level.

The separation between acquisition segments, candidate events, and task-specific labels is important. Acquisition segments describe what was continuously observed. Candidate events describe where the pipeline should look for operationally relevant samples. The supervised target definition is then chosen by the task. This design preserves a common data foundation while allowing each model family to answer a different operational question without silently changing the meaning of an event.

3.7 External Holdout Split

The final evaluation uses an external-holdout protocol. One complete raw file, `fc-uplink-fade.csv`, is reserved for test evaluation. It is not used for model fitting, hyperparameter selection, early stopping, architecture selection, normalization-statistic fitting, or decision-rule calibration. All remaining datasets form the development pool.

The split is therefore performed at two levels. At the file level, the held-out dataset is assigned entirely to the test split:

$$\mathcal{D}_{\text{test}} = \{\text{fc-uplink-fade.csv}\}.$$

The development set is the complement,

$$\mathcal{D}_{\text{dev}} = \mathcal{D} \setminus \mathcal{D}_{\text{test}}.$$

At the event level, each development dataset is split chronologically into training and validation events. Events are first ordered by their earliest available input timestamp. The first 85% of development events are assigned to training and the remaining 15% to validation, subject to a minimum-event guard. If a development dataset contains too few events to support a meaningful validation subdivision, all its events are kept in training rather than creating an unstable validation split. In the canonical autoregressive and survival dataset builders the minimum is three events; in the long-fade configuration it is two events.

This event-aware split prevents windows from the same physical fade episode from appearing in both training and validation. It also prevents the external test file from leaking into the development process. The validation split is used for model selection: selecting grid-search configurations, choosing early-stopping epochs, comparing candidate architectures, and calibrating only those decision rules that are explicitly treated as validation-calibrated. The test split is evaluated after this selection step. If a test-set sweep is ever performed for diagnostic purposes, it must be reported as exploratory and not as a validation-calibrated operating point.

The same principle is applied across tasks, although the exact number of samples differs because each task defines a different supervised target and therefore a different set of valid decision timestamps. Autoregressive forecasting produces past-context and future-trajectory windows. Current-level persistence produces duration-regression samples. Long-fade detection produces grouped-event classification samples. Survival persistence produces observed and right-censored residual-duration samples. In all cases, however, the final test data come from the same held-out acquisition file.

This protocol is stricter than a random sample split. A random split would place nearby, highly correlated timestamps on both sides of the split and would inflate performance estimates. The external-holdout protocol is closer to the intended use case: a model or decision rule is developed on existing acquisition files and then applied to a separate file. It should still be interpreted as file-level generalization within the available Fucino data, not as proof of generalization to all teleports,

satellite links, seasons, or operating conditions.

3.8 Perfect Switch Reference

All model-derived switch decisions are evaluated against a model-independent reference called the Perfect Switch. The Perfect Switch is not a trained model. It is a reference decision derived from the observed signal and the configured operational threshold. It is an oracle in the evaluation sense: it can use the future observed signal to decide whether a switch should have been active, but it is never available to a real online model.

The reference starts from an instantaneous outage mask. In the canonical experiments the threshold is $\theta = 10.0$ dB and comparisons are inclusive:

$$M_i = \mathbb{K}\{S(t_i) \geq \theta\}.$$

This binary mask alone is not the Perfect Switch. It only states whether the currently observed signal is in the threshold region. The Perfect Switch also incorporates the operational persistence requirement: a switch should not be activated for isolated threshold crossings that disappear immediately. With 30-second sampling and `switch_time = 10`, the reference requires eleven consecutive above-threshold samples, corresponding to ten 30-second intervals, i.e. approximately five minutes of persistence.

Equivalently, for each contiguous run of $M_i = 1$, the raw Perfect Switch is set to one only if the run contains at least `switch_time + 1` samples. Shorter above-threshold runs remain zero. This produces the stored column `perfect_switch_raw`. A legacy right-extension step is then applied to enforce the minimum switch-time behavior used in the previous implementation. Finally, if the switch is already active and the observed signal is still above threshold, the switch is held active even if the intermediate rule would otherwise return zero. This final held signal is the column used as the reference in the experiments:

$$\text{perfect_switch} \equiv \text{perfect_switch_min_time}.$$

The pipeline also stores diagnostic columns for auditing this transition:

- `outage_mask`, the instantaneous threshold mask;
- `perfect_switch_min_island_old`, the legacy right-extension intermediate signal;

- `perfect_switch_adjusted`, a separate adjusted diagnostic variant.

These columns help inspect the transition from threshold mask to reference switch, but the main switch metrics compare model decisions against `perfect_switch`.

The Perfect Switch is task-scoped in terms of timestamp grid. Each task can produce decisions only at its native decision timestamps, so the reference is aligned to those timestamps before computing task-level switch metrics. The underlying threshold convention and post-processing rule, however, are shared across tasks. This is why any change to the Perfect Switch definition is a methodological change: it affects every reported switch metric.

3.9 Shared Switch Post-Processing

The reference framework operates after a task-specific prediction has already been converted into a raw binary decision. The predictive quantity may be a forecast trajectory, a duration estimate, a class probability, or a survival curve, but that distinction belongs to the task formulation. At this stage, all models expose the same intermediate decision column:

`model_switch_raw.`

The mapping from a native model output to `model_switch_raw` is task-specific and is defined with the individual formulations in chapter 4. The data-and-reference layer does not choose those task thresholds or probability operating points; it only receives the raw decision sequence and applies the same temporal policy to all methods.

After this task-specific conversion, all methods use the same temporal post-processing. This is intentionally separated from model training: changing it changes the evaluation decision rule, but it does not require retraining the underlying forecasting, regression, classification, or survival model.

The first post-processing step enforces a minimum active switch duration on the raw decision sequence. Each contiguous island of ones is inspected. If the island is shorter than `switch_time`, it is extended to the right until it reaches that length, or until the available event grid ends. This rule does not fill arbitrary zero gaps between two switch islands. It can only make two islands touch if the right-extension of the first island reaches the second one.

The second step is stateful and uses only information available at the current decision time. If the post-processed switch was active at the previous timestamp and

the currently observed signal is still above the operational threshold, the switch is kept active even if the raw model decision has temporarily returned to zero. This avoids an unrealistic behavior in which the system switches off while the fade is still ongoing. Once the observed signal leaves the threshold region, the switch is no longer forced by this holding rule.

The post-processed decision produced by these two steps is the one used for the main reported switch metrics:

`model_switch.`

This separation prevents a common source of ambiguity. Parameters used inside a task-specific conversion rule, such as forecast-horizon requirements or probability thresholds, are different from `switch_time`, which controls only the temporal post-processing of an already binary decision sequence.

Raw and intermediate switch columns are still saved for diagnostics. For model outputs, `model_switch_raw` is the direct task-specific binary decision before post-processing; `model_switch_min_island_old` is the right-extended minimum-duration version; `model_switch_min_time` is the held version after the above-threshold rule; and `model_switch_adjusted` is a separate legacy diagnostic variant. These columns are useful for understanding whether an error comes from the model output itself or from the operational post-processing rule. Unless explicitly stated otherwise, the reported switch metrics use the held minimum-time version, saved as both `model_switch_min_time` and `model_switch`.

3.10 Evaluation Alignment

The data and reference layer provides the common objects needed for operational evaluation: the observed signal, the instantaneous outage mask, the task-aligned Perfect Switch, and the intermediate switch columns generated by the shared post-processing rule. It does not define the native losses of the models, select hyperparameters, or choose task-specific probability and duration thresholds. Those experimental choices are specified in chapters 4 and 5.

For each task, model outputs are first converted into `model_switch_raw` according to the task formulation. The shared post-processing then produces `model_switch`, which is compared with the aligned `perfect_switch`. The same tables also retain diagnostic fields for the outage mask, raw and adjusted Perfect Switch variants, and the raw model decision. These fields make it possible to audit whether a disagreement

comes from the predictive model, from the task-specific conversion rule, or from the operational post-processing layer.

Metric formulas and comparison denominators are intentionally left to the experimental setup chapter. The role of the present chapter is narrower: it defines the signal representation and the reference decision objects that all tasks must use. This keeps the thesis definitions separated: chapter 2 introduces general concepts, this chapter defines the data and reference layer, chapter 4 defines task-specific model outputs and raw switch rules, and chapter 5 defines how the resulting runs are selected and compared.

Chapter 4

Proposed Modeling Approaches

The modeling strategy is organized around four predictive formulations rather than around standalone architectures. All four approaches share the same operational objective: at decision time t_i , use only the signal history available up to that time to decide whether the backup link should be activated. They differ in the intermediate quantity estimated before this final decision: the future signal trajectory, a local persistence duration, the probability of a long fade event, or a residual-duration survival function.

4.1 Overview of the Four Modeling Formulations

The four formulations represent different levels of abstraction of the same nowcasting problem. Autoregressive forecasting retains the most detailed description of the future by predicting a trajectory. Current-level persistence reduces the future to the time for which the signal remains close to its current level. Long-fade detection formulates the problem as an event-level binary classification task. Survival persistence retains uncertainty about the time to recovery by estimating a residual-duration distribution while allowing for right-censored observations.

This organization separates the predictive task from the architecture used to implement it. The following matrix summarizes the target, the model families used in the corresponding experiments, and the conversion that connects each native output to the common switch decision. The detailed definitions and assumptions are given in sections 4.3 to 4.6.

The next sections define the common prediction setting and then describe each formulation in detail. For each one, the input representation, target construction, model families, native metrics, and output-to-switch rule are specified before the formulations are compared in a common operational evaluation framework.

Table 4.1: Overview of the four modeling formulations and their implemented model families.

Formulation	Native output	Model families	Switch conversion
Autoregressive forecasting (Sec. 4.3)	Future signal trajectory	GRU sequence models [20, 21]; PatchTST [26]; XGBoost [12]; Chronos [28]	Forecast trajectory to switch
Current-level persistence (Sec. 4.4)	Local persistence duration	Multiscale learnable-shapelet MLP, Transformer, and convolution variants [35]; XGBoost [12]	Duration threshold
Long-fade detection (Sec. 4.5)	Probability of a long grouped fade event	XGBoost lag-and-scalar classifier [12]; TCN classifier [30]; multiscale shapelet convolution classifier [35]	Probability threshold with signal gate
Survival persistence (Sec. 4.6)	Residual-duration survival function	XGBoost-AFT [14]; discrete-time causal TCN [39, 30]	Survival probability at the operational horizon

4.2 Common Prediction Setting

After the common loading and preprocessing stage, each dataset is represented as an ordered univariate time series

$$\{(t_i, S_i)\}_{i=1}^N,$$

where t_i is the timestamp of the i -th observation and S_i is the corresponding signal value. The project deliberately uses a signal-only representation: no meteorological, orbital, station-side, or file-specific auxiliary variable is used as model input. This constraint makes the comparison between datasets and tasks consistent, because every model receives information derived from the same physical quantity.

For a decision made at index i , the generic past context of length L is

$$X_i = [S_{i-L+1}, \dots, S_i].$$

The last value S_i is the most recent observation available when the decision is made. All features used by the models must be observable at time t_i : they can be functions of this past context, or task-specific elapsed-time and event-state quantities

computed from past and current threshold crossings. Future samples may be used to construct supervised targets during dataset generation, but they are never allowed to enter X_i or any derived input feature.

Different tasks apply different transformations to the same basic context. Some models use the raw window directly, some use a local context-standardized version, and some use relative representations such as $X_i - S_i$ or $X_i - \theta$, where θ is the operational signal threshold. When scalar context features are used, they are computed only from the available signal history and online task state, and their standardization parameters are estimated on the training split only.

The term scalar context is used in a precise sense throughout the chapter. It does not denote external covariates. It denotes past-only scalar summaries derived from the same signal window and, when the task is event-based, from the event state already observable at t_i . The recurring signal summaries are trailing quantities:

$$\bar{S}_{i,L} = \frac{1}{L} \sum_{r=0}^{L-1} S_{i-r}, \quad \text{sd}_{i,L} = \text{std}(S_{i-L+1}, \dots, S_i),$$

together with the window minimum, maximum, range, and finite differences

$$\Delta_i^{(q)} = S_i - S_{i-q+1}, \quad q \in \{2, 3, 5\}.$$

The short-window summaries $\bar{S}_{i,5}$ and $\text{sd}_{i,5}$ are computed on the last five available samples of the context. These are rolling only in the trailing online sense: they are never centered around t_i , and they never use samples after t_i . When a model uses a tabular lag matrix, the lag columns are ordered from oldest to newest; lag_{L-1} corresponds to S_{i-L+1} after the selected task transformation, and lag_0 corresponds to the current sample S_i after the same transformation.

The output of a model at index i is denoted generically as z_i . Depending on the task, z_i can be a future trajectory, a duration estimate, a binary event probability, or a survival curve. Each task therefore defines an explicit conversion rule

$$r_i = c(z_i, S_i),$$

where $r_i \in \{0, 1\}$ is the raw model-derived switch before the shared post-processing defined in section 3.9. The final evaluated switch is obtained only after applying that common policy. This separation keeps the task-specific predictive problem distinct from the common operational decision layer.

4.3 Autoregressive Forecasting

Autoregressive forecasting is the most direct formulation of the nowcasting problem considered in this work. Instead of predicting a binary switch or a duration variable directly, the model predicts the future signal trajectory over a short operational horizon. The switch decision is then derived from this forecasted trajectory. This formulation is useful because it preserves an interpretable intermediate output: before evaluating whether a method triggers a backup-link switch, one can inspect whether it predicts the local evolution of the signal correctly.

This signal-forecasting view deliberately avoids building a fade definition into the predictive target. The model is not trained to decide whether an event is a fade, whether a fade will last longer than a fixed duration, or whether a particular threshold crossing belongs to an operationally relevant event. It is trained only to estimate future values of the measured signal. As a consequence, the autoregressive task does not require a threshold θ , an onset or recovery rule, an event-grouping rule, or a minimum fade duration in order to be defined. These quantities become necessary only later, when the predicted signal trajectory is converted into a switch decision and compared with reference switch policies. This is a key difference from the task formulations that follow, where the supervised target itself is a duration, event class, or survival quantity whose definition depends on a fade or recovery criterion.

4.3.1 Task Definition

For a valid decision index i , the autoregressive input is the past signal window

$$X_i^{\text{AR}} = [S_{i-L+1}, \dots, S_i] \in \mathbb{R}^L,$$

and the target is the future signal trajectory

$$Y_i^{\text{AR}} = [S_{i+1}, \dots, S_{i+h}] \in \mathbb{R}^h.$$

Both quantities are defined only from the signal and the forecasting horizon. A valid sample requires an observed context and a fully observed future horizon inside one continuous acquisition segment. In the experiments, the eligible decision indices are restricted to the candidate event windows built during data preparation. This is a sampling and evaluation-design choice: it focuses the training and comparison on operationally relevant periods, but it does not change the autoregressive target, which remains the future signal trajectory.

In the main experimental configuration, $L = 30$ and $h = 10$. With the canonical sampling interval of approximately 30 seconds, the model therefore observes about 15 minutes of recent signal history and predicts about 5 minutes ahead. A longer-context Chronos diagnostic configuration with $L = 120$ is also kept separate from the main autoregressive comparison.

The future trajectory is used only to construct the offline target. The model output is

$$\hat{Y}_i^{\text{AR}} = [\hat{S}_{i+1}, \dots, \hat{S}_{i+h}],$$

and is converted back to the raw signal scale before forecast metrics or switch conversion are computed.

4.3.2 Tested Models

Four autoregressive model families are used. Their general modeling principles are introduced in chapter 2; this section only specifies how they are instantiated for the autoregressive task. For trainable neural models, the input tensor has shape $N \times L \times 1$, where the last dimension is the single **Signal** channel. Two input variants are considered. In the raw variant, X_i^{AR} and Y_i^{AR} are used in their original scale. In the context-standardized variant, each context is normalized with statistics computed from the context itself, and the corresponding target is represented in the same local scale during training. Predictions from this variant are always inverse-transformed to raw signal units before forecast metrics or switch decisions are computed.

This two-variant design is used for every trainable autoregressive model family in the experiments: GRU, PatchTST, and XGBoost. Chronos is evaluated separately as a raw zero-shot foundation-model baseline, because it performs its own internal scaling.

For the context-standardized variant, the dataset stores

$$\mu_i = \frac{1}{L} \sum_{r=0}^{L-1} S_{i-r}, \quad s_{\text{ctx},i} = \max\{\text{std}(X_i^{\text{AR}}), \epsilon\}.$$

Here $s_{\text{ctx},i}$ is the context standard deviation bounded below by $\epsilon > 0$ for numerical stability. No rolling statistics, slope features, scalar context vector, threshold mask, or event-position variable is appended to the autoregressive models. GRU, PatchTST, and Chronos receive only the ordered univariate sequence. The XGBoost autoregressive baseline receives the same values flattened into lag columns $\text{lag}_{L-1}, \dots, \text{lag}_0$, but no additional summary features are added.

GRU sequence models. Two GRU variants with unidirectional encoders are evaluated. Unidirectionality is an architectural choice in these experiments; as discussed in section 2.2.2, a bidirectional encoder restricted to the same past context would also respect the decision-time information boundary. Let $\widetilde{X}_i$ denote the model input in either raw or context-standardized scale. The sequence-to-vector variant encodes the full past context with a unidirectional GRU,

$$(A_{i,1}, \dots, A_{i,L}), a_i = \text{GRU}_{\text{enc}}(\widetilde{X}_i),$$

where a_i is the final hidden representation of the last recurrent layer. The complete future trajectory is then produced by one linear output head:

$$\widehat{Y}_i^{\text{AR}} = W_o a_i + b_o.$$

This is a direct multiple-output forecaster. It shares the encoder across the context but does not generate the forecast recursively; all horizons are predicted from the same encoded state.

The encoder-decoder variant uses the encoder hidden state to initialize a second GRU that runs over the future horizon. The decoder starts from the last observed input value and generates one value at a time:

$$d_{i,k} = \text{GRU}_{\text{dec}}(\xi_{i,k}, d_{i,k-1}), \quad \widehat{S}_{i+k} = W_d d_{i,k} + b_d.$$

During training, the decoder input $\xi_{i,k}$ can be either the previous prediction or the corresponding previous true target value according to the configured teacher-forcing ratio. During validation, testing, and switch conversion, teacher forcing is disabled, so future true values never enter the decoded trajectory.

Both GRU variants are optimized with mean squared error over the h -step target,

$$\mathcal{L}_{\text{AR}} = \frac{1}{Nh} \sum_i \sum_{k=1}^h \left(\widehat{S}_{i+k} - S_{i+k} \right)^2,$$

computed in the training scale of the selected variant.

PatchTST. PatchTST receives the same univariate context, partitions it into local temporal patches, processes the resulting tokens with Transformer encoder blocks, and maps the encoded representation to the h -step forecast. In this formulation, PatchTST tests whether patch-level attention is useful for representing short fade rises, plateaus, and recoveries within the recent signal history. The implemented model is

a deterministic PatchTST-style direct forecaster rather than a full NeuralForecast wrapper: it extracts complete patches from the single-channel signal context, applies a learned linear patch embedding and a learned positional embedding, encodes the patch sequence with a Transformer encoder, flattens the encoded tokens, and projects them to the full future signal trajectory. Channel independence is therefore satisfied trivially, because the task uses only the **Signal** input channel.

More explicitly, for patch length P and stride s , the input is unfolded into N_p complete patches $p_{i,j} \in \mathbb{R}^P$. Each patch is mapped to a token

$$e_{i,j} = W_p p_{i,j} + b_p + r_j,$$

where r_j is a learned positional embedding. The Transformer encoder returns contextualized patch tokens H_i , and the output head applies a direct multi-output projection:

$$\hat{Y}_i^{\text{AR}} = W_o \text{vec}(H_i) + b_o.$$

The model is trained with the same trajectory MSE used for the GRU forecasters. Unlike the GRU encoder-decoder, it does not autoregress over the predicted future values; the whole horizon is produced in one forward pass.

XGBoost autoregressive baseline. The XGBoost baseline converts each context into a tabular lag vector

$$\phi_i^{\text{AR}} = [\tilde{S}_{i-L+1}, \dots, \tilde{S}_i],$$

again using either the raw or context-standardized variant. It then trains one independent boosted-tree regressor for each forecast horizon:

$$\hat{S}_{i+k} = F_k(\phi_i^{\text{AR}}), \quad k = 1, \dots, h.$$

Each F_k is an XGBoost regressor with a squared-error regression objective. The horizon-specific regressors have separate parameters and random seeds, so the method does not impose a joint trajectory structure across the ten predicted future samples. Its purpose is to test whether a strong nonlinear learner over fixed lag features can compete with explicit sequence models for short-horizon signal prediction. Because the features are fixed lags, the baseline is causal by construction as long as ϕ_i^{AR} is built only from the past context.

Chronos foundation model: zero-shot forecasting. Chronos is used only in zero-shot mode. The pretrained model receives a univariate context and returns

sampled future trajectories

$$\hat{Y}_i^{(b)} \sim p_{\text{Chronos}}(Y_i \mid X_i), \quad b = 1, \dots, B.$$

The deterministic trajectory used for MAE, RMSE, and switch conversion is the sample median at each horizon,

$$\hat{S}_{i+k} = \text{median}_{b=1, \dots, B} \hat{S}_{i+k}^{(b)}.$$

Prediction quantiles are also saved as distributional diagnostics, but the switch pipeline uses the median trajectory for consistency with the deterministic forecasters. No Chronos parameter is trained or fine-tuned, and no train or validation data are loaded to adapt the model. A main run uses $L = 30$, while an additional diagnostic run uses $L = 120$ to test whether a longer context improves the foundation-model forecast.

4.3.3 Switch Conversion

The autoregressive model does not output a switch directly, and the threshold used below is not a training label. It belongs to the downstream decision layer: after the future signal trajectory has been predicted, the forecast is converted into one raw switch decision at the prediction time t_i . Let θ be the operational signal threshold and let q be the configured minimum number of forecasted future samples that must exceed the threshold. The raw autoregressive switch is

$$r_i^{\text{AR}} = \mathbb{I} \left[\sum_{k=1}^h \mathbb{I}(\hat{S}_{i+k} \geq \theta) \geq q \right].$$

In the main experiments, $\theta = 10.0$ dB and $q = h = 10$. Therefore the model switches at time t_i only if all ten predicted future signal values are at or above the operational threshold. This is a deliberately conservative rule: it requires the forecast to indicate persistence throughout the full prediction horizon before the backup-link decision is activated.

Changing θ , q , or the post-processing rule would change the operational interpretation of the same predicted trajectory, but it would not change the autoregressive supervised learning problem. This separation is useful when the desired fade definition is uncertain or may be revised, because the forecast remains a signal-level object rather than a model-specific fade label.

After this raw decision is produced, the shared switch post-processing defined in section 3.9 is applied. The autoregressive models are therefore compared against

the Perfect Switch using the same final decision layer as the other task formulations. Forecast MAE and RMSE remain useful diagnostics, but the operational comparison is based on the derived switch sequence.

4.4 Current-Level Persistence

Current-level persistence reformulates the problem as a duration regression task. Instead of predicting the future signal trajectory, the model estimates how long the signal will remain above a level defined relative to the current observation. This is not equivalent to forecasting the end of the whole fade event. It asks a local question: given the recent signal pattern and the current level S_i , for how long does the signal persist above $S_i - \delta$?

This formulation was included because the operational switch decision is ultimately related to persistence. A short threshold crossing may not justify switching to a backup link, while a signal condition expected to persist for several minutes may. Current-level persistence therefore compresses the future trajectory into a single scalar duration.

4.4.1 Task Definition

For a valid decision index i , the raw context is again

$$X_i = [S_{i-L+1}, \dots, S_i].$$

The task-specific recovery level is defined as

$$\lambda_i = S_i - \delta,$$

where $\delta > 0$ controls how far below the current value the signal is allowed to fall before the current-level persistence episode is considered to have ended. In the experiments, $\delta = 0.5$ dB. The target recovery index is

$$\tau_i = \inf \{j > i : S_j < \lambda_i\},$$

provided that such an index is observed before the end of the current acquisition segment. The observed remaining persistence duration is

$$D_i^{\text{CLP}} = t_{\tau_i} - t_i.$$

If no recovery is observed before the acquisition segment ends, the candidate is

right-censored. In the current duration-regression implementation these censored candidates are stored for diagnostics but excluded from ordinary supervised regression. They are not assigned an artificial finite duration, because doing so would bias the target downward.

The primary sequence representation is relative to the current value:

$$X_i^{\text{rel-current}} = X_i - S_i = [S_{i-L+1} - S_i, \dots, 0].$$

This removes the absolute level from the sequence and emphasizes local shape: rising history, recovery history, recent oscillations, and short-term stability. Absolute level information is reintroduced through scalar context features computed from X_i , such as the current signal, the recovery level, window summary statistics, recent slopes, and short-window moments. These features are standardized using training-split statistics only.

In the implemented current-level dataset, the sequence input is the current-relative array, and the scalar vector is the scalar-context feature array. For one sample, the scalar vector is ordered as

$$s_i^{\text{CLP}} = [S_i, S_i - \delta, \bar{S}_{i,L}, \text{sd}_{i,L}, \min(X_i), \max(X_i), \max(X_i) - \min(X_i), \Delta_i^{(2)}, \Delta_i^{(3)}, \Delta_i^{(5)}, \bar{S}_{i,5}, \text{sd}_{i,5}].$$

Thus the learnable sequence component sees only deviations from the current sample, while the scalar branch receives the absolute level, the recovery level $S_i - \delta$, full-window trailing summaries, and short-window trailing summaries. No fade-threshold variable, event-duration variable, future recovery time, or target duration is included as an input feature.

The regression target is log-compressed:

$$y_i^{\text{CLP}} = \log(1 + D_i^{\text{CLP}}).$$

Models are trained to predict y_i^{CLP} . For evaluation and switch conversion, predictions are mapped back to seconds:

$$\widehat{D}_i^{\text{CLP}} = \exp(\hat{y}_i^{\text{CLP}}) - 1.$$

The logarithmic target reduces the influence of very long persistence episodes while preserving the ordering of durations.

4.4.2 Tested Models

The current-level persistence branch uses learnable shapelet models and an XGBoost duration-regression baseline. Shapelet-based modeling is introduced in chapter 2; here the important point is that the learned shapelets operate on $X_i^{\text{rel-current}}$, so they search for local patterns around the current signal level rather than around an absolute threshold.

All current-level models are log-duration regressors. They predict

$$\hat{y}_i^{\text{CLP}} = g_{\vartheta}(X_i^{\text{rel-current}}, s_i),$$

where s_i is the optional train-standardized scalar context. Neural shapelet models are trained with a smooth- L_1 loss in log-duration space,

$$\ell(e_i) = \begin{cases} \frac{1}{2}e_i^2, & |e_i| < 1, \\ |e_i| - \frac{1}{2}, & |e_i| \geq 1, \end{cases} \quad e_i = \hat{y}_i^{\text{CLP}} - y_i^{\text{CLP}}.$$

This reduces the influence of extreme duration errors while preserving a quadratic region around small errors. XGBoost uses a squared-error objective on the same y_i^{CLP} target.

When scalar context is enabled, each neural model receives a one-channel sequence $X_i^{\text{rel-current}} \in \mathbb{R}^L$ and the standardized 12-dimensional vector s_i^{CLP} . The scalar vector is passed through a learned affine-ReLU-dropout embedding before concatenation with the shapelet representation. With $L = 30$, the XGBoost baseline receives a 42-column table: 30 current-relative lags followed by the same 12 scalar features.

Table 4.2: Current-level persistence models.

Model	Representation	Purpose
Shapelet MLP [35]	Minimum multiscale shapelet distances, optionally joined with scalar context.	Tests whether best-match local patterns are sufficient for duration regression.
Shapelet Transformer [35, 25]	The same minimum-distance vector interpreted as pattern-response tokens.	Tests interactions among learned pattern responses before regression.
Shapelet convolution [35]	Full multiscale shapelet response maps over temporal positions.	Preserves where a pattern occurs inside the context window.
XGBoost regressor [12]	Flattened current-relative lags plus scalar context.	Provides a non-neural tabular baseline for the same log-duration target.

All predictions are evaluated after conversion back to seconds through $\exp(\hat{y}_i^{\text{CLP}}) - 1$.

4.4.3 Switch Conversion

The current-level models output a duration estimate, not a switch. Let $T_{\min}$ be the operational minimum duration required to justify switching. The raw current-level switch is

$$r_i^{\text{CLP}} = \mathbb{I} \left[S_i \geq \theta \wedge \widehat{D}_i^{\text{CLP}} \geq T_{\min} \right].$$

In the experiments, $\theta = 10.0$ dB and $T_{\min} = 300$ seconds. The signal gate $S_i \geq \theta$ prevents a long predicted persistence around a non-critical level from triggering a backup-link switch. The duration condition then asks whether the current level is expected to remain persistent for at least the operational switch horizon. The shared post-processing policy is applied after this raw switch is produced.

4.5 Long-Fade Detection

Long-fade detection is a binary event-classification formulation. It is closely related to current-level persistence because both approaches ask whether the current fade condition is likely to persist long enough to justify an operational switch. The difference is the target. Current-level persistence predicts a continuous duration relative to the current signal level $S_i - \delta$. Long-fade detection instead predicts whether the grouped fade event to which the current timestamp belongs is a long event.

4.5.1 Task Definition

Let θ be the fade threshold and let $T_{\min}$ be the minimum event duration used to define an operationally relevant long fade. In the experiments, $\theta = 10.0$ dB and $T_{\min} = 300$ seconds. A threshold crossing occurs at index i when

$$S_i \geq \theta.$$

The long-fade task does not define an event as a single uninterrupted run of above-threshold samples. Such a definition would split one physical fade into several fragments whenever the signal temporarily oscillates around the threshold. Instead, threshold crossings are grouped with the same operational convention used in the

dataset builder. A grouped event starts at the first crossing. Further crossings whose timestamps are within three hours from that event start remain in the same grouped event. The event span extends from the first to the last threshold-crossing sample in the group. Samples below threshold inside this span are kept as part of the grouped event, so temporary recoveries do not create separate labels.

For a grouped event e , let a_e and b_e denote the first and last crossing indices in the group. Its duration is represented on the canonical sample grid as

$$D_e = (b_e - a_e + 1)\Delta t,$$

where Δt is the sampling interval. The event-level long-fade label is

$$y_e^{\text{LFD}} = \mathbb{1}[D_e \geq T_{\min}].$$

Every valid decision timestamp i inside the grouped event span receives the same label:

$$y_i^{\text{LFD}} = y_e^{\text{LFD}}, \quad i \in [a_e, b_e].$$

This design makes the task a grouped-event classifier rather than a pointwise-threshold classifier. The model is not asked whether the signal is above threshold at the next sample; it is asked whether the current grouped fade belongs to an event whose total duration reaches the operational minimum.

The input at a decision index is again restricted to past and current signal values. The main sequence representation is relative to the threshold:

$$X_i^{\text{rel-threshold}} = [S_{i-L+1} - \theta, \dots, S_i - \theta].$$

This representation centers the input around the operational decision boundary: positive values are above the fade threshold and negative values are below it. The model may also receive scalar context features computed only from the available past window and from online event-state information, such as the current signal, recent slopes, local window statistics, elapsed time since the first crossing, and the number of observed samples since the grouped event started. Quantities that require knowledge of the future event end, such as the final event duration or the final position fraction, are not allowed as model inputs.

The exact long-fade scalar vector is

$$s_i^{\text{LFD}} = [S_i, S_i - \theta, \bar{S}_{i,L}, \text{sd}_{i,L}, \min(X_i), \max(X_i), \max(X_i) - \min(X_i), \Delta_i^{(2)}, \Delta_i^{(3)}, \Delta_i^{(5)}, \bar{S}_{i,5}, \text{sd}_{i,5}, E_i^{\text{fade}}, P_i^{\text{fade}}],$$

where E_i^{fade} is the elapsed time in seconds since the first crossing of the grouped event, and P_i^{fade} is the number of observed samples since that event start. Both are online quantities because they depend only on the already observed beginning of the grouped event and the current timestamp. The tabular XGBoost input `X_lag_scalar` is the concatenation

$$[S_{i-L+1} - \theta, \dots, S_i - \theta, s_i^{\text{LFD}}],$$

with the lag columns named from lag_{L-1} to lag_0 . In the main $L = 30$ setup this gives 30 threshold-relative lags plus 14 scalar features. The TCN and shapelet-convolution classifiers receive the threshold-relative sequence as a sequence tensor and the same standardized scalar vector through a separate scalar branch.

4.5.2 Tested Models

All long-fade models output a probability

$$\hat{p}_i^{\text{LFD}} = P(y_i^{\text{LFD}} = 1 \mid X_i),$$

or a logit transformed into this probability. Neural classifiers are trained with weighted binary cross-entropy with logits,

$$\mathcal{L}_{\text{LFD}} = -\frac{1}{N} \sum_i \left[\alpha y_i^{\text{LFD}} \log \hat{p}_i^{\text{LFD}} + (1 - y_i^{\text{LFD}}) \log(1 - \hat{p}_i^{\text{LFD}}) \right],$$

where α is computed from the training split as the ratio between negative and positive samples, with a safe fallback when no positive sample is available. XGBoost uses the corresponding binary logistic boosted-tree objective and handles class imbalance through the positive-class scale parameter.

All three models are causal at decision time because their inputs contain only the past context, the current signal, and online event-state quantities already observable at t_i .

4.5.3 Switch Conversion

The long-fade classifier output is converted into a switch by thresholding the predicted probability and gating the decision with the current signal level. Let $\pi_{\min}$ be the configured probability threshold. The raw switch is

$$r_i^{\text{LFD}} = \mathbb{I} \left[S_i \geq \theta \wedge \hat{p}_i^{\text{LFD}} \geq \pi_{\min} \right].$$

The signal gate is important because the model can produce a long-fade probability

Table 4.3: Long-fade detection classifiers.

Model	Input representation	Purpose
XGBoost lag-and-scalar classifier	Threshold-relative lags plus train-standardized online scalar context.	Tests whether fixed lag features and simple signal summaries are sufficient for long-event classification.
Temporal convolution classifier	Threshold-relative sequence plus scalar-context embedding.	Uses dilated residual convolutions to combine local changes with wider temporal context.
Multiscale shapelet-convolution classifier	Multiscale distance-to-shapelet response maps plus scalar context.	Tests whether explicit local pattern matching improves over generic convolutional sequence features.

for timestamps inside the grouped event span even when the signal is temporarily below threshold. In those cases, the probability may still be informative about the event class, but the operational switch is activated only when the current signal is in the fade region. The default probability threshold used in the experiments is $\pi_{\min} = 0.5$. After this raw decision is formed, the same shared switch post-processing policy is applied before comparison with the Perfect Switch.

4.6 Survival Persistence

Survival persistence models the remaining duration of the current fade event as a time-to-event problem. It is related to current-level persistence because both tasks estimate a future duration, but the definition of the target is different. Current-level persistence asks how long the signal will remain above $S_i - \delta$. Survival persistence asks how long the current fade event will remain active before a stable recovery is observed.

The task is therefore naturally expressed with survival analysis notation. For a decision time t_i , let R_i be the residual duration of the current fade event. The desired model output is a conditional survival function

$$S(u \mid X_i) = P(R_i > u \mid X_i),$$

where u is a future time horizon. This output is more informative than a single duration estimate: it can answer questions such as whether the current fade is likely to persist for more than 60 seconds, 300 seconds, or 900 seconds. The formulation

also supports right-censored observations, a central feature of survival data [38]. In this project, censoring occurs when the acquisition segment ends before stable recovery from the fade can be observed.

4.6.1 Task Definition

Survival-persistence samples are generated only inside detected fade events. Events are constructed within continuous acquisition segments using a state-machine definition. The event starts when the activation condition is satisfied,

$$S_i \geq \theta_{\text{on}},$$

and remains active until a stable recovery is confirmed. A candidate recovery starts when the recovery condition is satisfied,

$$S_i < \theta_{\text{off}}.$$

The candidate becomes the event end only if the following recovery window is stable enough: in the implemented configuration, the recovery window is 300 seconds long, contains at least a minimum number of observations, and has at least 80% of its samples satisfying the recovery condition. If a new activation crossing occurs during the candidate recovery interval, the candidate is aborted and the event remains active. This rule prevents short oscillations around the threshold from breaking one physical fade into several artificial events.

For an observed event e , let a_e be the event start index and let c_e be the first timestamp of the stable recovery interval. The survival event end is t_{c_e} . The confirmation time may occur later, because future samples are needed to verify that the recovery interval is stable, but this confirmation delay is not added to the physical target duration. For a sample at index i inside the event,

$$R_i = t_{c_e} - t_i.$$

Samples at the event end itself are excluded so that all observed survival times remain strictly positive.

If the acquisition segment ends before a stable recovery can be confirmed, the sample is right-censored. In that case the exact residual duration is unknown, but it is known to be longer than the remaining observed follow-up in the segment. Let $t_{\text{seg,end}}$ be the last timestamp of the acquisition segment. The stored survival label is then

$$R_i > t_{\text{seg,end}} - t_i.$$

The canonical dataset represents this through lower and upper bounds:

$$y_{i,\text{lower}} = \begin{cases} R_i, & \text{if the event end is observed,} \\ t_{\text{seg,end}} - t_i, & \text{if right-censored,} \end{cases}$$

$$y_{i,\text{upper}} = \begin{cases} R_i, & \text{if the event end is observed,} \\ \infty, & \text{if right-censored.} \end{cases}$$

Observed samples therefore have a degenerate interval $[R_i, R_i]$, while right-censored samples have an interval $[t_{\text{seg,end}} - t_i, \infty)$. Censored samples are kept in the dataset; they are not dropped and they are not converted to artificial finite targets. This is a key methodological difference from ordinary duration regression.

The saved inputs are model-independent. For every survival sample, the dataset stores three sequence representations:

$$X_{\text{raw},i} = [S_{i-L+1}, \dots, S_i],$$

$$X_{\text{rel-threshold},i} = [S_{i-L+1} - \theta_{\text{on}}, \dots, S_i - \theta_{\text{on}}],$$

$$X_{\text{rel-current},i} = [S_{i-L+1} - S_i, \dots, 0].$$

It also stores a train-standardized scalar context vector. In the implemented dataset, this vector is ordered as

$$s_i^{\text{SP}} = [S_i, S_i - \theta_{\text{on}}, \bar{S}_{i,L}, \text{sd}_{i,L}, \min(X_i), \max(X_i), \max(X_i) - \min(X_i), \\ \Delta_i^{(2)}, \Delta_i^{(3)}, \Delta_i^{(5)}, \bar{S}_{i,5}, \text{sd}_{i,5}, E_i^{\text{event}}, P_i^{\text{event}}, \\ A_i, U_i, B_i, Q_i].$$

Here E_i^{event} and P_i^{event} are the elapsed seconds and sample count since the event start, A_i indicates whether the current sample is above the activation threshold, U_i is the time since the last above-threshold sample, B_i is the current run length of consecutive below-threshold samples, and Q_i is the fraction of above-threshold samples in the trailing event-local window of up to five samples. These variables are online event-state variables. They do not require the future event end or the future recovery confirmation.

Future fields such as event end time, recovery confirmation time, event duration, remaining duration, censoring flag, final event position, or future threshold

crossings are excluded from all survival model inputs. They are used only for target construction, censoring metadata, evaluation, or audit tables.

4.6.2 Tested Models

Two survival model baselines are considered: XGBoost with an accelerated failure time objective and a discrete-time causal TCN. Both models use the same canonical survival labels, including right-censored samples. They differ in how they represent the input history and in how they parameterize the conditional survival function.

The input choice is explicit for each baseline. XGBoost-AFT is tabular: selected sequence representations are flattened into lag columns and optionally concatenated with scalar context. The discrete-time TCN is sequential: it receives one or more signal channels of length L , with the standardized scalar vector entering through a separate branch when enabled.

XGBoost-AFT. XGBoost-AFT receives a feature vector ϕ_i^{SP} built from scalar context, flattened signal history, or their concatenation. No target bound, event-end field, censoring flag, or observed remaining time is part of this feature vector. The model assumes that the logarithm of the residual duration follows a location model,

$$\log R_i = F(\phi_i^{\text{SP}}) + s_{\text{AFT}}\varepsilon_i,$$

where F is the boosted-tree ensemble, $s_{\text{AFT}} > 0$ is the configured scale, and ε_i follows the selected AFT error distribution. The stored lower and upper label bounds are passed directly to XGBoost: observed samples use $[R_i, R_i]$, while right-censored samples use $[c_i, \infty)$, where $c_i = t_{\text{seg, end}} - t_i$ is the observed follow-up time.

After training, the booster outputs the location prediction $\hat{\mu}_i = F(\phi_i^{\text{SP}})$. The survival probability at a horizon u is obtained from the AFT error cumulative distribution F_ε :

$$\hat{S}(u \mid X_i) = 1 - F_\varepsilon\left(\frac{\log u - \hat{\mu}_i}{s_{\text{AFT}}}\right).$$

The same location prediction is also converted into a predicted median residual duration for diagnostic interpretation. The switch pipeline uses the survival probability at the operational horizon.

Discrete-time causal TCN. The discrete-time TCN uses the same canonical survival dataset but converts the continuous residual-duration labels to discrete event and censoring masks inside the model adapter. Let the time grid be defined by intervals $(b_{k-1}, b_k]$, with an optional final open-ended interval. For each sample,

the adapter constructs an event mask $e_{i,k}$ and an at-risk mask $a_{i,k}$. If the event is observed in bin k^* , then $e_{i,k^*} = 1$ and the at-risk mask is active up to that bin. If the sample is right-censored, no event bin is marked and the at-risk mask is active only for the bins that are fully observed before the censoring time.

The neural model receives one of the canonical sequence representations, or a multichannel stack of them, and optionally the 18-dimensional scalar context. A stack of left-padded causal residual convolutions encodes the sequence. The output head produces one hazard logit per discrete time bin,

$$g_{i,k} = g_{\vartheta,k}(X_i, s_i), \quad \hat{h}_{i,k} = \sigma(g_{i,k}).$$

Here $\hat{h}_{i,k}$ estimates the conditional hazard

$$P(R_i \in (b_{k-1}, b_k] \mid R_i > b_{k-1}, X_i).$$

The survival curve at the bin edges is reconstructed as the cumulative product of the predicted non-event probabilities:

$$\hat{S}(b_k \mid X_i) = \prod_{r=1}^k (1 - \hat{h}_{i,r}).$$

Training minimizes a masked discrete-time survival negative log-likelihood, implemented as binary cross-entropy over the at-risk bins,

$$\mathcal{L}_{\text{DT}} = \frac{\sum_i w_i \sum_k a_{i,k} \text{BCE}(g_{i,k}, e_{i,k})}{\sum_i w_i}.$$

For right-censored samples, the loss contains only bins known to have been survived. This keeps censored observations informative without converting them into artificial observed recoveries. The stored dataset remains model-independent because the discrete masks are derived inside the TCN adapter rather than saved as canonical labels.

4.6.3 Switch Conversion

The survival models output a curve rather than a binary switch. The operational conversion evaluates that curve at the switch-relevant horizon $u^* = 300$ seconds. Let $\rho_{\min}$ be the configured survival probability threshold. The raw switch is

$$r_i^{\text{SP}} = \mathbb{K} \left[S_i \geq \theta \ \wedge \ \hat{S}(u^* \mid X_i) \geq \rho_{\min} \right].$$

In words, the model recommends switching only if the current signal is in the fade region and the predicted probability that the event will persist for at least another five minutes is high enough. The default uncalibrated probability threshold is 0.5. An additional exploratory setting with threshold 0.65 was retained for analysis, but it should be interpreted as a test-set diagnostic rather than as a validation-calibrated operating point.

After this raw survival-derived switch is produced, the shared post-processing policy is applied and the final sequence is compared with the task-scoped Perfect Switch. This keeps the survival task aligned with the same operational evaluation layer used for the other formulations.

4.7 Comparison Between Formulations

The four formulations are not interchangeable supervised problems. They share the same signal-only input policy and the same final switch-evaluation layer, but they estimate different intermediate quantities. This matters because native predictive accuracy and operational switch quality are not equivalent: a model may forecast or classify well and still switch poorly near the decision boundary.

Table 4.4 summarizes the question addressed by each formulation, together with its main advantage and limitation.

Because these questions generate decisions at different native timestamp sets, cross-task comparison requires explicit alignment. The experimental protocol therefore reports native task metrics and coverage-aware cross-task switch metrics separately. The final comparison is a comparison of complete pipelines: past signal window, task-specific output, raw switch conversion, and shared post-processing against the same Perfect Switch reference.

Table 4.4: Conceptual comparison of the four formulations.

Formulation	Question answered	Main advantage	Main limitation
Autoregressive forecasting	What will the next signal values be?	Keeps an inspectable future trajectory and leaves the threshold outside the training target.	Optimizes generic forecast error, not switch-critical error.
Current-level persistence	How long will the current signal level persist?	Targets short-term persistence directly.	Does not model the end of the whole fade event and is sensitive to the current reference level.
Long-fade detection	Is the current grouped event a long fade?	Aligns the target with the minimum operational duration.	Discards residual-duration information within the event.
Survival persistence	What is $P(R_i > u \mid X_i)$ for the current event?	Keeps censored samples and provides horizon-dependent persistence probabilities.	Depends strongly on the event-start, stable-recovery, and censoring policy.

Chapter 5

Experimental Setup

Once the task formulations are fixed, the main experimental risk is no longer architectural ambiguity but unfair comparison. Models can look better or worse simply because they were trained on different splits, selected with different criteria, or evaluated through different switch rules. The protocol below makes these choices explicit. All experiments use the same raw signal sources and the same external-holdout principle, while each task still produces its own supervised dataset because the target variable and the set of valid decision timestamps are different.

The setup follows three constraints. First, the test dataset is held out as a complete external file and is never used for model selection. Second, all fitted transformations, such as feature standardization, are estimated on the training split only. Third, all final operational comparisons are made after converting model outputs into binary switch decisions and applying the shared post-processing rule defined in section 3.9.

5.1 Dataset

The common operational threshold used in the final experiments is $\theta = 10.0$ dB. Unless stated otherwise, samples are represented on a 30 s grid and the decision context contains the last $L = 30$ signal observations, corresponding to approximately 15 min of history. The operational persistence horizon used for switch decisions is $T_{\min} = 300$ s, i.e. 5 min. This horizon is used as the forecast horizon interpretation for autoregressive switch conversion, as the duration threshold for current-level persistence, as the minimum long-fade duration for long-fade detection, and as the survival horizon for survival persistence.

The external test source is the complete file `fc-uplink-fade.csv`. It is excluded from all training, validation, grid search, early stopping, and model-selection steps.

The remaining signal files form the development pool. Within the development pool, train and validation splits are constructed chronologically at event level when event identities are available, so that strongly related windows from the same event are not split between training and validation.

5.1.1 Autoregressive Forecasting Dataset

The autoregressive dataset uses past windows and future signal trajectories. The main configuration uses $L = 30$ and prediction length $h = 10$, so each target covers the next 300 s. The resulting external-holdout dataset contains 19,412 training windows, 4,783 validation windows, and 7,708 test windows under the inclusive $S_i \geq 10.0$ dB threshold convention. A separate diagnostic Chronos experiment also uses $L = 120$ while keeping $h = 10$; this longer context is treated as an additional Chronos run, not as the common autoregressive configuration for all models.

5.1.2 Current-Level Persistence Dataset

The current-level persistence dataset uses the same $L = 30$ context length and $\delta = 0.5$ dB. Its target is the residual time for which the future signal remains above $S_i - \delta$. Since this target requires observing a future recovery below a sample-specific level, a small number of candidate windows is right-censored and excluded from ordinary duration-regression training. The final uncensored regression dataset contains 19,598 training windows, 4,798 validation windows, and 7,692 test windows. The excluded censored candidate windows account for approximately 2.12%, 2.74%, and 3.22% of the train, validation, and test candidate windows respectively.

5.1.3 Long-Fade Detection Dataset

The long-fade detection dataset is built only inside grouped fade events. It uses $L = 30$, $\theta = 10.0$ dB, and a long-fade label defined by event duration at least $T_{\min} = 300$ s. The resulting dataset contains 4,589 training samples from 62 grouped events, 790 validation samples from 15 grouped events, and 2,569 test samples from 25 grouped events. This dataset is much smaller than the autoregressive and current-level datasets because it does not generate samples outside grouped fade events.

5.1.4 Survival-Persistence Dataset

The survival-persistence dataset also uses $L = 30$, but keeps censored samples instead of discarding them. Each sample is generated inside a detected fade event and stores lower and upper time bounds for the residual event duration. Observed samples have equal lower and upper bounds, while censored samples have an infinite upper bound. The final survival dataset contains 1,729 training samples, 729 validation samples, and 921 test samples. The validation split contains 6 censored samples from one event, while the external test split contains 70 censored samples from two events.

5.1.5 Small-Gap Imputation Policy

Small-gap imputation is enabled only through the shared conservative policy. It may fill short gaps up to 90 s, corresponding to at most two inserted 30 s samples, by linear interpolation in time. The policy is applied before task-specific dataset construction where explicitly configured, but it does not bridge acquisition-segment boundaries or long missing intervals.

5.2 Train, Validation, and Test Protocol

The experiments use a three-way protocol: training, validation, and external test. The training split is used to fit model parameters. The validation split is used for early stopping, hyperparameter selection, architecture selection, and any comparison among alternative variants of the same task. The external test split is used only once a model or configuration has already been selected.

The distinction between validation and test is especially important in this work, because the models are evaluated not only through their native predictive losses but also through derived switch decisions. A threshold or architecture that looks attractive on the test set would no longer provide an unbiased estimate of future operational behavior. For this reason, the final selected models are chosen using validation-only criteria. Exploratory operating points reported outside this rule are explicitly treated as diagnostics rather than as calibrated deployment thresholds.

For trainable neural models, early stopping or fixed-epoch training is governed by validation-only information during the selection phase. After the best configuration is chosen, the published run is either retrained on the full development set, i.e. training plus validation, or the selected validation-stage checkpoint is promoted. For reproducibility, this finalization choice is not left implicit: each published run stores in its metadata whether the test artifact was obtained from a train-validation refit

or from a promoted validation-stage checkpoint, together with the selected epoch, configuration, random seed, and dataset fingerprint when available. The external test split is evaluated only after the selection step has been completed. XGBoost models follow the same principle: validation selects the hyperparameters, and the selected configuration is then published before test evaluation. Chronos is the only exception, because it is evaluated in zero-shot mode and therefore loads no training or validation data.

All splits preserve the signal-only policy. Time, event identifiers, dataset identifiers, and quality flags are stored as metadata for alignment, diagnostics, and grouping, but they are not used as model inputs unless a feature is explicitly described as an online scalar derived from the past signal window or from the current event state available at the decision time.

5.3 Hyperparameter Search

Hyperparameter search is performed separately for each task and model family. Each search is defined by a YAML configuration file and produces a structured trial table, selected-parameter file, final run folder, model artifact folder, and metadata file. The grid-search runners assign independent trials across all available CUDA devices when possible. If CUDA is unavailable, the code falls back to MPS when supported, and finally to CPU.

Table 5.1: Summary of the main model-selection searches.

Task	Model family	Trials	Selection metric
Autoregressive	GRU S2V, raw/context-standard	72 each	validation RMSE
Autoregressive	GRU encoder-decoder, raw/context-standard	216 each	validation RMSE
Autoregressive	PatchTST, raw/context-standard	864 each	validation RMSE
Autoregressive	XGBoost, raw/context-standard	1152 each	validation RMSE
Current-level	Shapelet	216 each	validation MAE
persistence	MLP/Transformer/Convolution		
Current-level	XGBoost duration regressor	1152	validation MAE
persistence			
Long-fade	XGBoost/TCN/Shapelet-	8928 total	validation AUPRC
detection	Convolution classifiers		
Survival	XGBoost-AFT	1920	validation AFT
persistence			negative log-likelihood
Survival	Discrete-time TCN	1620	validation Integrated
persistence			Brier Score

The search dimensions and validation-only selection rules are summarized in table 5.2. All error-based selection metrics are computed after returning predictions to their reported physical scale.

Table 5.2: Main hyperparameter dimensions and validation-only selection rules.

Task and model family	Tuned dimensions	Selection rule
Autoregressive GRU	input variant, recurrent capacity, number of layers, dropout, learning rate, weight decay, teacher forcing for encoder-decoder	validation RMSE in raw signal scale
Autoregressive PatchTST	input variant, patch length and stride, Transformer width, heads, layers, dropout, learning rate, weight decay	validation RMSE in raw signal scale
Autoregressive XGBoost	input variant, tree depth, estimators, learning rate, subsampling, column sampling, child weight, regularization	validation RMSE in raw signal scale
Current-level persistence	shapelet representation size, neural head capacity, scalar-context embedding, dropout, learning rate; XGBoost tree parameters	validation MAE in seconds
Long-fade detection	classifier family, sequence or lag representation, model capacity, tree or neural regularization	validation AUPRC
Survival persistence	AFT tree parameters; TCN binning, input representation, sample weighting, architecture scale, seed	AFT negative log-likelihood or Integrated Brier Score

5.4 Forecast, Regression, Classification, and Survival Metrics

The metric definitions are introduced in section 2.4.3. This section specifies how those metrics are used in the experimental protocol.

For autoregressive forecasting, MAE and RMSE are computed after all predictions have been returned to raw signal scale. They are reported globally and by horizon step. Validation RMSE is used for neural and XGBoost model selection, because large forecast errors near the operational threshold can strongly affect the derived switch decision.

For current-level persistence, the models are trained on $\log(1 + D_i^{\text{CLP}})$, but native evaluation is performed after mapping predictions back to seconds. Validation MAE

in seconds is used for model selection, and test MAE/RMSE in seconds are retained as diagnostic duration metrics.

For long-fade detection, the native output is a probability that the grouped event is a long fade. Validation AUPRC is used for model selection because the classification problem is imbalanced. AUROC and thresholded classification metrics are reported as secondary diagnostics.

For survival persistence, XGBoost-AFT is selected by validation AFT negative log-likelihood, while the discrete-time TCN is selected by validation Integrated Brier Score. Concordance index, Brier scores, and survival likelihood values are reported to diagnose whether the predicted residual duration distribution is meaningful before it is converted into a switch.

5.5 Switch Metrics

The switch metrics are also defined in section 2.4.3. In the experiments, they are computed after task-specific raw decisions have been converted into the common post-processed switch sequence. The evaluated sequence is always compared with the task-scoped Perfect Switch on the same timestamp grid.

The reported switch tables include pointwise metrics, active switch duration, the number of switch islands, detected and missed Perfect Switch events, false switch events, onset delay when defined, and the amount of change introduced by post-processing. These quantities are reported because the operational cost of backup-link activation depends on the duration and fragmentation of switch periods, not only on pointwise classification agreement.

5.6 Use of Shared Switch Post-Processing

The post-processing rule itself is defined with the reference framework in section 3.9. In the experiments, the same configured rule is applied unchanged after each task-specific raw switch conversion. The primary evaluated sequence is `model_switch`, which is the held minimum-time sequence introduced in section 3.9. The auxiliary diagnostic columns are:

- `model_switch_raw`, the immediate task-specific decision;
- `model_switch_min_island_old`, after minimum-duration extension;
- `model_switch_min_time`, after the above-threshold holding rule;

- `model_switch_adjusted`, the legacy adjusted diagnostic variant.

Only `model_switch` is used as the main operational sequence; the other columns are retained to audit how much post-processing changes the immediate model decision.

5.7 Intra-Task Comparison Protocol

Intra-task comparisons evaluate models that solve the same supervised problem. They therefore share the same dataset, split, native target, Perfect Switch grid, and switch-conversion rule. This is the cleanest comparison level because the only intended difference is the model family or input representation.

For autoregressive forecasting, the comparison includes GRU sequence-to-vector, GRU encoder-decoder, PatchTST, XGBoost, and Chronos zero-shot runs. Forecast metrics are compared where available, and switch metrics are computed after applying the autoregressive conversion rule defined in chapter 4.

For current-level persistence, the comparison includes the three learnable shapelet regressors and the XGBoost duration regressor. The models are selected by validation duration error and then evaluated through the common duration-to-switch conversion.

For long-fade detection, the comparison includes XGBoost, TCN, and shapelet-convolution classifiers. The models are selected by validation AUPRC and then evaluated through the common probability-to-switch conversion.

For survival persistence, XGBoost-AFT and the grid-selected discrete-time TCN are compared through survival metrics and through the survival-to-switch conversion at the configured operational horizon.

Each intra-task comparison also produces event-level plots. The plots show the raw signal, the operational threshold, the Perfect Switch, and the post-processed model switch sequence. For long-fade and survival models, the visual window is extended to 90 minutes before and after the event timestamp, while switches outside the task-native decision grid are displayed as zero. This makes event plots easier to compare visually without changing the metric denominator.

5.8 Cross-Task Comparison Protocol

Cross-task comparison is more difficult than intra-task comparison because the tasks do not necessarily produce decisions at the same timestamps. The autoregressive task produces one decision for each valid forecast window. Current-level persistence produces decisions for valid duration-regression samples. Long-fade detection and

survival persistence produce decisions only inside their own event definitions. Directly comparing only the common intersection of all timestamps would hide this difference in coverage.

The primary cross-task protocol therefore aligns all methods to a common reference grid derived from the external-holdout Perfect Switch. If a method has no native decision at a reference-grid timestamp, the missing value is filled as zero. This denominator is called `reference_grid_missing_as_zero`. It evaluates both the correctness of the decisions that a method makes and the operational consequence of not producing decisions at some reference-grid timestamps.

A second denominator, `strict_common_timestamp_intersection`, is also computed as a diagnostic. It keeps only timestamps where all compared methods have native decisions. This view is useful for asking which method behaves better when all methods are active on the same timestamps, but it is not sufficient as the main comparison because it removes the coverage problem.

The cross-task analysis therefore reports both switch metrics and native decision coverage. Coverage is the fraction of the reference grid on which a method actually produced a native decision before missing values were filled. This is essential for interpreting models such as long-fade and survival methods, whose task definitions intentionally restrict the decision set.

5.9 Hardware and Reproducibility

The implementation is based on Python. Neural models are implemented with PyTorch, while tree-based baselines use XGBoost. Experiment configuration is stored in YAML files, and generated artifacts are organized by task, selection identifier, and run identifier.

The source-code repository is linked in section 1.4. Its structure and artifact schemas are described in chapter A.

All model runners use centralized device selection. The preferred device order is CUDA, then Apple MPS when supported, then CPU. Grid searches consist of independent trials and therefore distribute trials across all available CUDA devices, for example `cuda:0`, `cuda:1`, and `cuda:2`, when the machine exposes multiple GPUs. Final run metadata records the selected device, configuration fingerprint, dataset fingerprint when available, model family, architecture, variant, seed, and output paths.

The main random seed is 42 unless otherwise stated. Neural training stores the best checkpoint and training metadata. XGBoost runs store the fitted model and

the resolved configuration. Chronos stores run metadata and predictions but no trained checkpoint, because no project-specific training is performed. All published runs save prediction files so that switch comparisons can be regenerated without retraining the models.

The result structure separates four artifact types. First, dataset-preparation artifacts store split summaries, metadata, and quality checks. Second, grid-search artifacts store trial tables and selected configurations. Third, run artifacts store model predictions, native metrics, plots, and run metadata. Fourth, comparison artifacts store intra-task switch evaluations, model summaries, and cross-task switch plots. This separation is used to make later thesis tables reproducible from saved outputs rather than from manually edited numbers.

Chapter 6

Experimental Results and Analysis

This chapter reports the empirical results obtained with the task formulations defined in chapter 4 and the experimental protocol defined in chapter 5. The analysis follows the same order as the modeling pipeline: first the prepared data are summarized, then each task is evaluated in its native predictive space, and finally all model outputs are converted into switch decisions and compared against the common Perfect Switch reference.

The test set is always the external file `fc-uplink-fade.csv`. Native metrics are useful for diagnosing whether a model learned its assigned target, but the main operational interpretation is based on the switch metrics, because the practical objective is the decision of whether the backup link should be activated.

Throughout this chapter, arrows in performance-table headers indicate the preferred direction of each metric: $\uparrow$ means that higher values are better, whereas $\downarrow$ means that lower values are better. Descriptive quantities without a universally preferred direction are shown without an arrow.

6.1 Preliminary Data Analysis

The raw collection contains eight signal files acquired between July 2020 and July 2021. After parsing timestamps, selecting the signal column, removing invalid rows, and enforcing the signal-only schema, the usable collection contains 2,835,814 valid observations out of 3,042,803 raw rows. Four files contain additional columns: the weather-beacon file contains meteorological variables, the two `fc-*` files contain one extra column, and `uplink_beacon_data_fucino_GX4.csv` contains three additional columns. These fields are ignored in all experiments, consistently with the signal-only policy.

The sampling analysis confirms that the data are close to a regular 30 s grid.

Six files have median inter-sample spacing of 30 s, while the GX4 uplink beacon file has median spacing of 31 s. The complete collection contains 462 continuous acquisition segments when gaps larger than 90 s are used as segment boundaries. These boundaries are important because input windows and duration targets must not assume continuity across long gaps. Short gaps are handled only by the conservative imputation policy described in section 5.1.

Most samples are close to the non-fade operating region, while threshold exceedances are rare and concentrated in events. For example, the median signal is between approximately 0.3 dB and 1.5 dB across the source files, whereas the final operational threshold is $\theta = 10.0$ dB. The largest observed values are much higher, reaching 69.0 dB in the Fucino uplink sources. This imbalance is the reason why all tasks are built around candidate fades, grouped fade events, or residual durations rather than around uniformly sampled background windows.

Table 6.1: Final supervised datasets used in the experimental phase.

Task	Main target	Train	Validation	Test	Event coverage
Autoregressive forecasting	future signal trajectory	19,412	4,783	7,708	61/15/24 events
Current-level persistence	residual persistence above $S_i - \delta$	19,598	4,798	7,692	61/15/24 events
Long-fade detection	long-event class label	4,589	790	2,569	62/15/25 grouped events
Survival persistence	residual fade-event duration	1,729	729	921	117/23/55 survival events

Table 6.1 shows that the four task datasets do not have the same number of samples, even though they originate from the same raw signals and use the same external holdout source. This is expected. The autoregressive and current-level datasets are built from candidate event windows; long-fade detection is restricted to timestamps inside grouped fade events; survival persistence is restricted to timestamps inside state-machine fade events and keeps censored observations. Consequently, cross-task comparison cannot rely only on native losses. The common switch-conversion and reference-grid analyses later in this chapter are used to compare the operational behavior of models trained under these different sample definitions.

6.2 Autoregressive Forecasting

The autoregressive experiments predict the next $h = 10$ signal samples from a past context of $L = 30$ samples. Validation RMSE is reported for model selection; final forecast and switch metrics use the external test file. Forecast errors on both splits are reported in raw signal scale, while switch metrics are test-only. The switch decision is obtained from the predicted trajectory: a switch is requested at time i only when all 10 predicted future samples are at least 10.0 dB, followed by the shared switch post-processing defined in section 3.9.

6.2.1 Forecast Accuracy

Table 6.2 summarizes validation and test forecast errors. The validation column is absent for Chronos because it is used only in zero-shot mode and does not load train or validation data.

Table 6.2: Autoregressive validation RMSE used for model selection and forecast accuracy on the external holdout split.

Method	Val. RMSE ↓	Test MAE ↓	Test RMSE ↓	Variant
GRU S2V	3.124	1.343	2.852	raw
GRU Seq2Seq	3.560	1.326	3.040	raw
XGBoost	3.863	1.376	3.088	raw
PatchTST	4.090	1.515	3.088	raw
XGBoost	3.221	1.465	3.363	context std.
PatchTST	3.811	1.475	3.512	context std.
GRU S2V	3.686	1.443	3.558	context std.
GRU Seq2Seq	3.702	1.460	3.671	context std.
Chronos zero-shot	—	1.540	3.899	raw

The best test RMSE is obtained by the raw GRU sequence-to-vector model (2.852), while the lowest test MAE is obtained by the raw GRU encoder-decoder (1.326). This difference is expected because RMSE penalizes large peak errors more strongly than MAE. The raw XGBoost and raw PatchTST models have almost identical test RMSE, despite using very different model classes. Chronos performs worse than the trained models on this external holdout, which is consistent with its zero-shot use: it is not adapted to the particular scale, noise, and event morphology of the Fucino data.

The context-standardized variants do not improve external test performance. Although context standardization can simplify optimization, it removes part of the absolute signal-level information that is operationally important near the 10.0 dB threshold. For this task, the raw representation is therefore the more reliable choice.

Figure 6.1 shows the horizon-wise error profile for PatchTST raw. Both MAE and RMSE generally increase with the forecast step, confirming that the last points of the 300s horizon are more uncertain than the immediate future. This behavior matters for switch conversion because the operational rule requires all 10 predicted samples to remain above the threshold.

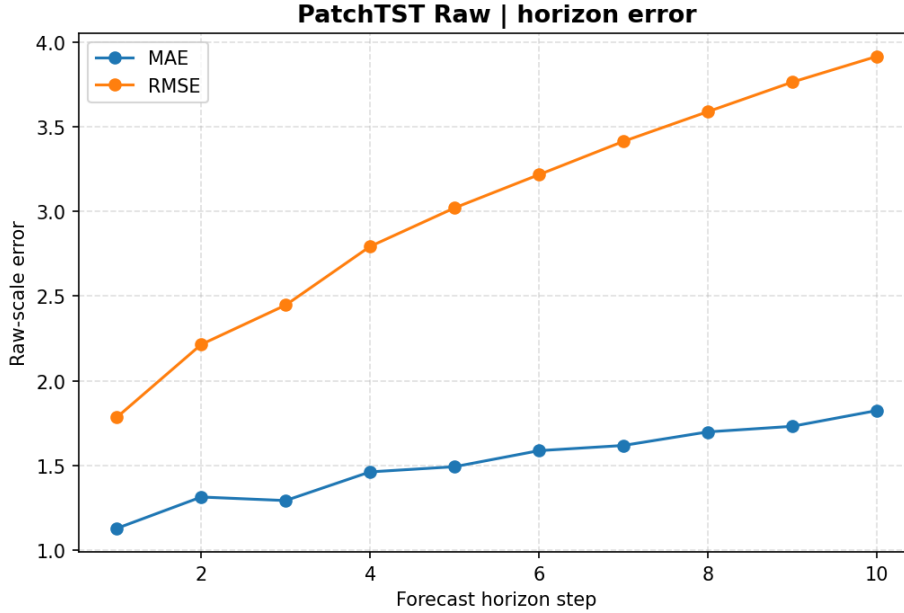


Figure 6.1: Horizon-wise forecast error for the PatchTST raw autoregressive model on the external holdout split.

6.2.2 Switch Performance

Table 6.3 reports the switch metrics obtained after applying the trajectory-to-switch rule and the shared post-processing. The ranking differs from the pure forecast-error ranking. PatchTST raw has the best F1 and IoU, even though it is not the best model in RMSE.

Point-wise precision, recall, F1, and IoU do not fully describe the operational behavior of a switch policy. Two models can have similar F1 while producing different active durations or a different number of switch activations. For this reason, fig. 6.2 reports the total post-processed switch duration and the number of contiguous switch islands on the external holdout split. The Perfect Switch contains 459 active samples, corresponding to 229.5 min, grouped into 18 switch islands. PatchTST raw is the closest autoregressive model in this diagnostic view, with 494 active samples and 20 islands. Several other models reach high recall by keeping the switch active for much longer or by producing many more islands, which explains their lower precision.

Table 6.3: Autoregressive switch performance against Perfect Switch on the external holdout split.

Method	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	IoU $\uparrow$
PatchTST raw	0.779	0.839	0.808	0.678
XGBoost raw	0.692	0.943	0.798	0.664
GRU Seq2Seq raw	0.619	0.980	0.759	0.611
Chronos zero-shot raw	0.567	0.961	0.713	0.554
XGBoost context-standard	0.512	0.991	0.675	0.510
GRU S2V raw	0.509	0.993	0.674	0.508
PatchTST context-standard	0.496	0.991	0.661	0.493
GRU S2V context-standard	0.496	0.987	0.660	0.493
GRU Seq2Seq context-standard	0.497	0.985	0.660	0.493

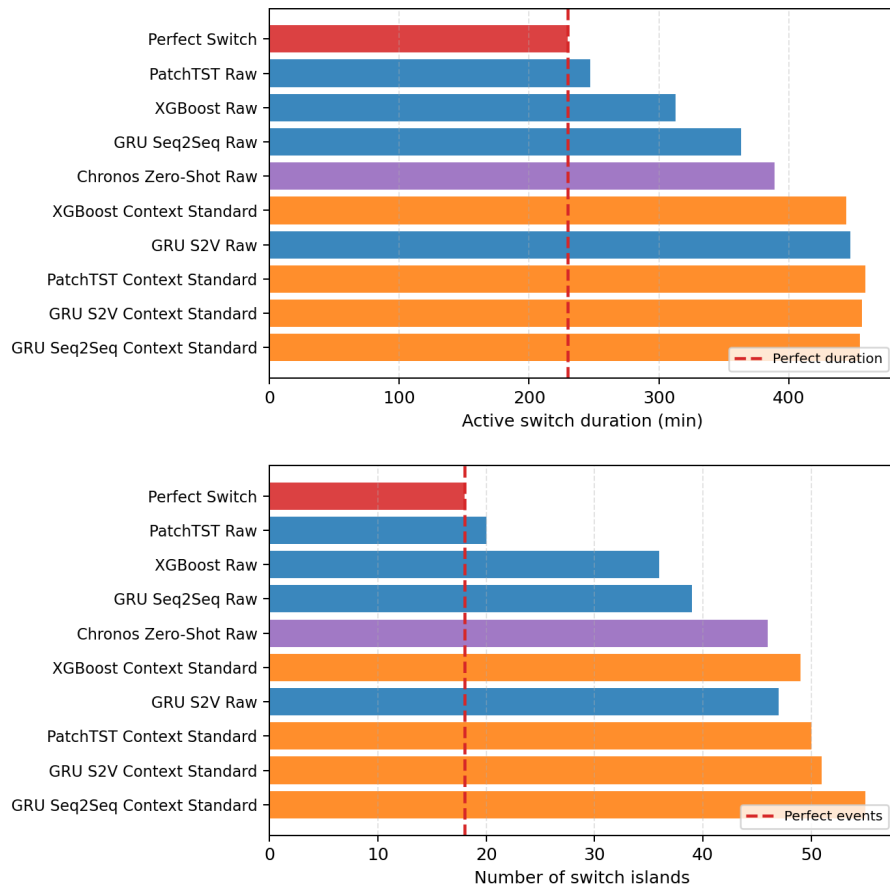
Autoregressive switch duration and event-count behaviour

Figure 6.2: Autoregressive switch duration and event-count behavior on the external holdout split. Duration counts post-processed positive switch samples, converted to minutes using the 30 s sampling time. The dashed red line marks the Perfect Switch value.

The common pattern is high recall and lower precision. In operational terms, most Perfect Switch points are detected, but several models keep the switch active for longer than the reference. The raw PatchTST model is the most balanced among the autoregressive models: it sacrifices some recall relative to the GRU models, but reduces false positives enough to obtain the highest F1 and IoU. XGBoost raw is close behind and is notable because it reaches competitive switch performance without an explicit neural sequence architecture.

The confusion counts clarify this trade-off. PatchTST raw detects 385 of 459 Perfect Switch samples and misses 74, while adding 109 false-positive switch samples over 7708 evaluated timestamps. This corresponds to a balanced accuracy of 0.912 and a false-positive rate of 0.015. Its advantage is therefore not that it detects every switch sample, but that it preserves most operational intervals while keeping unnecessary switch time limited. The context-standardized variants have the opposite tendency: they often increase recall, but they lose precision because local standardization removes part of the absolute-level information needed near the 10.0 dB threshold.

6.2.3 Interpretation and Event-Level Examples

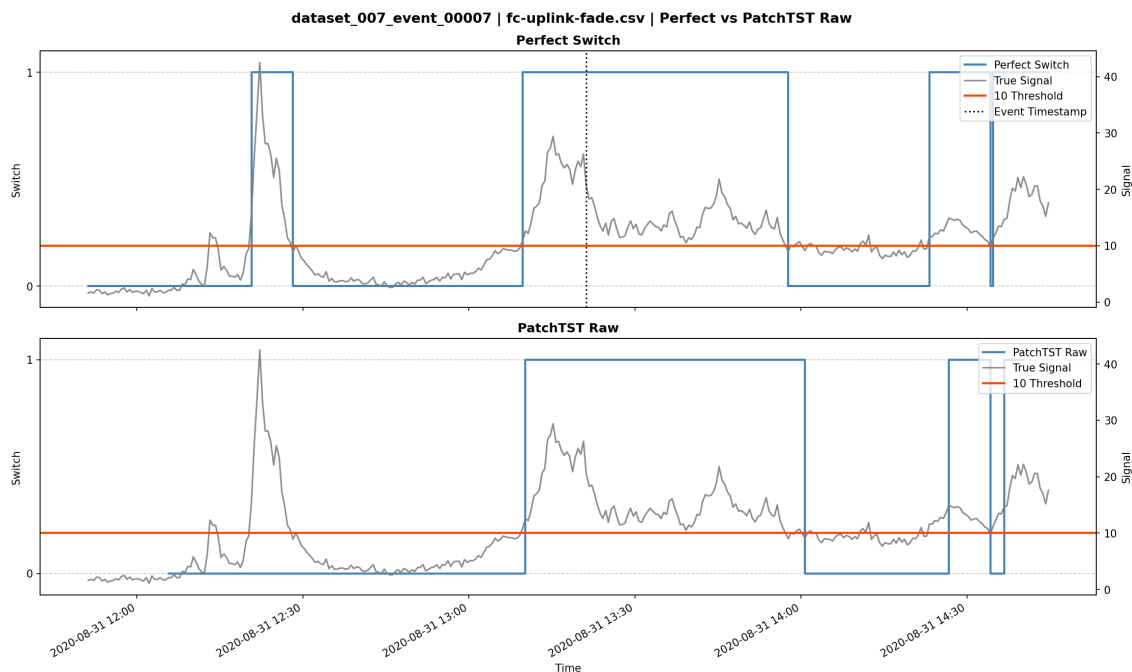


Figure 6.3: Example autoregressive switch comparison for PatchTST raw on a multi-interval external-holdout event. The model-derived switch is compared with the Perfect Switch reference after applying the common post-processing rule.

Figure 6.3 shows event `dataset_007_event_00007`, which is more informative than a single isolated fade because the Perfect Switch turns on and off several times. The upper panel contains the Perfect Switch reference, while the lower panel contains the PatchTST raw decision. The model captures the long central switch interval and the late reactivation, but misses the first short activation. This illustrates why switch metrics are not determined only by point forecast error: a model can tolerate moderate forecast deviations if it still preserves the operational threshold crossing structure.

Figure 6.4 compares all autoregressive switch decisions on event 00003 of dataset 007, a different, single-interval example from fig. 6.3. The upper panel shows the raw signal and the 10.0 dB threshold, while each lower row shows the final post-processed switch sequence for one method. This compact view makes clear that most autoregressive models identify the main fade, but they slightly differ in onset timing, release timing, and the amount of extra switch activity around the event.

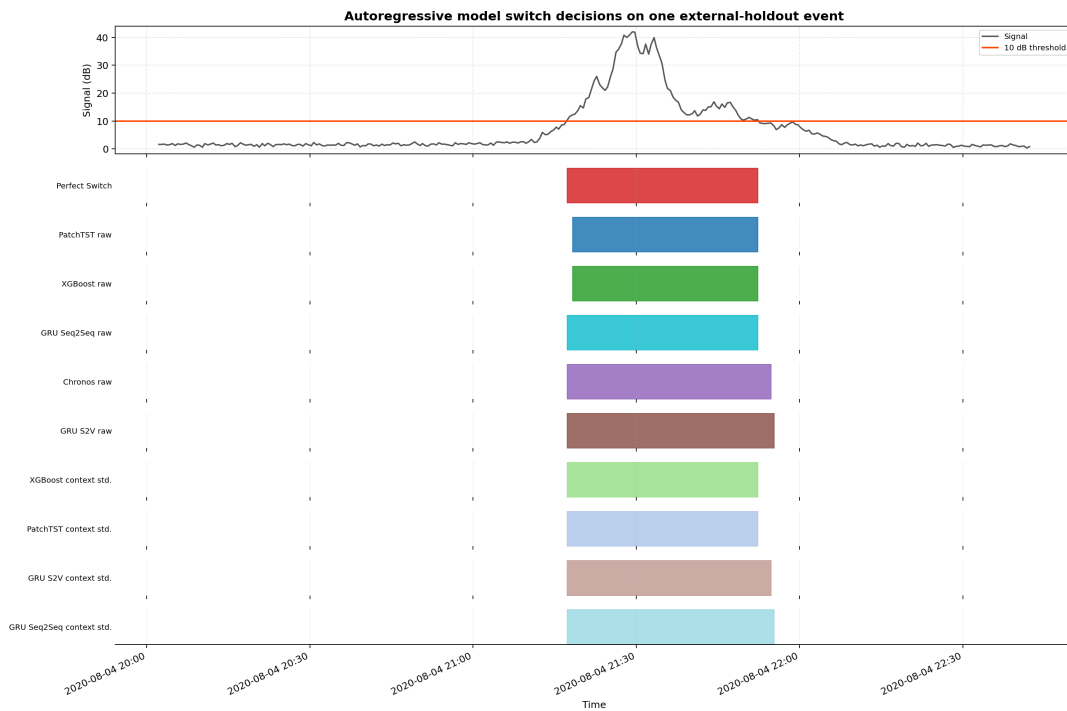


Figure 6.4: Mosaic of post-processed switch decisions for all autoregressive methods on one external-holdout event. Each row reports the final switch used for evaluation, not the raw thresholded model output.

6.3 Current-Level Persistence

The current-level persistence task predicts how long the signal will remain above $S_i - \delta$, with $\delta = 0.5$ dB. This is a scalar duration regression problem, not a future-trajectory forecast. A switch is requested only when the current signal is at least 10.0 dB and the predicted remaining persistence is at least 300 s; the same switch post-processing used for the other tasks is then applied.

6.3.1 Duration Prediction Accuracy

Table 6.4 reports the validation and test duration-regression metrics. Errors are expressed in seconds because the operational threshold is 300 s. The table also reports the median absolute error, which is important because the target distribution is highly right-skewed: a small number of very long persistence intervals can dominate MAE and especially RMSE.

Table 6.4: Current-level persistence duration-regression results on the external holdout split. All error metrics are expressed in seconds.

Method	Val. MAE (s) ↓	Test MAE (s) ↓	Test RMSE (s) ↓	Test median AE (s) ↓
XGBoost scalar context	1220.9	586.9	2584.6	27.2
Shapelet convolution scalar context	4613.2	1835.6	6491.0	141.5
Shapelet MLP scalar context	5116.5	2063.8	7629.7	160.6
Shapelet Transformer scalar context	5278.9	2056.6	7685.3	169.4

XGBoost is clearly the strongest model for this formulation. Its test median absolute error is only 27.2 s, which means that typical short-horizon decisions are often accurate. However, its test MAE is 586.9 s and its RMSE is 2584.6 s, showing that large errors still occur on long persistence intervals. The learnable-shapelet models are substantially worse: their test MAE is around 30 min to 34 min, and their RMSE exceeds 108 min. These errors are too large for the scalar duration prediction to be considered reliable by itself.

Figure 6.5 summarizes the same behavior in minutes. The median errors are much smaller than the mean errors, confirming that the hard cases are concentrated in the long tail of the duration distribution rather than uniformly distributed over all windows.

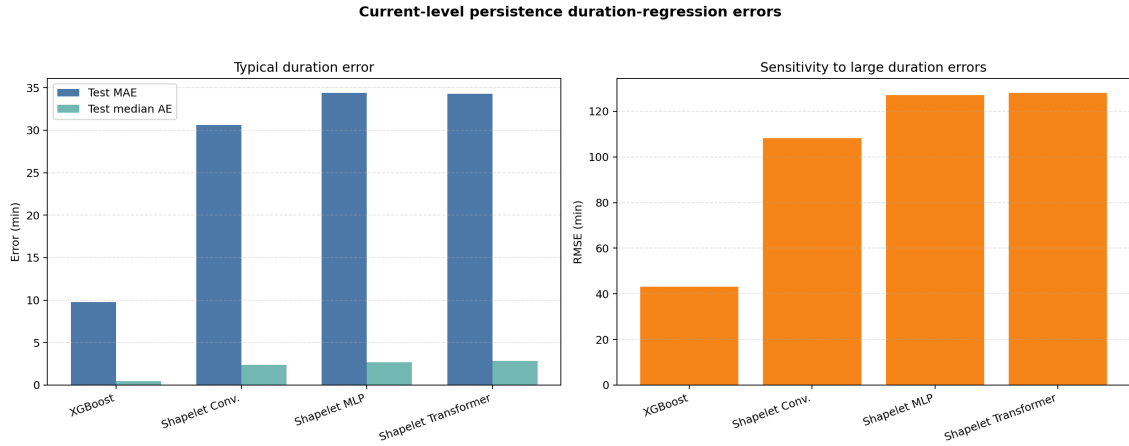


Figure 6.5: Current-level persistence duration-regression errors on the external holdout split. MAE and median absolute error describe typical behavior, while RMSE highlights sensitivity to long-duration mistakes.

Figure 6.6 shows the predicted and true durations for the best current-level model. The dense region near the origin is well aligned, but the dispersion increases for longer intervals. This explains why the median absolute error is low while MAE and RMSE remain large.

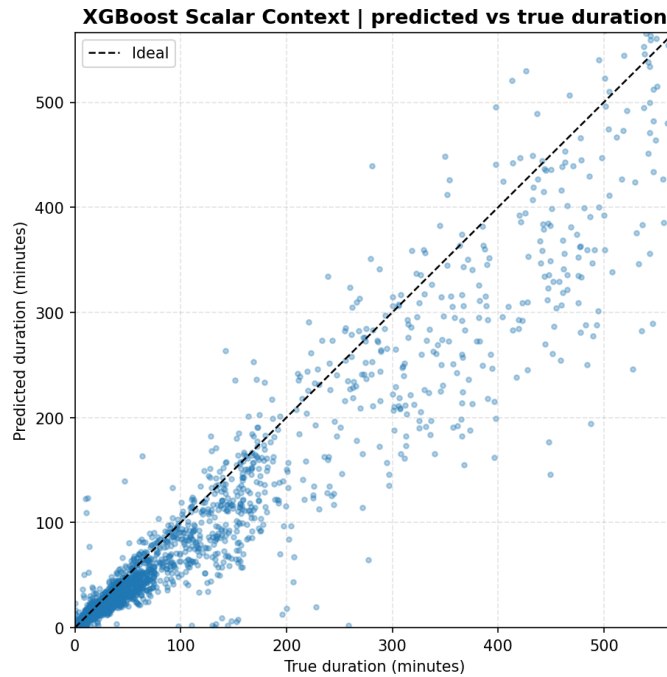


Figure 6.6: Predicted versus true current-level persistence duration for the XGBoost scalar-context model on the external holdout split.

6.3.2 Switch Performance

Table 6.5 reports the resulting switch metrics. Despite the imperfect duration regression, XGBoost gives a useful switch policy: it reaches an F1 score of 0.902 and produces no false-positive switch samples on the external holdout split. The three learnable-shapelet models instead produce an all-zero final switch sequence, so they obtain zero recall and zero F1. This is not caused by missing predictions. The signal gate is active at 672 test timestamps, but the maximum persistence predicted at those timestamps is only 148.8s for the shapelet MLP, 204.6s for the shapelet Transformer, and 167.3s for the shapelet convolution model. All remain below the 300s duration threshold. The same models do predict durations above 300s at other timestamps, but only when the current signal is below 10.0dB. Consequently, the two conditions required by the decision rule are never true simultaneously. This is an operationally important negative result: the models fit the regression target to some extent, but they are not usable switch controllers under the configured rule.

Table 6.5: Current-level persistence switch performance against Perfect Switch on the external holdout split.

Method	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	IoU $\uparrow$	Active min.
XGBoost scalar context	1.000	0.822	0.902	0.822	189.5
Shapelet MLP scalar context	0.000	0.000	0.000	0.000	0.0
Shapelet Transformer scalar context	0.000	0.000	0.000	0.000	0.0
Shapelet convolution scalar context	0.000	0.000	0.000	0.000	0.0

Figure 6.7 makes this contrast explicit. The Perfect Switch has 461 active samples, corresponding to 230.5 min, and 17 switch islands. XGBoost produces 379 active samples and 11 islands. It therefore under-covers part of the Perfect Switch, but its conservative behavior avoids the excessive switch activity observed in several autoregressive variants. The shapelet models produce no active switch interval under the configured decision rule.

This result should be read together with the decision rule. A current-level model is allowed to request a switch only when the current signal is already at or above the 10.0dB threshold and the predicted persistence exceeds 300s. The XGBoost model therefore behaves as a conservative switch policy: it produces 379 true-positive switch samples, no false-positive samples, and 82 false negatives. The resulting precision of 1.000 is operationally attractive because every requested switch is aligned with the reference, but the recall loss indicates that some useful backup-link intervals would

be missed. This is a different failure mode from autoregressive forecasting, where the main issue is excess active duration rather than missed coverage.

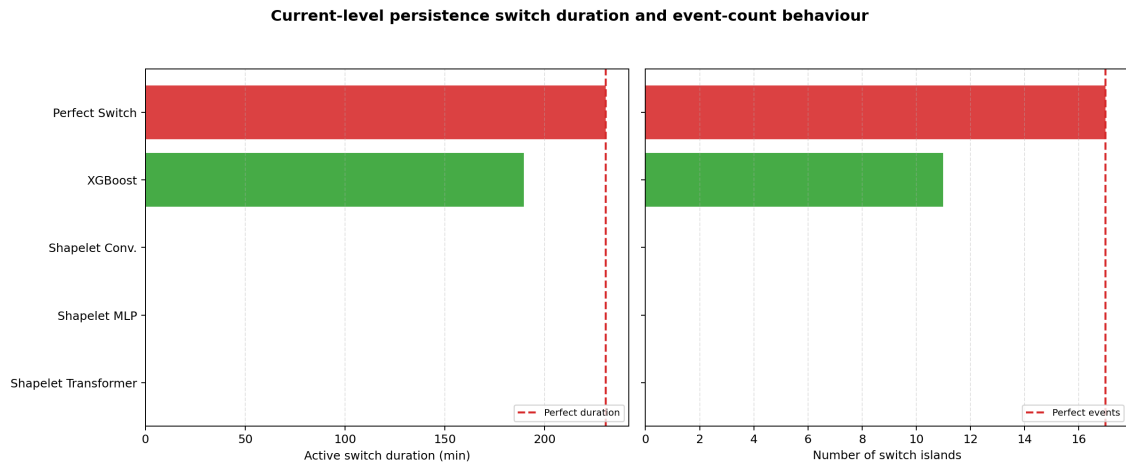


Figure 6.7: Current-level persistence switch duration and event-count behavior on the external holdout split. Duration counts post-processed positive switch samples, converted to minutes using the 30 s sampling time.

6.3.3 Interpretation and Event-Level Examples

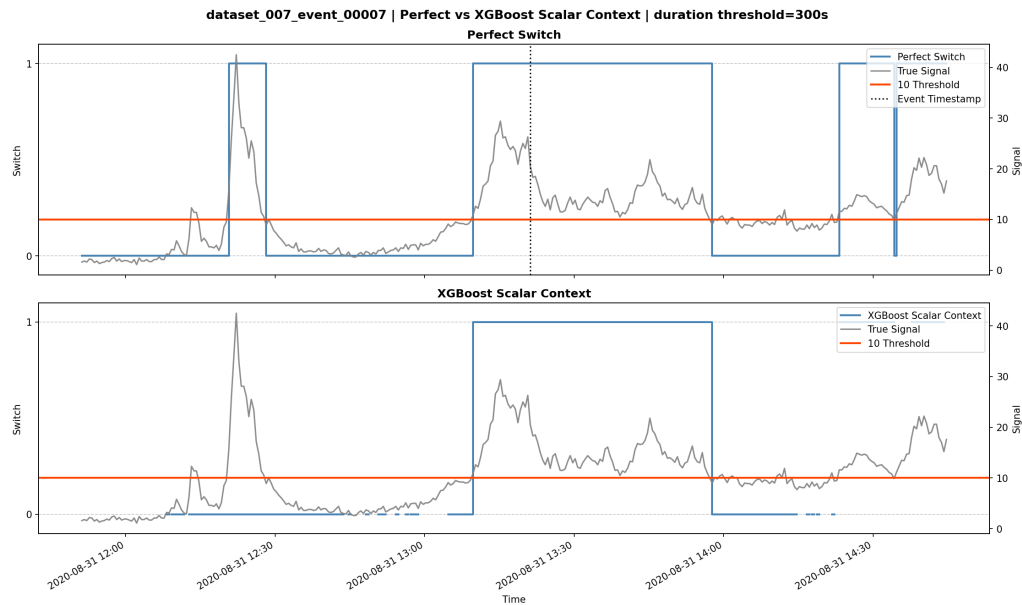


Figure 6.8: Example current-level persistence switch comparison for XGBoost scalar context on the selected multi-interval external-holdout event. The model-derived switch is compared with the Perfect Switch reference after applying the common post-processing rule.

Figure 6.8 shows the current-level XGBoost model on the task-native decision grid for the same source event identifier used in fig. 6.3. Because each task has its own valid decision timestamps, the plotted Perfect Switch can differ slightly between task-native figures. The model decision follows the long central interval well and avoids the short early activation. This is consistent with the global behavior of the task: the model is conservative and does not create false-positive switch samples, but it can ignore short or marginal intervals that the Perfect Switch keeps. The example also clarifies the role of the formulation: the model does not forecast the whole future trajectory; it only estimates whether the current level is likely to persist long enough to justify the switch.

6.4 Long-Fade Detection

The long-fade detection task is a binary classification problem defined inside grouped fade events. At each eligible timestamp, the model estimates whether the grouped event will last at least 300s. The native output is therefore a class probability, not a duration or a future signal trajectory. For switch evaluation, a switch is requested when the predicted long-fade probability is at least 0.5, followed by the same shared switch post-processing used in the other tasks.

6.4.1 Classification Metrics

Table 6.6 reports the native classification metrics. All three models reach very high validation and test AUPRC. This indicates that the long-fade label is easy to separate once the sample is known to belong to a grouped fade event. The result must therefore be interpreted with care: the task is useful as an event-level persistence classifier, but its native metrics are less demanding than the final switch decision because the dataset excludes most non-event background timestamps.

Table 6.6: Long-fade detection classification results on the external holdout split.

Method	Val. AUPRC ↑	Test AUPRC ↑	Test AUROC ↑	Test F1@0.5 ↑
TCN classifier	1.000	1.000	1.000	0.999
XGBoost lag + scalar	1.000	1.000	1.000	1.000
Shapelet convolution classifier	1.000	1.000	0.999	0.997

The TCN and XGBoost classifiers are almost indistinguishable in native classification space. The shapelet convolution model is slightly weaker on the test split,

but it remains close to the other two models. The stronger differences appear only after the probabilities are converted into switch sequences and compared with the Perfect Switch reference.

6.4.2 Switch Performance

Table 6.7 reports the switch metrics obtained after thresholding the predicted probabilities and applying the shared post-processing. All models recover nearly all Perfect Switch samples, but this comes at the cost of many false-positive switch samples. The TCN has the best F1 and IoU, closely followed by XGBoost.

Table 6.7: Long-fade detection switch performance against Perfect Switch on the external holdout split.

Method	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	IoU $\uparrow$	Active min.
TCN classifier	0.540	1.000	0.702	0.540	445.0
XGBoost lag + scalar	0.540	1.000	0.701	0.540	445.5
Shapelet convolution classifier	0.521	0.998	0.685	0.521	460.5

Figure 6.9 shows why the switch scores are lower than the native classification scores. The Perfect Switch contains 481 active samples, corresponding to 240.5 min, and 19 switch islands on the long-fade evaluation grid. The model-derived switches contain almost twice as many active samples and more than twice as many islands. This means that the models correctly identify long fades, but they keep the switch active over a broader portion of the grouped-event windows than the Perfect Switch requires.

For the TCN classifier, the confusion matrix contains no false negatives: all 481 Perfect Switch samples are detected. The cost is 409 false-positive samples, a false-positive rate of 0.196, and 890 active switch samples in total. This explains why the recall is 1.000 but the precision is only 0.540. The result is not a modeling failure in the native classification task; it is a consequence of converting a grouped-event label into an instantaneous switch policy. Once a grouped fade is classified as long, the model tends to remain positive over a wide portion of that grouped event, including regions where the Perfect Switch has already turned off.

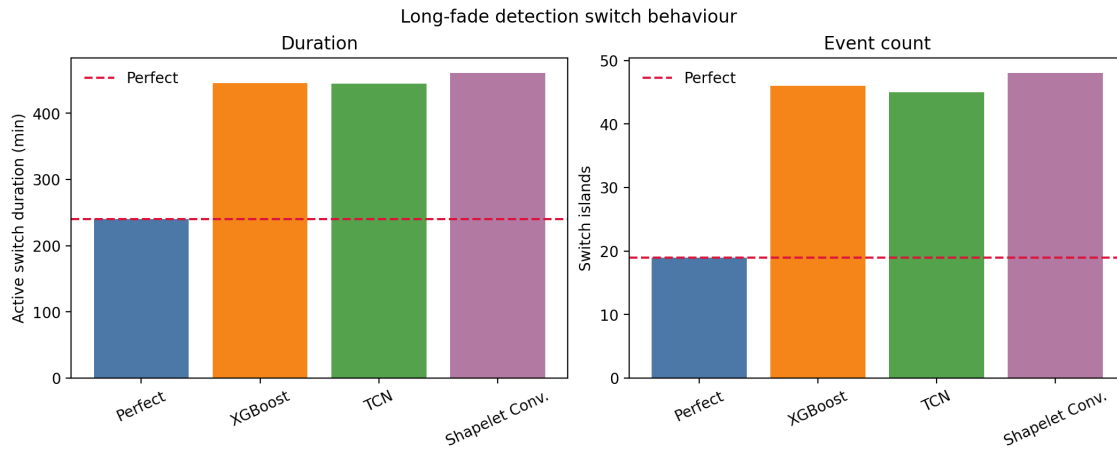


Figure 6.9: Long-fade detection switch duration and event-count behavior on the external holdout split. Duration counts post-processed positive switch samples, converted to minutes using the 30s sampling time. The dashed red line marks the Perfect Switch value.

6.4.3 Interpretation and Event-Level Examples

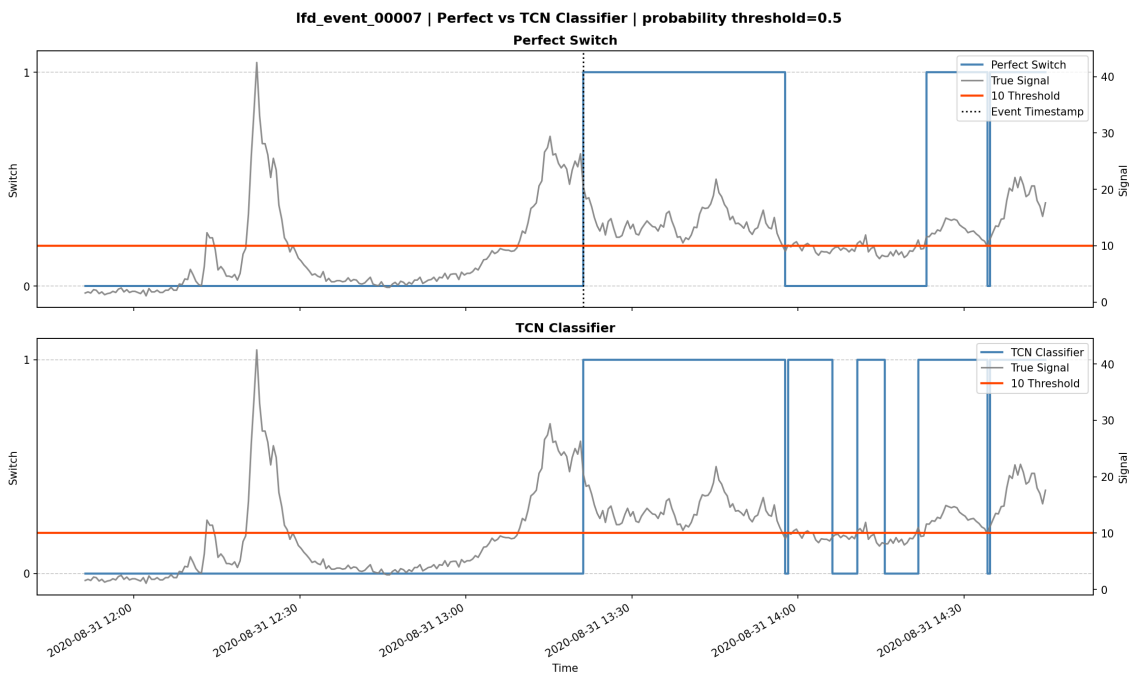


Figure 6.10: Example long-fade detection switch comparison for the TCN classifier on the selected multi-interval external-holdout event. The model-derived switch is compared with the Perfect Switch reference after applying the common post-processing rule.

Figure 6.10 shows a task-native long-fade event for the TCN classifier. The event number is local to the long-fade dataset and should not be interpreted as the same global event number used in the previous task-native figures. The model recognizes the grouped event as a long fade and keeps the switch active over most of the grouped region. This makes the formulation robust to temporary oscillations around the threshold, but it is less precise than the Perfect Switch when the operational reference contains several separate on/off intervals. The example therefore illustrates the main trade-off of long-fade detection: it is a reliable high-recall detector of problematic events, but it is too coarse to be the most selective switch controller. The wider signal window provides context; switch values outside the native event segment are displayed as zero and are not evaluated.

Overall, long-fade detection is highly effective at recognizing whether a grouped fade is long, but it is less precise as a direct switch policy. It therefore behaves as a high-recall formulation: useful when missed long fades are especially costly, but less attractive when unnecessary switch duration is penalized heavily.

6.5 Survival Persistence

The survival persistence task estimates the distribution of the remaining fade duration rather than a single deterministic target. The evaluated models output survival probabilities $S(u \mid X_i)$, where u is a future duration horizon. The operational switch rule uses the 300s horizon: a switch is requested when $S(300\text{s} \mid X_i) \geq 0.65$ and the current signal is at least 10.0 dB. The probability threshold 0.65 is retained as an exploratory operating point from the switch-analysis stage; it should not be interpreted as a fully validation-calibrated threshold.

6.5.1 Survival Metrics

Table 6.8 reports the native survival metrics on the external holdout split. The two models are evaluated on the same censored survival dataset. The XGBoost-AFT model uses continuous lower and upper survival bounds, whereas the discrete-time TCN predicts hazards on a fixed duration grid. For this reason, likelihood values are model-specific diagnostics; the comparison below focuses on Harrell’s C-index, integrated Brier score, and horizon-specific Brier and calibration errors.

The two models show different strengths. The C-index point estimate of XGBoost-AFT exceeds that of the discrete-time TCN by (0.014). To assess the robustness of this difference while preserving dependence among timestamps of the same event, a paired event-level bootstrap resampled the 55 external-holdout events 2,000 times.

Table 6.8: Survival persistence metrics on the external holdout split.

Method	C-index $\uparrow$	IBS $\downarrow$	Brier@300s $\downarrow$	Calib. error@300s $\downarrow$
XGBoost-AFT	0.672	0.117	0.218	0.123
Discrete-time TCN	0.658	0.103	0.214	0.105

The resulting 95% percentile interval for the C-index difference, XGBoost-AFT minus TCN, was $([-0.021, 0.049])$. Because this interval includes zero, the observed gap does not provide evidence of a statistically significant ranking advantage and should be interpreted only as a small numerical difference on this test set. Conversely, the discrete-time TCN has the lower integrated Brier score and lower 300 s calibration error, indicating better probabilistic accuracy over the evaluated time grid. Neither result alone determines switch quality, because the operational decision uses only one point on the survival curve and then applies binary post-processing.

Figure 6.11 illustrates the TCN output using six test windows selected near the 5th, 23rd, 41st, 59th, 77th, and 95th percentiles of the predicted $S(300\text{ s} \mid X_i)$ distribution. This replaces a purely chronological selection, which produced several almost indistinguishable curves from adjacent timestamps. The legend reports, for each window, its selection quantile, predicted survival probability at the operational horizon, predicted median residual duration, and observed residual duration. The selected $S(300\text{ s} \mid X_i)$ values range from (0.39) to (0.81), and therefore show materially different operational predictions around the (0.65) switch threshold. These curves are deterministic predictions for different windows, not repeated estimates of a common population curve; consequently, a pairwise statistical-significance test between them is not defined. Residual overlap instead indicates that the TCN assigns similar hazard profiles to some inputs, especially at long horizons where all predicted survival probabilities approach zero.

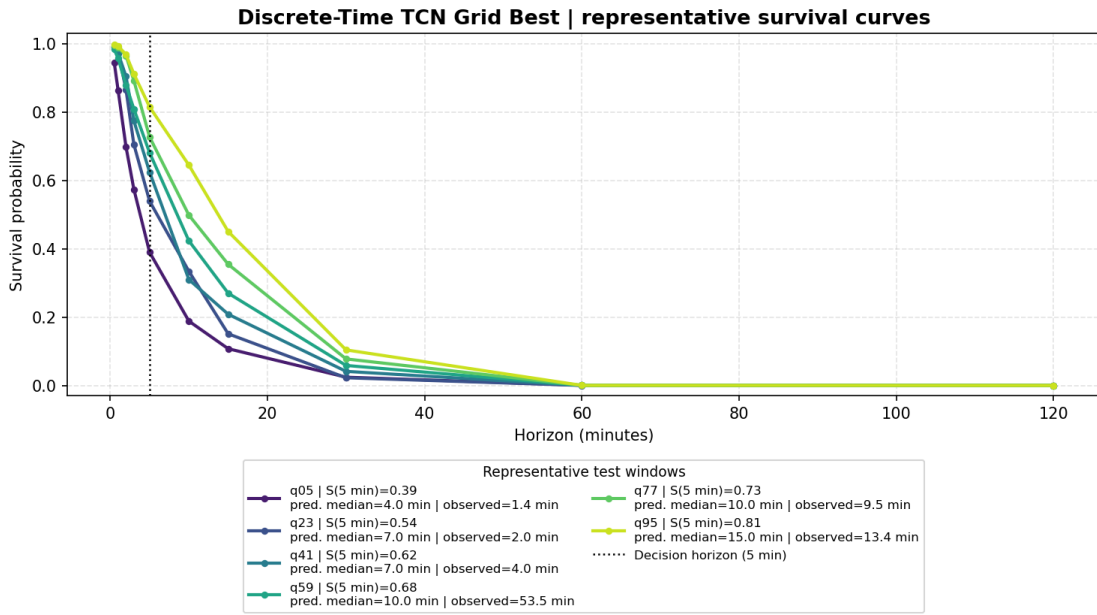


Figure 6.11: Representative survival curves produced by the discrete-time TCN on the external holdout split. Curves are selected by quantiles of $S(300 \text{ s} | X_i)$; colors identify the windows described in the legend, and the dotted line marks the 300 s operational horizon.

6.5.2 Switch Performance

Table 6.9 reports the switch metrics after converting survival probabilities at 300 s into binary decisions. XGBoost-AFT is the stronger switch model: it reaches higher precision, F1, and IoU than the discrete-time TCN. The TCN has higher recall, but it does so by keeping the switch active much more often.

Table 6.9: Survival persistence switch performance against Perfect Switch on the external holdout split.

Method	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	IoU $\uparrow$	Active min.
XGBoost-AFT	0.828	0.902	0.863	0.759	299.0
Discrete-time TCN	0.657	0.971	0.784	0.644	405.5

Figure 6.12 shows the same difference from an operational perspective. The Perfect Switch contains 549 active samples, corresponding to 274.5 min, and 22 switch islands on the survival evaluation grid. XGBoost-AFT is close in duration but produces more switch islands, while the TCN overextends the active duration and still produces more islands than the reference.

The XGBoost-AFT switch has 495 true positives, 103 false positives, and 54

false negatives. Compared with current-level persistence, it is less conservative: it accepts some unnecessary switch time in order to recover more of the Perfect Switch. Compared with long-fade detection, it is much more selective: the active duration is close to the reference, and the F1 score rises to 0.863. This supports the methodological motivation for survival persistence. Estimating $P(R_i > u \mid X_i)$ provides a quantity that is already aligned with the operational question, rather than requiring a trajectory or event-classification output to be indirectly converted into a switch.

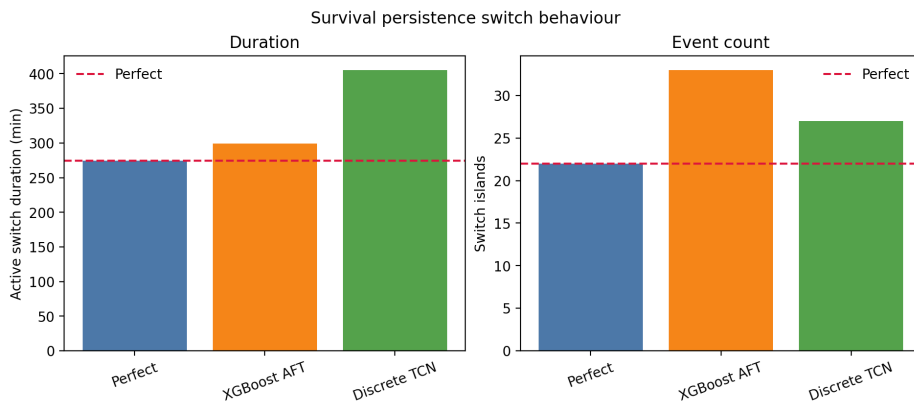


Figure 6.12: Survival persistence switch duration and event-count behavior on the external holdout split. Duration counts post-processed positive switch samples, converted to minutes using the 30s sampling time. The dashed red line marks the Perfect Switch value.

6.5.3 Interpretation and Event-Level Examples

Figure 6.13 shows a false-positive example for XGBoost-AFT on the task-native event `sp_event_00007`. This identifier is local to the survival dataset and does not denote the event used in figs. 6.3 and 6.8. On the 18 native decision timestamps, Perfect Switch is always zero, whereas the final model switch is active for 12 samples. The example therefore illustrates an unnecessary activation, not a successfully recovered reference interval. The wider signal window also contains a later peak outside this native event segment; the displayed switches are padded with zeros there and must not be interpreted as predictions for that peak. This distinction matters when interpreting task-native plots: only the native timestamps contribute to the event metrics, while fig. 6.18 provides the aligned cross-task view.

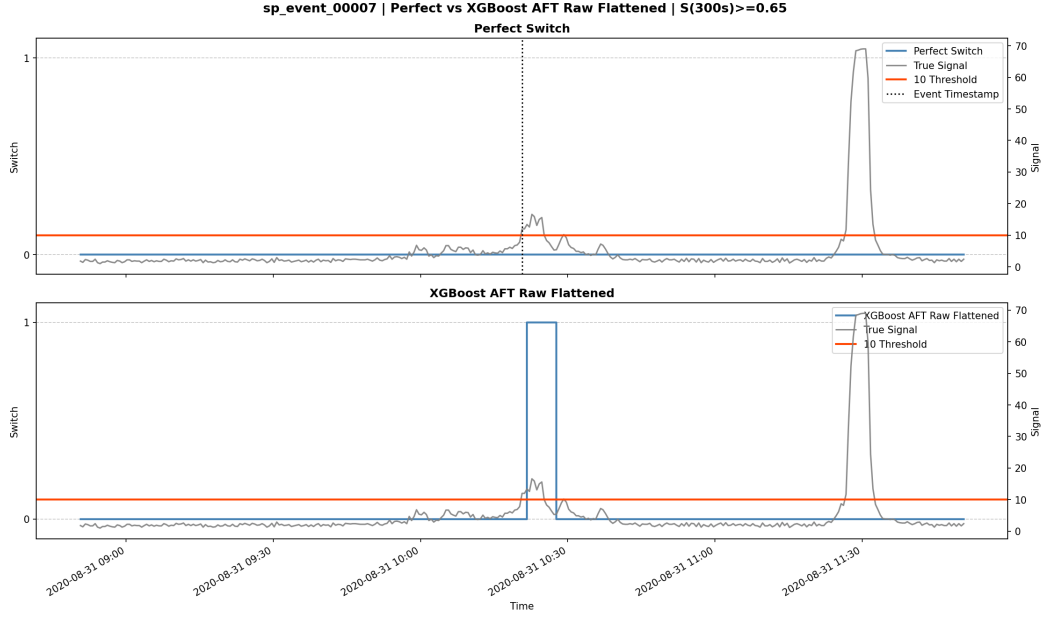


Figure 6.13: False-positive survival switch example for XGBoost-AFT on `sp_event_00007`. The model uses $S(300s | X_i) \geq 0.65$, the current-signal gate, and common post-processing. Perfect Switch remains zero on the native event; switches outside that segment are displayed as zero for context only.

Overall, survival persistence is one of the most operationally meaningful formulations because its output is a residual-duration distribution. In the current experiments, XGBoost-AFT is the best survival switch model, while the TCN provides better integrated probabilistic Brier performance but a less precise binary switch sequence at the selected probability threshold.

6.6 Cross-Task Switch Comparison

The previous sections compare models inside each task. A direct cross-task comparison requires an additional alignment step, because the tasks do not make decisions on exactly the same timestamps. The main comparison uses the common Perfect Switch reference grid and treats missing native decisions as switch 0. This is intentionally conservative: a method is penalized if its task definition does not produce decisions over part of the reference grid. A second strict-intersection comparison was also generated, but it is used only as a diagnostic because it keeps only timestamps where all methods have native decisions.

Table 6.10 reports the best method from each task on the reference-grid comparison. Current-level XGBoost obtains the highest F1 and perfect precision, but it does so with lower recall than the survival and long-fade models. XGBoost-AFT

is the strongest survival method and is close to current-level XGBoost in F1 while reaching higher recall. Among autoregressive models, PatchTST raw gives the best balance. Long-fade detection reaches full recall, but its precision is lower because it keeps the switch active over a larger part of grouped fade regions.

Table 6.10: Best switch method from each task on the common reference grid. Missing native decisions are treated as switch 0.

Task	Best method	Coverage ↑	Precision ↑	Recall ↑	F1 ↑
Current-level persistence	XGBoost scalar context	89.0%	1.000	0.808	0.894
Survival persistence	XGBoost-AFT	9.2%	0.812	0.938	0.870
Autoregressive forecasting	PatchTST raw	89.2%	0.779	0.821	0.800
Long-fade detection	TCN classifier	28.7%	0.557	1.000	0.715

Figures 6.14 and 6.15 show the same comparison across all evaluated methods: F1 and precision in the first figure, recall and balanced accuracy in the second. Both use the same method order, ranked by F1. All values refer to the final switches after the common island-removal and holding rules; raw model decisions are retained only as diagnostics.

The F1 and precision panels highlight current-level XGBoost as the most conservative useful method, with no false-positive switch samples on the reference grid. PatchTST raw provides the strongest autoregressive balance. However, precision alone does not show how much reference switch time is missed, so these results must be read alongside recall.

The recall and balanced-accuracy panels show the other side of this trade-off. Survival XGBoost-AFT recovers more reference switch time while keeping precision relatively high. Long-fade detection models reach full or nearly full recall but introduce extra switch duration, explaining their lower precision in fig. 6.14. The best model therefore depends on the relative cost of missed coverage and unnecessary activation.

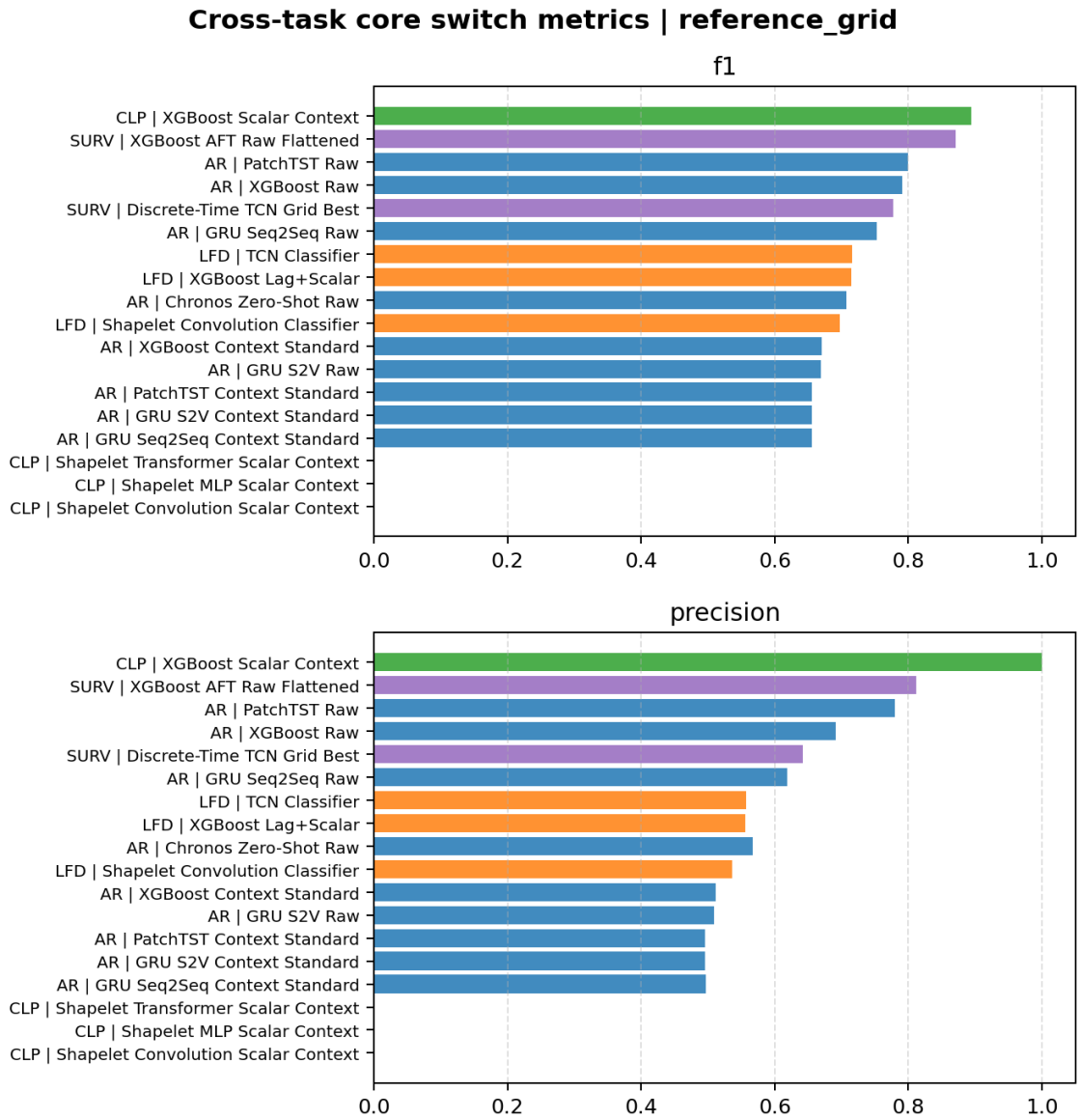


Figure 6.14: Cross-task F1 (top) and precision (bottom) on the common reference grid. Missing native decisions are mapped to switch 0, so the comparison includes both decision quality and task coverage.

Figure 6.16 shows why native decision coverage is a critical part of the interpretation. Autoregressive and current-level models operate on most of the reference grid because their datasets are based on candidate event windows. Long-fade detection is defined only inside grouped fade events, and survival persistence is defined only while the sample is inside a detected survival fade event.

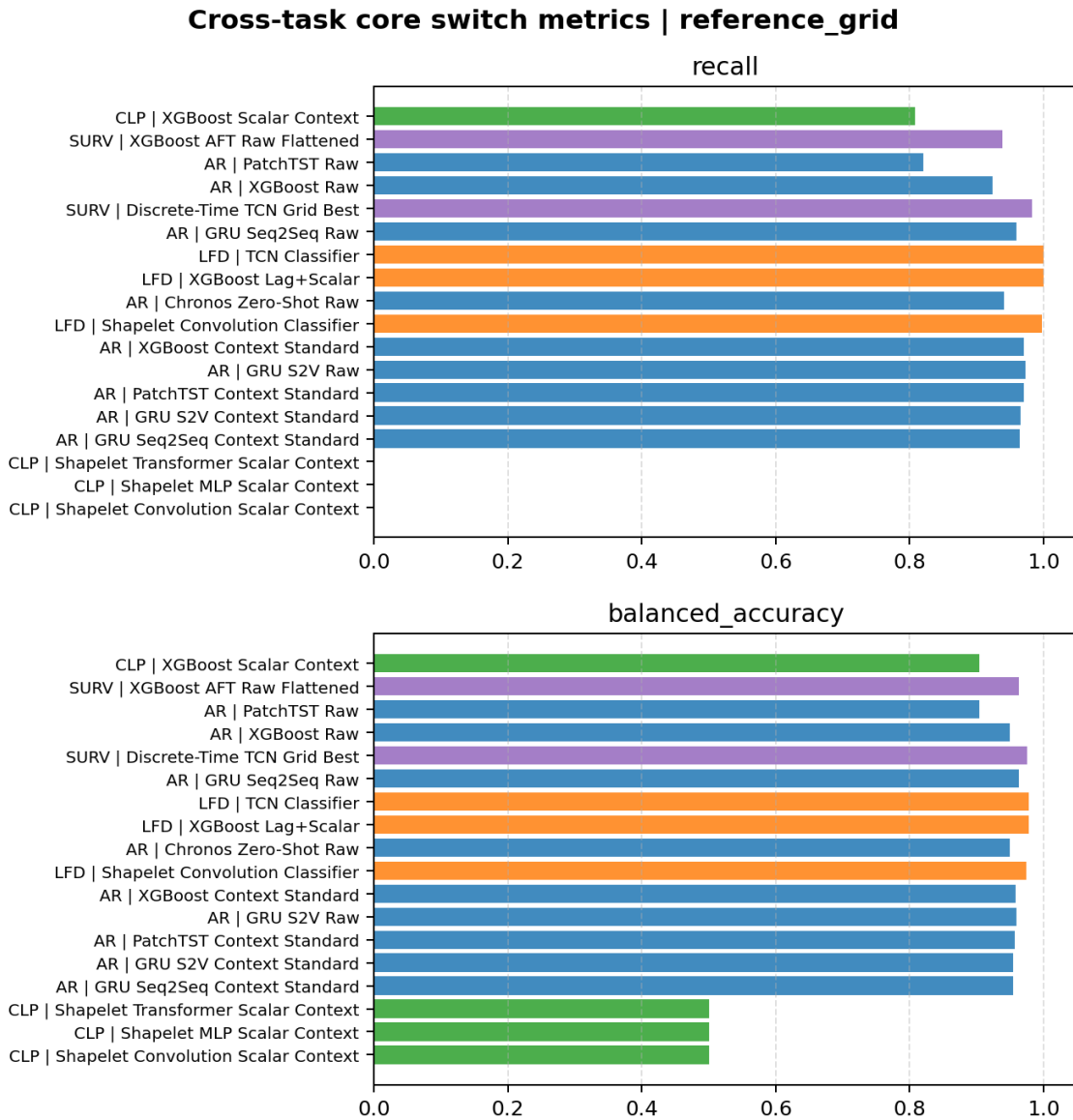


Figure 6.15: Cross-task recall (top) and balanced accuracy (bottom) on the common reference grid, with the same method order and missing-decision policy as fig. 6.14.

Consequently, survival metrics on the reference grid evaluate a smaller native decision domain, even though the predictions are meaningful inside that domain. The strict-intersection diagnostic removes this coverage effect by retaining only the 771 timestamps where all task formulations provide native decisions. On this reduced grid, current-level XGBoost remains the best conservative method with precision 1.000, recall 0.819, and F1 0.900. XGBoost-AFT obtains precision 0.814, recall 0.938, and F1 0.872, and XGBoost raw is the strongest autoregressive method with F1 0.866.

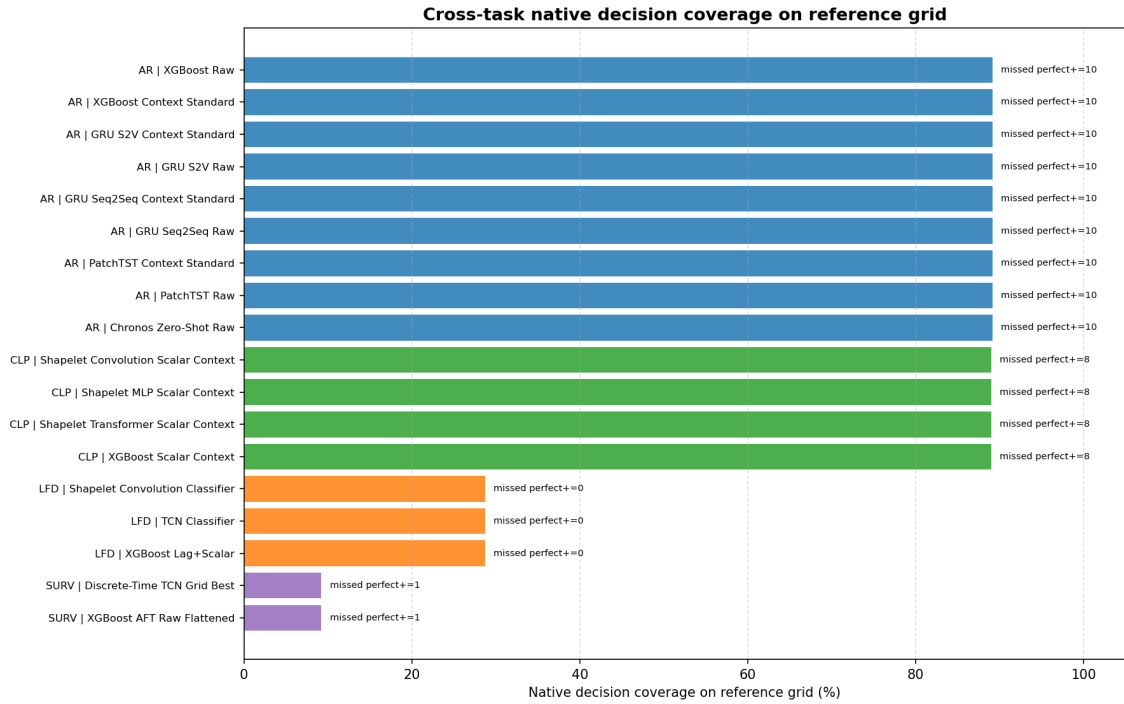


Figure 6.16: Native decision coverage for each method on the common Perfect Switch reference grid. Lower coverage means that the corresponding task produces decisions on a smaller subset of reference timestamps.

The long-fade TCN reaches recall 1.000 but only F1 0.778. Therefore, the main ranking is not only an artifact of missing decisions on the reference grid: the same qualitative pattern remains when the comparison is restricted to the shared native decision domain.

Figure 6.17 reports event-level F1 values for the selected reference events. The heatmap shows that no formulation dominates uniformly. Some events are handled well by most models, while others expose clear differences in onset timing, release timing, or excessive switch activation. This supports the use of both global and event-level metrics: global F1 summarizes average behavior, whereas event-level views reveal which fade morphologies remain difficult.

The three all-zero rows corresponding to the current-level shapelet models are genuine results rather than missing values or a coverage artifact. As discussed in Section 6.3.2, none of their test predictions satisfies both the 10.0 dB signal gate and the 300 s predicted persistence requirement. Their final switch sequence is therefore zero over the complete test set. Every selected event containing a positive Perfect Switch interval consequently has no true-positive shapelet decision and an event-level F1 of zero. The heatmap makes explicit that this failure is systematic across events, not confined to a few difficult fade morphologies.

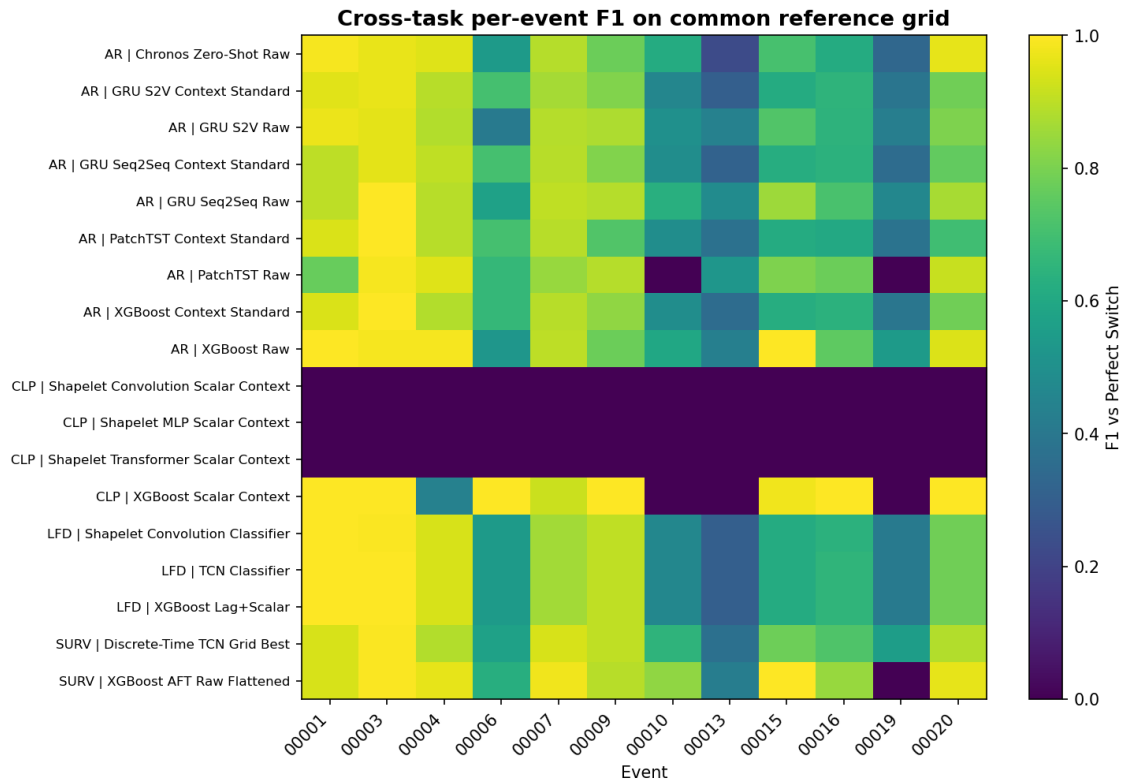


Figure 6.17: Event-level F1 heatmap for the cross-task reference-grid comparison. Rows are model-derived switch methods and columns are selected external-holdout events.

Figure 6.18 gives a concrete example on a globally aligned reference event, `dataset_007_event_00007`, also used in figs. 6.3 and 6.8. This is the figure to use when comparing methods from different task formulations on exactly the same timestamps. The all-method timeline makes the precision-recall trade-off visible: conservative models activate later or for shorter intervals, while high-recall models cover more of the fade but may extend beyond the Perfect Switch interval.

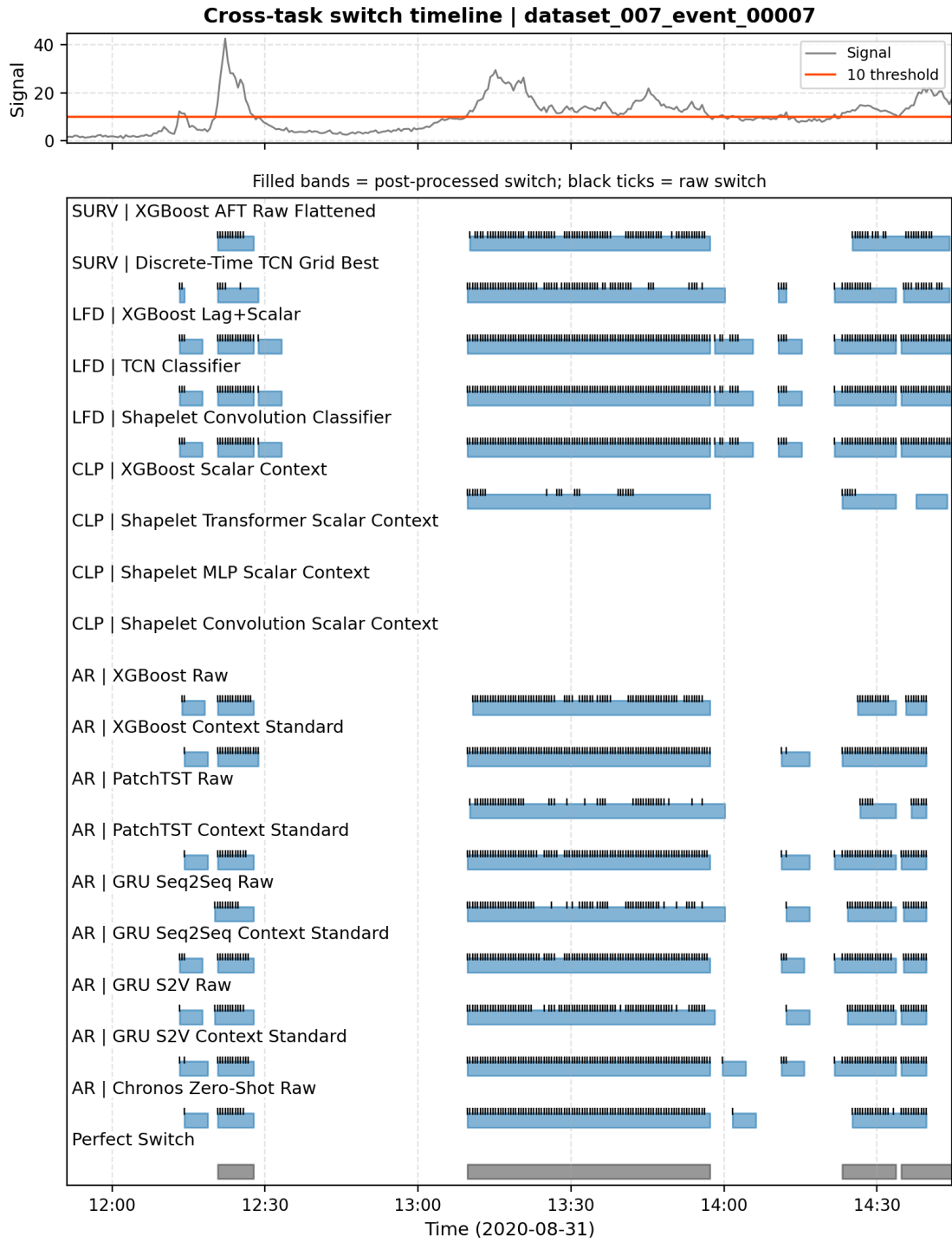


Figure 6.18: Cross-task event timeline for the selected multi-interval external-holdout event. The upper panel shows the raw signal and threshold. Each name labels the switch band immediately below it; filled bands show final post-processed decisions and black ticks show raw decisions.

6.7 Summary of Main Empirical Findings

The experimental results lead to four main findings.

First, native predictive accuracy and switch quality are related but not equivalent. In the autoregressive task, the raw GRU models obtain strong forecast losses, but PatchTST raw gives the best switch F1 because its errors produce a more useful threshold-crossing pattern. In survival persistence, the discrete-time TCN has the best integrated Brier score, but XGBoost-AFT gives the better switch sequence. In long-fade detection, near-perfect AUPRC does not translate into a precise switch controller because the grouped-event label overextends the active decision. These cases confirm that final evaluation must be performed after switch conversion, not only in the native predictive space.

Second, tabular gradient-boosted models are strong baselines. XGBoost is competitive in autoregressive forecasting, is clearly the best current-level persistence model, and XGBoost-AFT is the best survival switch model. This is important methodologically: more complex neural sequence models are useful, but they do not automatically dominate well-tuned lag and scalar representations on this dataset.

Third, the different task formulations express different operating policies. Current-level XGBoost is conservative and precise; survival XGBoost-AFT is a strong residual-duration formulation with higher recall; PatchTST raw provides the best autoregressive switch balance; long-fade detection is a robust high-recall classifier but tends to overextend switch activity. The learnable-shapelet current-level models are the clearest negative result: after the operational conversion they never request a switch. These behaviors are more informative than a single global ranking.

Fourth, the shared post-processing is essential for a fair operational comparison. All final switch sequences are evaluated after the same island and holding rules, and all are compared with the same Perfect Switch construction. The remaining differences therefore come from the task formulation and model outputs, not from task-specific switch logic. This also means that the reported switch metrics should be read as operational metrics of the post-processed decision sequence, not as metrics of the raw model threshold crossings.

Overall, the most promising formulations are current-level persistence with XGBoost and survival persistence with XGBoost-AFT, followed by autoregressive PatchTST raw. Current-level persistence gives the cleanest switch sequence in the reported reference-grid comparison, but its underlying duration regression has large tail errors. Survival persistence is conceptually closer to the operational question because it estimates residual fade duration directly. PatchTST raw remains valuable because it predicts the future signal trajectory and can support analyses beyond the

binary switch decision. Long-fade detection is useful as a high-recall event classifier, but less precise as a standalone switch controller.

Chapter 7

Conclusions

Satellite fade nowcasting is useful when it supports timely switching, not merely accurate intermediate predictions. The conclusions below identify what the experiments establish about that objective and what remains unresolved.

7.1 Contribution and Operational Lessons

The contribution of this thesis is a decision-level comparison of four fade nowcasting formulations, rather than a new model architecture. By mapping different predictive outputs to a common switch reference and making decision coverage explicit, the study distinguishes accurate intermediate predictions from useful operational decisions. Its central empirical result is that the best predictor according to a task’s native loss is not necessarily the best switch controller.

On the external holdout, current-level XGBoost, survival XGBoost-AFT, and raw PatchTST produced useful but different switching policies: conservative activation, higher fade coverage, and the strongest autoregressive switch balance, respectively. These results support the use of signal-only models for switching, but do not establish a universal winner. In particular, the retained survival operating point was chosen exploratorily on test data and still requires validation-only calibration.

The negative results are equally informative. The current-level shapelet models never activated the switch on the evaluated test grid, while the long-fade classifiers’ strong classification scores did not prevent excessive switch activity. Neither architectural complexity nor high native accuracy was sufficient. The practical lesson is to select the predictive target and assess its decision rule together: an output that is poorly aligned with the required switch behavior cannot be judged useful from its native score alone.

7.2 Answers to the Research Questions

Can learned models improve switch decisions? The experiments support their usefulness: signal-history models can recover reference switch intervals while limiting unnecessary activation. They do not, however, establish superiority over a deployed engineering controller. Well-tuned tabular models remained competitive with neural sequence models, so greater architectural complexity is not itself evidence of operational improvement.

Which formulation is most useful? The choice depends on the cost of missed fade coverage relative to unnecessary switch time. Residual-duration survival modeling directly addresses whether an ongoing fade will persist; current-level regression can support a conservative policy but estimates a different quantity. Forecasting remains useful when the future trajectory is also required. Grouped-event classification is better suited to a high-recall warning than precise switch timing. These differences make target definition an operational choice, not merely a modeling detail.

Do native metrics explain operational usefulness? Not sufficiently. The survival TCN obtained the better integrated Brier score, whereas XGBoost-AFT produced the better switch sequence at the evaluated operating point. Similarly, low forecast error did not identify the strongest autoregressive switch model. Native scores remain necessary to assess the predicted quantity, but decision timing, active duration, and fragmentation must be assessed after conversion and post-processing.

Why are a common reference and post-processing necessary? They prevent differences in reference construction or decision smoothing from being mistaken for model improvements. A common timestamp grid is also needed: task-native scores alone compare decisions over different domains. Reporting coverage alongside switch quality makes this distinction explicit and shows where a method would still require a fallback decision in operation.

7.3 Limitations

The main limitation is the size and heterogeneity of the available data. The external holdout design is stricter than a random split and is appropriate for testing generalization, but it also means that the final conclusions are based on one held-out acquisition source. Additional acquisition periods, terminals, satellite links, and weather conditions would be needed to establish broader operational robustness.

The thesis deliberately used a signal-only policy. This made the comparison consistent across files and avoided dependence on columns that were not available in all datasets. At the same time, it excluded potentially relevant physical covariates, such as weather measurements, link geometry, or system state variables. A model that combines signal history with consistently available physical information could improve anticipation of fades, especially before the signal crosses the operational threshold.

Another limitation concerns threshold and calibration choices. The switch reference was built around a 10.0 dB threshold and a 300 s operational duration. These values were fixed to make the experiments comparable, but real deployments may prefer different operating points. The survival probability threshold used for the retained switch analysis was exploratory and not a fully validation-calibrated operating point. Future work should calibrate all decision thresholds on validation data only and then evaluate the chosen operating point on test data.

Finally, the cross-task comparison required aligning task-native decisions to a common reference grid. This is necessary for fairness, but it also exposes a structural issue: different task definitions do not naturally produce decisions at exactly the same timestamps. The reported coverage values should therefore be interpreted as part of the result, not as a nuisance. A task that is accurate only on a small native domain may still be useful, but it cannot replace a controller that must decide at every operational timestamp.

7.4 Future Work

The first extension should be validation-only calibration of switch-conversion rules. Autoregressive trajectory thresholds, current-level duration thresholds, long-fade probability thresholds, and survival probability thresholds should be tuned on validation data using explicitly chosen operational costs. This would turn the current comparison from a fixed-rule benchmark into a calibrated decision study.

Second, the survival formulation should be developed further. XGBoost-AFT showed that residual-duration modeling is promising, but richer survival models could improve calibration and event-level timing. Candidate directions include better discrete-time neural survival models, DeepHit-style competing risk formulations if multiple event types are introduced, and validation of survival probability calibration across operational horizons.

Third, the dataset should be expanded and enriched. More external holdout sources would allow blocked or source-level cross-validation, reducing the risk that

the conclusions depend on one held-out file. If auxiliary variables become available consistently, future work could compare the current signal-only framework with a physically enriched framework while preserving the same no-leakage and switch-evaluation protocol.

Fourth, future experiments should connect the switch metrics to explicit operational costs. Precision, recall, F1, IoU, active duration, and event counts describe different aspects of switch behavior, but an operator may assign asymmetric costs to false activations, missed fades, delayed onsets, and excessive switch fragmentation. A cost-based decision layer would make the choice between conservative and high-recall models more transparent.

Fifth, the framework should be moved from offline evaluation toward an online nowcasting prototype. The current experiments reproduce the information available at each decision time, but they are still executed as batch analyses. A deployable system would need streaming ingestion, stateful preprocessing, incremental gap and quality checks, stateful switch post-processing, latency measurement, and clear fallback behavior when input quality is poor. Such a prototype would also make it possible to compare the learned decision rules against engineering baselines under realistic operating constraints, rather than only after the acquisition has been processed offline.

The broader conclusion is that satellite fade nowcasting should be treated as a decision-oriented time-series problem. Accurate forecasts, duration estimates, and survival curves are valuable only insofar as they support timely and reliable switch decisions. The framework developed in this thesis provides a reproducible basis for that evaluation and shows that task formulation, post-processing, and operational metrics are central components of the modeling problem, not secondary details.

Appendix A

Implementation Details

This appendix summarizes implementation details that are useful for reproducing the experiments but would interrupt the main methodological discussion. The repository was organized so that each task has its own configuration files, dataset builder, model runners, saved model artifacts, and result folders, while sharing common preprocessing, switch construction, and evaluation utilities.

A.1 Repository Structure

The codebase separates common infrastructure from task-specific logic. Common utilities live under `src/data/`, `src/switching/`, `src/utils/`, and `src/tuning/`. Task-specific implementations live under `src/tasks/<task_name>/`. Experiment entry points are grouped in `scripts/experiments/<task_name>/`, while analysis and comparison scripts are grouped in `scripts/analysis/`.

Table A.1 gives the practical role of the main repository folders. The table is intended as a reproducibility map: it shows where configuration, immutable raw data, intermediate preprocessing outputs, final model-ready datasets, fitted models, and result artifacts are stored.

The main task namespaces are `autoregressive`, `current_level_persistence`, `long_fade_detection`, and `survival_persistence`. This separation avoids storing task-specific models or outputs under a generic folder and makes it possible to trace every result back to its task formulation.

A.2 Dataset Artifact Schema

Every processed dataset is stored together with metadata describing the selection identifier, split definition, context length, threshold assumptions, and preprocessing

Table A.1: Main repository folders used by the experimental pipeline.

Folder	Purpose
<code>configs/</code>	YAML configuration files for data preparation, model runs, grid searches, Perfect Switch construction, and switch comparisons.
<code>data/raw/</code>	Original input files. These files are treated as read-only.
<code>data/interim/</code>	Cleaned signals, candidate events, event windows, and intermediate reusable preprocessing artifacts.
<code>data/processed/</code>	Final task-specific arrays and metadata used by model runners.
<code>src/tasks/</code>	Task-scoped dataset, model, evaluation, plotting, and utility code.
<code>src/switching/</code>	Model-agnostic Perfect Switch, switch conversion, post-processing, metrics, and plotting utilities.
<code>scripts/experiments/</code>	Executable training, zero-shot evaluation, dataset-building, and grid-search scripts.
<code>scripts/analysis/</code>	Data-analysis and post-training comparison scripts.
<code>models/</code>	Saved checkpoints, fitted estimators, model configurations, and run metadata.
<code>results/</code>	Predictions, metrics, tables, figures, comparison summaries, and result indexes.

fingerprint. The canonical split is development versus external holdout: training and validation are drawn from development sources, whereas `fc-uplink-fade.csv` is used only as the final test source.

Table A.2 summarizes the arrays and metadata saved by each task-specific dataset builder. The table separates numerical arrays from metadata because the metadata is required to align predictions back to timestamps, events, source datasets, and switch-evaluation windows.

The split metadata is as important as the numerical arrays. It preserves the alignment between model inputs, event timestamps, source files, event identifiers, and later switch comparisons. Model scripts do not reconstruct splits independently; they load these artifacts and validate their fingerprints where necessary.

A.3 Run and Result Artifact Schema

Each trained or evaluated model writes a run folder under `results/runs/<task_name>/` and, when applicable, a model folder under `models/<task_name>/`. Run identifiers encode the task, model family, variant, selection identifier, and relevant setup. A typical run folder contains:

Table A.2: Main processed dataset artifacts by task.

Task	Array artifacts	Metadata artifacts
Autoregressive forecasting	<code>train.npz</code> , <code>val.npz</code> , and <code>test.npz</code> with raw and context-standardized past windows and future trajectories.	Split metadata parquet files with window identifiers, event identifiers, source dataset names, timestamps, quality flags, and scaling parameters.
Current-level persistence	Split NPZ files with raw context, relative-to-current context, scalar context features, and persistence duration targets.	Metadata with target duration, censoring flags, acquisition segment identifiers, scalar feature names, and train-only scaler parameters.
Long-fade detection	Split NPZ files with sequence context, relative-to-threshold context, scalar context, and binary long-fade labels.	Metadata with grouped-event identifiers, event labels, timestamps, dataset identifiers, and quality flags.
Survival persistence	Split NPZ files with past contexts, scalar features, residual-duration labels, and AFT lower/upper bounds.	Metadata with event identifiers, censoring indicators, acquisition segment identifiers, observed durations, and survival target bounds.

- `predictions/`: validation and/or test predictions in parquet format;
- `metrics/`: native task metrics such as forecast errors, classification scores, duration errors, or survival metrics;
- `figures/`: diagnostic plots generated by the run;
- `tables/`: compact run summaries;
- `metadata.yaml`: configuration, dataset fingerprint, selected device, seed, and file paths.

Switch comparisons are stored separately from native model runs because the same predictions can be converted into switch sequences under different decision rules. Model-vs-reference comparisons are written under `results/comparisons/switch_eval/<task_name>/<selection_id>/`. Cross-task comparisons are written under `results/comparisons/cross_task_switch/<selection_id>/`. This separation keeps model training artifacts distinct from downstream operational evaluation.

A.4 Reproducibility Checklist

The experiments are intended to be reproducible from configurations and saved artifacts. The following information is recorded or fixed for each run:

- the YAML configuration path and configuration fingerprint;
- the processed dataset path and dataset metadata fingerprint;
- the task formulation, input representation, target definition, context length, prediction horizon or duration horizon, and threshold assumptions;
- train, validation, and external test split metadata;
- random seed and hardware device used by the model runner;
- model hyperparameters, selected validation score, and final test metrics;
- saved predictions used for switch conversion;
- switch-conversion rule, post-processing settings, and Perfect Switch reference path.

The Python environment is specified by `environment.yml`. Experiments use the active environment rather than creating a new virtual environment inside the repository. GPU acceleration is used when available, with CUDA preferred, then Apple MPS when supported, and CPU as a fallback. Grid searches distribute independent trials over available CUDA devices when possible.

Appendix B

Hyperparameter Grids

This appendix reports the configurations selected by the validation-only model-selection searches summarized in section 5.3. The tables list the archived selection score used by each search. Final external test metrics and switch metrics are reported in chapter 6; they should be considered the primary performance results.

For current-level persistence, some duration metrics were refreshed without retraining after dataset-alignment updates. The configurations below are the selected model artifacts; the refreshed validation and test metrics used for the final discussion are those in chapter 6.

B.1 Search Design Summary

Table B.1 summarizes the model-selection searches used in the thesis. All searches use validation-only criteria. The external holdout split is not used to choose hyperparameters, thresholds, or checkpoints. The purpose of this table is to make the scale of the model-selection phase explicit before listing the selected configurations. It reports the criterion used to rank trials, the number of evaluated trials, and the main hyperparameter families varied by each search.

For tree-based grids, the configuration files use explicit numeric ranges defined by start, stop, and step values. For neural grids, categorical lists are used for architecture choices such as hidden dimensions, layer counts, dropout values, and learning rates. The resulting trial tables are archived under `results/grid_searches/`; the tables below report only the selected configuration for each model family or variant.

Table B.1: Summary of the hyperparameter searches. Trial counts refer to the archived searches used for the reported model artifacts.

Task	Search	Selection criterion	Trials	Main varied hyperparameters
Autoregressive	GRU S2V and encoder-decoder, raw and context-standard variants	validation RMSE, minimized	576	hidden size, number of layers, learning rate, weight decay, teacher forcing, architecture.
Autoregressive	PatchTST, raw and context-standard variants	validation RMSE, minimized	1728	patch length, stride, model dimension, attention heads, number of layers, dropout, learning rate.
Autoregressive	XGBoost multi-output horizon regressors	validation RMSE, minimized	2304	number of trees, depth, learning rate, subsampling, column sampling, minimum child weight, L_2 regularization.
Current-level persistence	Learnable-shapelet MLP, Transformer, and convolution heads	validation MAE in seconds, minimized	648	shapelet count, hidden dimensions, scalar encoder size, depth, dropout, learning rate.
Current-level persistence	XGBoost duration regressor	validation MAE in seconds, minimized	1152	number of trees, depth, learning rate, subsampling, column sampling, minimum child weight, L_2 regularization.
Long-fade detection	XGBoost, TCN, and shapelet-convolution classifiers	validation AUPRC, maximized	8928	tree parameters, TCN channels/depth/kernel/dropout, shapelet count/lengths, scalar hidden dimension, learning rate, batch size.
Survival persistence	XGBoost-AFT	validation AFT negative log-likelihood, minimized	1920	input representation, sample weighting, AFT distribution, AFT scale, tree depth, learning rate, subsampling, column sampling, regularization.
Survival persistence	Discrete-time TCN	validation integrated Brier score, minimized	1620	binning strategy, input representation, sample weighting, channel schedule, kernel size, dropout, learning rate, weight decay, batch size.

B.2 Autoregressive Forecasting

The autoregressive searches are run separately for the raw and context-standardized variants. Chronos is excluded from this table because it is evaluated in zero-shot mode and no hyperparameters are fitted. Table B.2 reports the best validation-selected configuration for each trainable autoregressive model and input variant. The selection score is the validation RMSE on the raw signal scale; the external test metrics are discussed in the results chapter rather than repeated here.

Table B.2: Best validation-selected autoregressive configurations.

Model	Variant	Selected configuration	Validation RMSE
GRU S2V	raw	hidden size 16; 1 layer; learning rate 0.001; weight decay 0; teacher forcing 0	5.520
GRU S2V	context-standard	hidden size 64; 1 layer; learning rate 0.001; weight decay 0.0001; teacher forcing 0	3.716
GRU encoder-decoder	raw	hidden size 64; 1 layer; learning rate 0.001; weight decay 0.0001; teacher forcing 0.5	4.023
GRU encoder-decoder	context-standard	hidden size 64; 3 layers; learning rate 0.0003; weight decay 0; teacher forcing 0.25	3.691
PatchTST	raw	patch length 10; stride 2; model dimension 128; 4 heads; 3 layers; dropout 0.1; learning rate 0.001	5.526
PatchTST	context-standard	patch length 5; stride 5; model dimension 32; 2 heads; 1 layer; dropout 0.1; learning rate 0.001	3.815
XGBoost	raw	300 trees; max depth 2; learning rate 0.03; subsample 1.0; column sample 0.8; minimum child weight 1; $L_2 = 2.0$	6.381
XGBoost	context-standard	300 trees; max depth 4; learning rate 0.03; subsample 1.0; column sample 0.8; minimum child weight 3; $L_2 = 2.0$	3.912

B.3 Current-Level Persistence

Table B.3 reports the selected duration-regression configuration for each current-level persistence model. The target for these models is the remaining time for which the signal stays above the current-level threshold $S_t - \delta$. The table lists validation MAE in seconds because this was the selection metric used by the duration-regression searches.

Table B.3: Best validation-selected current-level persistence configurations.

Model	Selected configuration	Validation MAE [s]
Shapelet MLP	hidden dimension 64; 2 hidden layers; dropout 0.1; 64 shapelets per length; scalar encoder dimension 32; learning rate 0.001	5590.139
Shapelet Transformer	model dimension 32; 1 Transformer layer; dropout 0; 32 shapelets per length; scalar encoder dimension 16; learning rate 0.001	5653.660
Shapelet Convolution	convolution channels 64; 2 convolution layers; dropout 0; 16 shapelets per length; scalar encoder dimension 32; learning rate 0.001	5408.826
XGBoost duration regressor	1200 trees; max depth 6; learning rate 0.11; subsample 0.8; column sample 0.8; minimum child weight 3; $L_2 = 1.0$	5465.665

B.4 Long-Fade Detection

The long-fade detection search is a classification search. It compares one tabular gradient-boosted model and two neural sequence models on the same event-level binary target. The selection criterion is validation AUPRC because the task is imbalanced and the positive class corresponds to operationally important long fade events. The table reports the selected configuration for each classifier family. Table B.4 therefore should be read as a summary of the best event-level classifiers, not as a switch-performance table. Switch quality is computed only after converting classifier probabilities into post-processed switch sequences.

Table B.4: Best validation-selected long-fade detection configurations.

Model	Selected configuration	Validation AUPRC
XGBoost classifier	200 trees; max depth 6; learning rate 0.1; subsample 0.8; column sample 0.9; $L_2 = 0.1$	0.999816
TCN classifier	128 hidden channels; 5 temporal blocks; kernel size 3; dropout 0; scalar hidden dimension 64; learning rate 0.0003; batch size 128	0.999972
Shapelet convolution classifier	shapelet lengths 10, 15, and 20; 48 shapelets per length; 32 convolution channels; 1 convolution layer; dropout 0; scalar hidden dimension 32; learning rate 0.003	0.999949

B.5 Survival Persistence

The survival-persistence searches are reported separately from the classification and regression tasks because their validation criteria are survival-specific. XGBoost-AFT is selected by validation AFT negative log-likelihood on continuous-time censored targets. The discrete-time TCN is selected by validation integrated Brier score after converting the same continuous-time labels into discrete survival bins during training. Table B.5 lists the selected survival models and their task-native validation criterion. The two scores are not directly comparable because XGBoost-AFT and the discrete-time TCN optimize different survival objectives.

Table B.5: Best validation-selected survival persistence configurations.

Model	Selected configuration	Selection metric	Score
XGBoost-AFT	raw flattened context; uniform sample weights; logistic AFT loss with scale 1.0; 1200 boosting rounds; max depth 3; learning rate 0.1; $L_2 = 5.0$	validation AFT negative log-likelihood	1.610
Discrete-time TCN	relative-to-current sequence with scalar context; hybrid survival bins; channels 32, 64, and 64; kernel size 3; dilations 1, 2, 4, 8, and 16; dropout 0.25; learning rate 0.0015	validation Integrated Brier Score	0.104

Appendix C

Additional Results

The main text reports compact tables and representative figures. The complete experiment outputs are stored as repository artifacts so that individual runs, event-level metrics, and plots can be inspected without expanding the thesis with very large tables. This appendix documents where those additional results are located and how they relate to the summaries in chapter 6.

C.1 Native Prediction Metrics

Native prediction metrics are stored in the corresponding run folders. These metrics should be interpreted as diagnostics for the underlying modeling task, not as replacements for the switch metrics. The main locations are summarized in table C.1. Table C.1 is included to make the archived outputs traceable: for each task it identifies the native metric family and the repository locations where the full run-level tables can be inspected.

For autoregressive models, the saved prediction files contain one row per forecast window and one column per forecast horizon. This makes it possible to recompute horizon-wise errors or alternative trajectory-to-switch rules without retraining the model. For duration and survival models, the prediction files contain the scalar duration estimate, probability estimate, or survival curve needed by the downstream switch-conversion scripts.

C.2 Switch Metrics and Event-Level Tables

Switch evaluation artifacts are separated from native model metrics. Each switch-comparison folder contains:

Table C.1: Repository locations for complete native prediction metrics.

Task	Main metric type	Artifact location
Autoregressive forecasting	MAE, RMSE, horizon-wise forecast errors, validation/test summaries.	<code>results/runs/autoregressive/</code> and <code>results/comparisons/model_selection/autoregressive/</code> .
Current-level persistence	Duration-regression MAE, RMSE, median absolute error, and model-selection summaries.	<code>results/runs/current_level_persistence/</code> and <code>results/comparisons/model_selection/current_level_persistence/</code> .
Long-fade detection	AUPRC, AUROC, F1 at the configured probability threshold, and classifier summaries.	<code>results/runs/long_fade_detection/</code> and <code>results/comparisons/model_selection/long_fade_detection/</code> .
Survival persistence	AFT negative log-likelihood, C-index, integrated Brier score, horizon-specific Brier scores, and calibration diagnostics.	<code>results/runs/survival_persistence/</code> and <code>results/comparisons/model_selection/survival_persistence/</code> .

- `switch_metrics_summary.csv`, with global precision, recall, F1, IoU, balanced accuracy, and related binary metrics;
- `switch_metrics_by_event.csv`, with event-level binary metrics;
- `global_switch_metrics.csv`, with active duration, number of switch islands, and duration agreement diagnostics;
- `event_switch_metrics.csv`, with the same behavior metrics computed per event;
- `predictions/model_vs_perfect_switch.parquet`, with the aligned model and Perfect Switch time series used by the metrics;
- `figures/event_switch_plots/`, with event-level timelines.

Table C.2 summarizes the folder layout for these switch artifacts. The distinction between task-native comparisons, task-level summaries, and cross-task comparisons is important because they use different timestamp grids and therefore answer different questions.

The cross-task comparison uses a common reference grid. Missing native decisions are mapped to switch 0, so the comparison reflects both switch quality and task

Table C.2: Switch-evaluation artifact locations.

Comparison type	Artifact location
Task-native model-vs-Perfect comparisons	<code>results/comparisons/switch_eval/<task_name>/<selection_id>/<comparison_id>/.</code>
Task-level model summaries	<code>results/comparisons/model_summary/<task_name>/<selection_id>/.</code>
Cross-task switch comparison	<code>results/comparisons/cross_task_switch/<selection_id>/.</code>
Perfect Switch reference	<code>results/switching/perfect_switch/<task_name>/<selection_id>/.</code>
Central result indexes	<code>results/index/.</code>

coverage. The task-native switch comparisons remain useful for understanding model behavior on the timestamps where each formulation is defined.

C.3 Additional Diagnostic Plots

The repository contains diagnostic plots beyond those shown in the main results chapter. These include forecast examples, worst forecast examples, horizon-wise error curves, duration-regression scatter plots, event-level switch timelines, cross-task heatmaps, and native decision-coverage plots. Table C.3 lists the main plot groups. Table C.3 is intended as a navigation aid for the generated visual audit trail. The plots are not all reproduced in the thesis, but they remain available for checking individual events, model failures, and coverage differences.

Table C.3: Main diagnostic plot locations.

Plot group	Location
Forecast diagnostics	<code>results/runs/autoregressive/<selection_id>/<run_id>/figures/.</code>
Duration-regression diagnostics	<code>results/runs/current_level_persistence/<selection_id>/<run_id>/figures/.</code>
Survival curves and survival diagnostics	<code>results/runs/survival_persistence/<selection_id>/<run_id>/figures/.</code>
Task-native switch timelines	<code>results/comparisons/switch_eval/<task_name>/<selection_id>/<comparison_id>/figures/event_switch_plots/.</code>
Cross-task timelines, heatmaps, and coverage plots	<code>results/comparisons/cross_task_switch/<selection_id>/<comparison_id>/figures/.</code>

These plots are intentionally kept as generated artifacts rather than copied exhaustively into the thesis. The main text includes representative examples that support the empirical discussion, while the artifact folders preserve the complete visual audit trail for further inspection.

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